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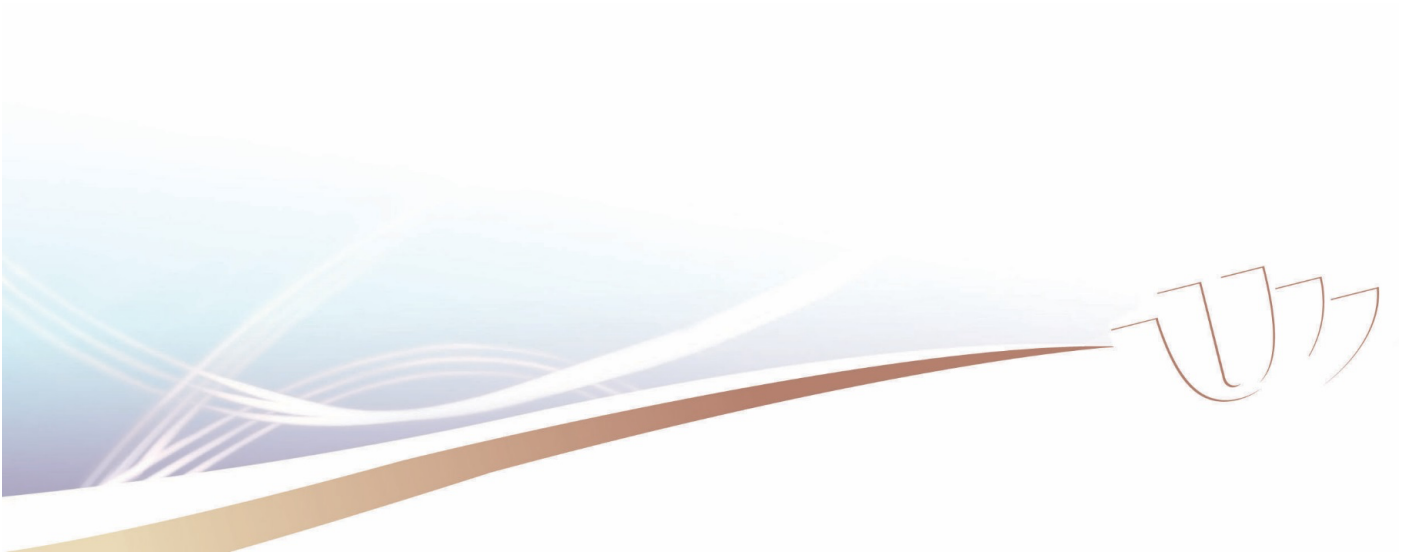
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Système de détection de la fraude financière à l'aide d'approches et de techniques d'intelligence artificielle

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Système de détection de la fraude financière à l'aide d'approches et de techniques d'intelligence artificielle

La fraude financière est devenue un défi majeur à l'ère numérique moderne, posant des menaces à la stabilité économique, à la confiance des clients et à l'intégrité des systèmes financiers. Les approches traditionnelles basées sur des règles et guidées par des experts pour la détection de la fraude ont montré des limites dans la résolution de la nature dynamique et évolutive des activités frauduleuses. Cette thèse présente une exploration approfondie d'un nouveau paradigme qui combine la puissance de l'Intelligence Artificielle (IA) basée sur la Connaissance et des techniques d'Apprentissage Automatique (ML) pour une détection efficace et robuste des fraudes financières.

L'objectif principal de cette recherche est double : d'une part, examiner la viabilité de l'utilisation des méthodes ML et DL pour détecter la fraude financière et d'autre part, évaluer l'efficacité d'une approche basée sur l'IA basée sur la Connaissance pour traiter les subtilités du paysage frauduleux en évolution. La recherche commence par proposer deux hypothèses : premièrement, que les méthodes ML et DL peuvent identifier efficacement les schémas frauduleux et deuxièmement, qu'une approche basée sur l'IA basée sur la Connaissance offre un cadre flexible pour intégrer des connaissances spécifiques au domaine pour améliorer la détection de la fraude.

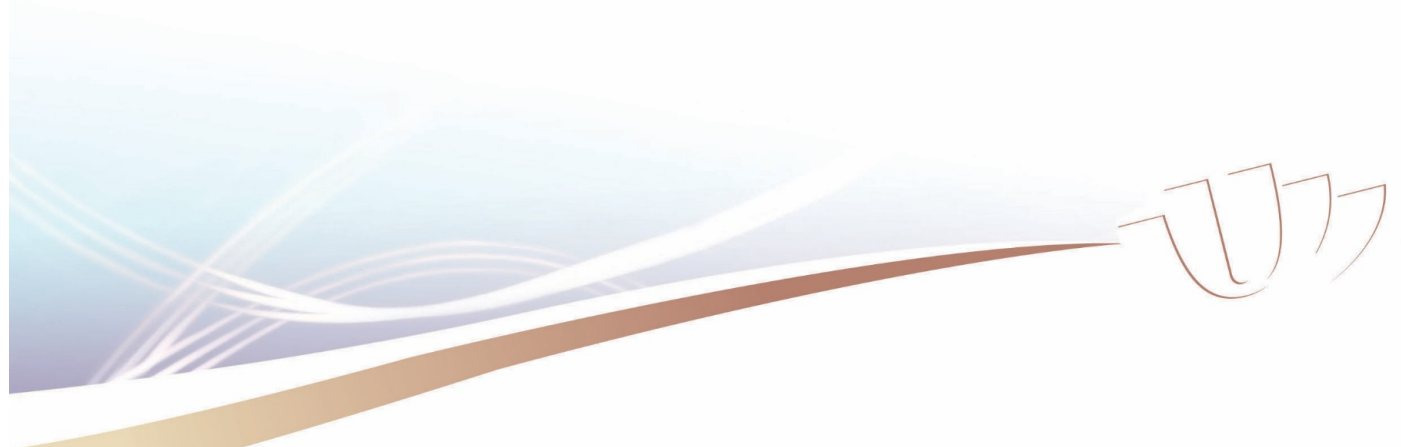
Pour tester ces hypothèses, un cadre expérimental approfondi est utilisé, impliquant divers algorithmes ML tels que XGBoost, Random Forest, les réseaux neuronaux, et plus encore. Un ensemble de données de fraude financière soigneusement sélectionné forme la base de ces expériences. Chaque algorithme subit un prétraitement rigoureux, un réglage des hyperparamètres, une formation et une évaluation, fournissant des informations précieuses sur leurs forces et leurs limites.

Parallèlement, une approche novatrice basée sur l'IA basée sur la Connaissance est développée. Cette approche tire parti d'un cadre basé sur une ontologie qui capture des connaissances spécifiques au domaine sur la fraude financière. L'ontologie est enrichie de règles et de relations qui permettent le raisonnement et la prise de décision. Cette approche basée sur la connaissance vise à combler le fossé entre les connaissances guidées par des experts et la détection automatisée, offrant ainsi une transparence et une interprétabilité dans les décisions de détection de la fraude.

Les résultats des deux approches, ML et IA basée sur la Connaissance, sont analysés de manière critique. Les forces et les faiblesses de chaque approche sont disséquées en termes d'exactitude, d'interprétabilité, de scalabilité et d'adaptabilité. Une comparaison entre les deux paradigmes est établie, mettant en évidence leurs avantages distincts et leur potentiel d'hybridation.

En fin de compte, cette thèse contribue à l'avancement de la détection de la fraude financière en mettant en lumière la synergie entre les méthodes ML et l'IA basée sur la Connaissance. Elle présente une enquête approfondie sur les capacités de chaque approche, offrant ainsi une feuille de route pour la création d'un système hybride qui exploite les points forts des deux paradigmes. Les conclusions de cette recherche fournissent des informations précieuses aux chercheurs, aux praticiens et aux décideurs souhaitant construire des systèmes efficaces et résilients de détection de la fraude financière dans une ère de menaces en constante évolution.

Mots-clés en français : Fraude financière, ontologie, menace, intelligence artificielle.



Financial fraud detection system using artificial intelligence approaches and techniques

This thesis addresses the very timely challenge of detecting and preventing more and more numerous financial frauds. The transition from physical/cash transactions to electronic/digital payments is highly explored by hackers and other cybercriminals. The primary targets of frauds are the banking sector, insurance, telecommunications, and e-commerce. Traditional financial fraud detection tools are insufficient in the current dynamic context due to the incredible creativity of fraudsters, who are permanently inventing new methods, and the lack of flexibility of regulatory systems.

In aim to efficiently address this challenge, we studied existing fraud detection approaches based on exploring various Artificial Intelligence (AI) techniques, as well as connectionist (machine learning and deep learning), knowledge-based, and blockchain. After a deep analysis of fraud methods and their dynamics and considering the availability of reliable data, we decided to build an ontology of frauds with the aim of having a "big picture" and an easy update.

This ontology was double validated, using an available dataset and exploring selected scenario of fraud. The first explored the clustering algorithm with similarity measures—the distance between each test data point and the centroid. The computed distances provided valuable insights into the effectiveness of the clustering algorithm and the quality of the assigned clusters.

This experiment allows for the evaluation of the model's performance in capturing the underlying patterns and characteristics of the dataset. It provides a quantitative measure of similarity, aiding in the assessment of the clustering algorithm's accuracy and the efficacy of the ontology-based approach. We also, gain valuable insights to capture fraud patterns. The obtained similarity scores obtained allowed for informed decisions and adjustments to improve the model's accuracy and reliability. The second method consisted of exploring three typical scenarios of attacks with ontology and anti-fraud rules.

The research findings indicate that ontology-based methods can enhance the accuracy and efficiency of financial fraud detection compared to alternative approaches in a dynamic context. In waiting for the dynamic models for ML and DL, the financial institutions can leverage the proposed framework to develop robust real-time fraud detection systems, reducing financial losses and enhancing customer trust.

Mots-clés en anglais : Financial Fraud, Ontology, Threat, Artificial Intelligence

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Abstract

This thesis addresses the very timely challenge of detecting and preventing more and more numerous financial frauds.

The transition from physical/cash transactions to electronic/digital payments is highly explored by hackers and other cybercriminals. The primary targets of frauds are the banking sector, insurance, telecommunications, and e-commerce. Large number of efficient fraud detection solutions have been developed, but the traditional financial fraud detection tools are insufficient in the current dynamic context due to the incredible creativity of fraudsters, who are permanently inventing new methods, and the lack of flexibility of regulatory systems.

In aim to efficiently address this challenge, we studied existing fraud detection approaches based on exploring various Artificial Intelligence (AI) techniques, including connectionist (machine learning and deep learning), knowledge-based, and blockchain, to detect frauds. After analysing fraud methods and their dynamics, and considering the availability of reliable data, we decided to build a financial-fraud ontology.

This ontology was double validated, using an available dataset and exploring selected scenario of frauds. The first explored the clustering algorithm with similarity to measure the distance between each test data point and the centroid. The computed distances provided valuable insights into the effectiveness of the clustering algorithm and the quality of the assigned clusters.

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The second method consisted of exploring three typical scenarios of attacks with ontology and anti-fraud rules. These scenarios transcend traditional algorithms, intertwining structured knowledge representation and adaptive rule systems. Through meticulous exploration, we unravel the complexities of fraud within specific contexts, deciphering not just the 'what' but the 'how' and 'why' behind each malicious act. In essence, this part stands at the intersection of pioneering a new era in fraud detection. By fusing ontology and anti-fraud rules, we present a comprehensive framework. This framework not only identifies fraud but comprehensively interprets its nuances, offering a profound insight into the modus operandi of digital mal efactors.

The findings from the second part not only advance the field of fraud detection but also bear significant implications for financial institutions, regulatory bodies, and cybersecurity experts to bridge the gap between artificial intelligence and domain expertise.

The research findings indicate that ontology-based methods can enhance the accuracy and efficiency of financial fraud detection compared to alternative approaches in a dynamic context. In waiting for the dynamic models for ML and DL, the financial institutions can leverage the proposed framework to develop robust real-time fraud detection systems, re-

ducing financial losses and enhancing customer trust.

Keywords: Financial Fraud, Ontology, Fraud Detection, Knowledge model, Reasoning rules, Machine learning, Dynamic model, Scenario, Clustering.

Résumé

La fraude financière est devenue un défi majeur à l'ère numérique moderne, posant des menaces à la stabilité économique, à la confiance des clients et à l'intégrité des systèmes financiers. Les approches traditionnelles basées sur des règles et guidées par des experts pour la détection de la fraude ont montré des limites dans la résolution de la nature dynamique et évolutive des activités frauduleuses. Cette thèse présente une exploration approfondie d'un nouveau paradigme qui combine la puissance de l'Intelligence Artificielle (IA) basée sur la Connaissance et des techniques d'Apprentissage Automatique (ML) pour une détection efficace et robuste des fraudes financières.

L'objectif principal de cette recherche est double : d'une part, examiner la viabilité de l'utilisation des méthodes ML et DL pour détecter la fraude financière et d'autre part, évaluer l'efficacité d'une approche basée sur les connaissances pour traiter les subtilités du paysage frauduleux en évolution. La recherche commence par proposer deux hypothèses : premièrement, que les méthodes ML peuvent identifier efficacement les schémas frauduleux et deuxièmement, qu'une approche basée sur les Connaissances offre un cadre flexible pour intégrer des connaissances spécifiques au domaine pour améliorer la détection de la fraude.

Pour tester ces hypothèses, un cadre expérimental approfondi est utilisé, impliquant divers algorithmes ML tels que XGBoost, Random Forest, les réseaux neuronaux, et plus encore. Un ensemble de données de fraudes financières soigneusement sélectionné forme la base de ces expériences. Chaque algorithme subit un prétraitement rigoureux, un réglage des hyperparamètres, un entraînement et une évaluation, fournissant des informations précieuses sur leurs forces et leurs limites.

Parallèlement, une approche novatrice basée sur les connaissances est développée. Cette approche est fondée sur une ontologie qui capture des connaissances spécifiques au domaine des fraudes financières. L'ontologie est enrichie de règles et de relations qui permettent le raisonnement et la prise de décision. Cette approche vise à combler le fossé entre les connaissances guidées par des experts et la détection automatisée, offrant ainsi une transparence et une interprétabilité dans les décisions de détection des fraudes.

Les résultats des deux approches, ML et ontologie, sont analysés de manière critique. Les forces et les faiblesses de chaque approche sont disséquées en termes d'exactitude, d'interprétabilité, de scalabilité et d'adaptabilité. Une comparaison entre les deux paradigmes est établie, mettant en évidence leurs avantages distincts et leur potentiel d'hybridation.

En fin de compte, cette thèse contribue à l'avancement de la détection de la fraude financière en mettant en lumière la synergie entre les méthodes ML et les ontologies. Elle présente une étude approfondie sur les capacités de chaque approche, offrant ainsi une feuille de route pour la création d'un système hybride qui exploite les points forts des deux paradigmes. Les conclusions de cette recherche fournissent des informations précieuses aux chercheurs, aux praticiens et aux décideurs souhaitant construire des systèmes efficaces et résilients de détection de la fraude financière dans une ère de menaces en constante évolution.

Mots-clés : Fraude financière, Ontologie, Détection des fraudes, Modèle de connaissances, Règles de raisonnement, Apprentissage automatique, Modèle dynamique, Scénarios, Clustering.

GENERAL INTRODUCTION

Fraud poses a serious risk to our day-to-day existence and is getting worse. The figures that are now available indicate a notable rise in financial fraud as a result of technological advancements and the shift from cash to digital or cashless financial transactions. This gives scammers a great chance to investigate other avenues for tricking clients, retailers, and big businesses. The crooks' strategies are incredibly inventive, adaptable, and dynamic. Individuals tend to preserve money to protect their financial stability, but that security can be virtually erased or endangered at any time. Fraud experts anticipate that this trend will continue or pick up speed as older baby boomers become more and more common targets for fraud. Based on approximations, financial fraud

1.1 Motivation

The evolution from physical or cash modes of transactions to electronic or digital payments is another contributing factor to financial fraud. With the widespread adoption of electronic payments, there has been a significant increase in cybercrime and fraud. Fraudsters can use various techniques, such as phishing, identity theft, and social engineering, to gain access to sensitive financial information and engage in fraudulent activities.

Increasing complexity of financial systems, This factor contributes to financial fraud as financial systems become more complex, it becomes harder to detect fraudulent activities. Fraudsters can hide their activities behind a maze of financial transactions, making it challenging to trace the origin of fraudulent transactions.

Unemployment is a key factor to consider in the rise of fraud. This thesis at some point used Nigeria as a key study area. In this case, in today's Nigeria, a number of cyber crimes and the like are perpetuated on a regular basis in diverse forms, among which are, but are not limited to, fraudulent electronic mails, cyber impersonation of characters, spamming and phishing, or hacking of automated teller machines (ATM), just to mention a few. It is instructive to note that the crimes are perpetuated through cyberspace, and assurances of huge financial gains are used by the fraudsters to lure their online victims. The high

rate of unemployment is key to the surging volume of notorious crime. The possibility of establishing a significant correlation between the unemployment rate in Nigeria and financial fraud is more palpable given the fact that "the unemployment level of the youth is twice that of the adult's unemployment.

Fraudsters take advantage of digitization processes; this enable fraudsters to commit financial fraud with increasing efficiency and sophistication. A significant amount of financial fraud takes place through banking and E-commerce, while at the same time people and businesses are migrating to the digital platform. Nowadays, people demand more online and mobile services from their financial institutions. Users are doing more transaction approvals online and de-emphasizing in person transactions, which makes it harder to verify identities. Also, in response to user demand, a new breed of online banks has been born and is doing business entirely in the online world. This makes fraudsters jobs easy by targeting users of such banks. Many of the newly invented digital payment technologies on which the companies rely are being adopted by fraudsters. The existing fraud prevention systems that many of the financial institutions have deployed are now struggling to keep up.

The lack of awareness and education about financial fraud is also a contributing factor. Many people are not aware of the various types of financial fraud, the complexity of frauds and how to protect themselves. This lack of knowledge is an added advantages to the fraudsters vulnerable to fraudulent activities.

The dynamic nature of recent fraud patterns makes them difficult to detect and prevent. In recent years, fraudsters patterns have dramatically changed, and as criminals take advantage of digitization of financial transaction, which enable them to commit financial fraud with increasing efficiency. A significant amount of financial fraud takes place through banking and E-commerce, while at the same time the pandemic has only hastened the emergence of digital money laundering. Sophistication is one of the dynamics of fraud, which means it is more difficult to detect and often undertaken by criminals who run large scale fraud operations. Recently, there has been an increase in the number of these organized and sophisticated attacks, such as deep fakes, 2D and 3D masks, and skimming.

Some of the sophisticated patterns used by the fraudsters involve data breaches. Once fraudsters gain a breakthrough to access personally identifiable information and use it against the user, by combining real and fake data . The criminals open bank accounts and cards, acting like legitimate customers. Fraudsters also use another pattern of account takeover: using passwords and credentials obtained by data breaches and social engineering. Skimming is one of the most sophisticated and dynamic patterns of fraud. This technique involves inserting chips inside ATM machines, gas stations, and points of sale to steal user card information. The device owner can then use this information to make fraudulent transactions, such as online purchases.

Nature of Related Data and Knowledge: The data is the sole entity for the identification of fraudulent patterns, such as unusual transactions or suspicious activity. This data can be gathered from various sources, such as transaction records, financial statements, customer databases, and other records. We analyze various data to understand the behavior of potential fraud activities as well as to provide us with insights into the nature of the fraud.

Furthermore, we used the knowledge acquired to understand the reasoning behind the nature and scope of several domains of financial fraud. This includes the types of fraud, the perpetrators, the victims, and the methods used. Also, we utilized the knowledge for insight into methods of detection, prevention, and mitigation of financial fraud. These have been used to improve strategies to improve the accuracy of the proposed system. In financial fraud detection, we can use several types of data to detect fraud, but the most accurate and qualitative data for financial fraud detection are transactional data. This transactional data includes information about individual or organizational transactions, such as the amount, date, and time of the transaction, as well as information about the parties involved in the transaction. All of these types have been used to implement ontological reasoning for financial fraud knowledge and elaborations of patterns.

Complexity and Velocity of Changes: The complexity of changes in financial frauds varies greatly; we analyzed and classified these depending on the type of fraud involved. For example, frauds that involve the use of complex and sophisticated techniques require a more detailed analysis than simpler frauds, such as credit card fraud or identity theft. Additionally, the speed at which the fraud is committed will also affect the complexity and velocity of the changes.

The present study uncovered a diverse array of social and economic ramifications stemming from financial fraud, which harm not only individuals but also enterprises, organizations, governments, and the economy at large. One result of this has been a decline in economic activity due to a lack of faith in financial institutions. Financial fraud has also diminished society's general well-being and weakened its social cohesion. As these actions undermine credibility and trust, another effect of financial fraud to take into account is the damage to a company's reputation. Fraud undermines social cohesion and confidence by undermining trust, both of which are necessary for fostering a cooperative and collaborative culture. Additionally, we took into account the negative economic effects at different phases.

At the core of financial fraud is the fraudster, who initiates the fraudulent scheme. The fraudster may be an individual or a group of individuals, and their motivations for committing financial fraud can be diverse, ranging from greed to desperation. The fraudster may employ various tactics to deceive or manipulate financial transactions, including forgery, false statements, identity theft, and insider trading. Another key actor in the dynamics of financial fraud is the victim. Victims of financial fraud can be individuals, businesses, or governments. The consequences of financial fraud on victims can be severe, including loss of assets, reputation, and trust. Victims of financial fraud may also face legal penalties or regulatory sanctions, depending on the nature and scope of the fraud.

Financial institutions, such as banks, insurance companies, and investment firms, also play a critical role in the dynamics of financial fraud. Financial institutions are often the targets of financial fraud as fraudsters seek to exploit vulnerabilities in their systems and processes. Financial institutions also have a responsibility to detect and prevent financial fraud as part of their duty to protect their customers and stakeholders. At the same time, the regulatory environment also contributes to the dynamics of financial fraud. Regulations and laws aim to provide a framework for preventing and detecting financial fraud, but can also create unintended consequences and loopholes that fraudsters may exploit. In some

cases, regulations may be inadequate or outdated, creating opportunities for financial fraud to occur.

All these factors contributing to the rise of financial fraud are stacked against several research approaches to prevent it. Until artificial intelligence (machine learning based) started to be used to combat it. While large customer datasets are required for a machine learning based approach for the purpose of analyzing user transaction patterns to determine whether a particular transaction on a user account is fraudulent or genuine, understanding the dynamics of financial fraud is critical for developing effective prevention and detection strategies, as well as for mitigating the impact of financial fraud on victims and society at large. This is why ontologies can be more flexible by providing accurate knowledge between fraud patterns and solutions to detect the fraud.

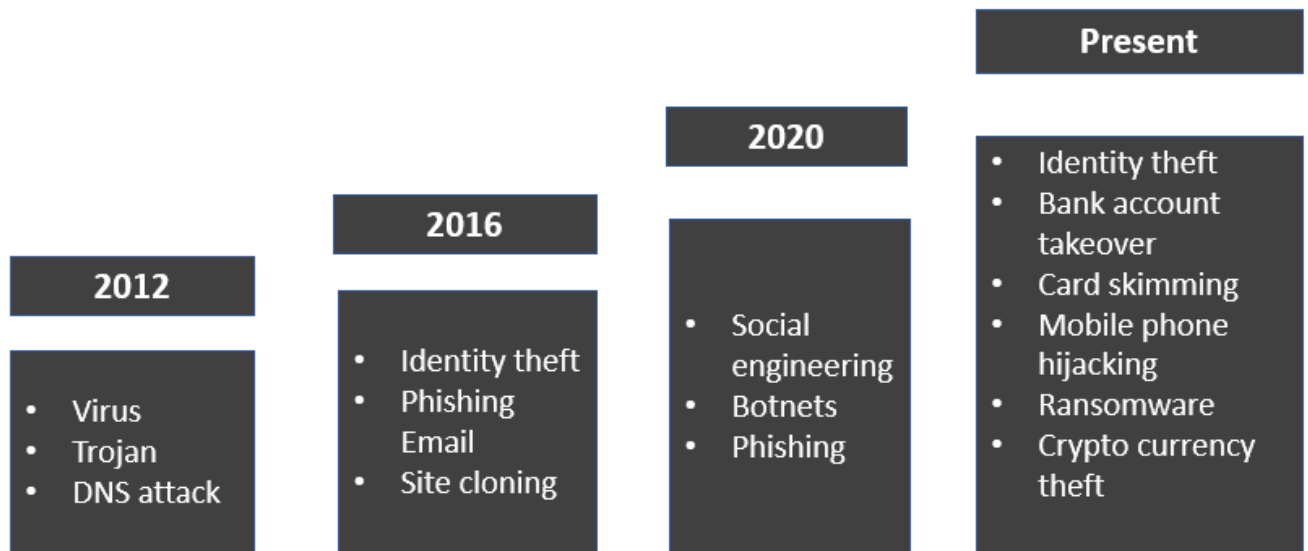


Figure 1.1: Dynamic of fraud chart.

1.2 Research Objective

The aim is to investigate machine learning methods used for detecting financial fraud; and second, to proposed effective knowledge based AI approach in tackling the intricacies of the ever-changing fraud landscape.

Financial fraud detection demands a high degree of accuracy and precision to minimize false positives and negatives. The research aims to develop methodologies and algorithms that enhance the accuracy of fraud detection, reducing the financial losses and reputational damage caused by fraudulent activities.

In today's digital financial landscape, we understood the requirements of a system that can keep pace. The research seeks to develop solutions that enable dynamic and real-time

fraud detection, responding to evolving fraud tactics as they happen.

Moreover, compliance with financial regulations and standards is of paramount importance. The research work aims to ensure that the developed fraud detection systems align with and facilitate compliance with relevant industry regulations and data protection laws.

Also, we endeavor to create knowledge-based system that is not only effective but also transferable across domains. This means that the methodologies and insights gained from this work can potentially be applied to other knowledge-intensive fields beyond financial fraud detection.

Ultimately, the aim is to bridge the gap between research and practical implementation. The research seeks to provide tangible, deployable solutions that financial institutions and organizations can use to bolster their fraud detection capabilities effectively.

Finally, the aim of this research work is to contribute significantly to the development of more robust, accurate, and adaptable financial fraud detection systems. By integrating ontology-based knowledge with machine learning, the research endeavors to enhance the capabilities of financial institutions and organizations to combat fraud effectively in an increasingly digital world.

The research hypotheses that guide this exploration are as follows:

1.3 Hypothesis

There are two possible ways to face the stated problem, which are connectionist AI and Knowledge Base AI. These hypotheses rely on a continuous process of preventing, authenticating, and verifying the patterns of fraudsters and legitimate financial transactions.

1.3.1 Machine learning techniques

The machine learning hypothesis is rooted in the premise that ML techniques can process and analyse vast amounts of transactional data to uncover hidden patterns indicative of fraud. ML models can adapt and learn from data, identifying complex relationships that might not be apparent through manual analysis. The goal is to investigate how ML algorithms, such as Random Forest, Support Vector Machine (SVM), and Neural Networks, can enhance financial fraud detection, leading to more reliable and efficient outcomes.

To test the hypothesis, various ML algorithms will be experimented and evaluated using real world financial transaction datasets. The models will undergo training and testing to measure their accuracy, precision, recall, F1 score, and other performance metrics. These experiments will provide insights into the strengths and limitations of each model in financial fraud detection.

If the hypothesis holds true, the outcomes are substantial as follows:

Machine learning models expedite the detection process, reducing the time and resources required to identify fraudulent activities. Moreover, machine learning models can adapt to changing fraud patterns, minimizing the risk of false negatives and keeping up with evolving tactics. Data driven Insights: Through feature importance analysis, ML models can provide insights into the factors driving fraud, aiding in proactive fraud prevention.

Machine learning models will autonomously identify the most influential features (variables) that contribute to fraudulent transactions, aiding investigators in understanding the underlying patterns and indicators of fraud.

Furthermore, machine learning models will demonstrate scalability, enabling the handling of vast datasets and high transaction volumes encountered in large financial systems without compromising efficiency. Also integrating machine learning with domain specific knowledge, like ontologies, will offer a more holistic and interpretable approach to fraud detection, combining the strengths of both methods for improved accuracy and explainability.

Finally, machine learning hypothesis reflects a significant leap in the field of financial fraud detection, emphasizing the potential of ML techniques to revolutionize the accuracy, efficiency, and adaptability of fraud detection systems. The outcomes of this research have the potential to shape the future landscape of financial security, fostering a safer and more robust financial ecosystem.

1.3.2 Knowledge base AI

Knowledge base AI refer to a set of methods and approaches that leverage knowledge representation and reasoning to enable intelligent systems to store, organize, retrieve, and reason with structured knowledge. These techniques are designed to capture and utilize explicit knowledge in a machine readable format, and to allow AI systems to make informed decisions, answer queries, and solve complex problems.

There are various knowledge based techniques used for financial fraud detection (e.g., ontology, case based reasoning, and expert systems). Knowledge based techniques can be very effective in detecting financial fraud. In these approaches, we collect and analyze information about types of financial fraud, methodologies and techniques used by fraudsters, and characteristics of fraudulent transactions. Reference to the submission by [37] one of the most widely used knowledge base techniques for financial fraud detection is ontology. Ontology approaches have been very effective for financial fraud detection and are useful in solving several challenges compared to a machine learning approach. Using ontology for financial fraud detection can give us a wide variety of knowledge about several domains of financial fraud, identify the concepts of financial fraud, and identify relationships between them. Finally, using ontology we can implement an intelligent rule based to detect financial fraud. There are various challenges with knowledge based fraud detection techniques that must be addressed, these challenges are discussed below:

A knowledge based financial fraud detection system leveraging domain specific knowledge in the form of an ontology will lead to improved accuracy in detecting fraud instances compared to solely data driven machine learning approaches.

The knowledge based approach will be more adaptable to dynamic and evolving fraud patterns, as the underlying ontology can be updated to incorporate new knowledge and insights into emerging fraud techniques. While utilizing an ontology for fraud detection will result in more interpretable and explainable fraud detection outcomes, allowing analysts to understand the reasons behind fraud predictions and enabling regulatory compliance.

Moreover knowledge based approach will have an advantage in detecting complex and multi dimensional fraud schemes that involve intricate relationships among various attributes, which may be challenging for traditional machine learning algorithms.

Incorporating collaborative knowledge from multiple experts in the field of financial fraud detection into an ontology will enhance the overall fraud detection capability, leveraging the collective intelligence of domain experts. At the same time, the use of a knowledge based approach will provide deeper insights into the domain specific characteristics and indicators of financial fraud, leading to more accurate detection of fraudulent activities.

Finally, integration of knowledge based and data driven approaches will result in a hybrid system that leverages the strengths of both methodologies, enhancing fraud detection accuracy while maintaining interpretability.

1.4 Research Challenge

There are various techniques proposed in artificial intelligence and other advanced detective approaches such as the research by [75] for Intelligent financial fraud detection practices in post pandemic era. [138] and the approach proposed by for Impression Allocation for Combating Fraud in E-commerce Via Deep Reinforcement Learning with Action Norm Penalty. The research work by [106] implement an approach for Deep learning fraud detection in credit card transactions.

Constructing a comprehensive and accurate ontology for financial fraud requires meticulous domain expertise and collaboration with domain experts. The challenge lies in the accurate elicitation of fraud related concepts, relationships, and rules, followed by the validation of the ontology's effectiveness in capturing real world scenarios. In addition, the financial landscape is highly dynamic, and fraud tactics evolve rapidly. To create a knowledge base that remains relevant and adaptable to emerging fraud patterns demands continuous updates and revisions. Balancing the need for timely updates with the stability of the ontology is a challenge.

Moreover, representing intricate fraud scenarios and domain specific rules in a structured format while ensuring clarity and interoperability can be complex. While the knowledge based systems heavily rely on explicit domain knowledge, which may lack the insights derived from data driven machine learning approaches. Bridging the gap between knowledge driven and data driven insights is a challenge in creating holistic fraud detection solutions.

Furthermore, financial datasets are often voluminous, complex, and noisy. Cleaning and preprocessing data for accurate analysis is challenging, involving data imputation, handling missing values, and managing imbalanced classes to avoid skewed model performance. At the same time extracting meaningful features from financial data that effectively capture fraud patterns is intricate. Balancing the selection of relevant features with reducing dimensionality to avoid overfitting is a continuous challenge.

Finally, to integrate knowledge based approach with machine learning model to create hybrid systems presents challenges in maintaining interpretability, scalability, and accuracy. Achieving synergy between the two paradigms is intricate. Thus to develop a solution that comply with ethical standards, data privacy regulations, and industry norms while effectively countering fraud poses challenges in balancing innovation and responsibility.

1.5 Organization of the Thesis

The organization of the thesis provides a brief explanation of the thesis' structure and organization.

Introduction: Briefly explain the background and motivation of the thesis. The section on financial fraud analysis looks at the critical elements in financial fraud and detection, including the level of sophistication, the role of data and knowledge, etc.

The research challenges are clearly stated, with a clear view of the hypothesis of the thesis. These hypotheses summarize the novelty of the approaches that will be achieved by combining two different methods. In this sense, the output of ontology is used as an instrument to build an intelligent rule based fraud detection system. The Research contributions have been identified to use and deploy an ontology for financial fraud (FROD). The ontology will identify the relationships between various entities in financial fraud, such as individuals, financial institutions, victims, criminals, and transactions.

Related work: We developed a comprehensive survey that concentrated on artificial intelligence techniques for financial fraud detection. We analyzed and classified several fraud detection solutions according to the used detection techniques, namely knowledge based, learning based, and blockchain based solutions.

Building of Ontology System for Financial Fraud Detection: In this stage of the thesis, an ontology to describe financial frauds and related knowledge is proposed and developed. The aim of this ontology is to provide a semantic framework for the detection of financial fraud. The theoretical ontology named FROD has been elaborated by exploring various sources of information. FROD ontology presents details of a theoretical ontology with various domains and concepts of financial fraud. The properties and relationships have been implemented.

Ontology Validation: This ontology was designed and built based on multiple layers; as a result, validation should be done layer by layer or component by component. The aim of this section is to determine a concise and acceptable ontology for experts and users. We identified a set of ontology quality criteria and ontology aspects that must be achieved, including accuracy, adaptability, clarity, completeness, computational efficiency, and consistency. To validate this financial fraud ontology, we proposed two distinct approaches namely.

Dataset based validation: This process involves using a dataset to test and validate the ontology and ensure that it accurately represents the real-world domain it is intended to model. One method of data-based validation is to use instance data to test the ontology. Instances consist of specific examples of entities and relationships within the domain being modeled. The main steps of a dataset-based validation approach are: 1: split the dataset into two parts; 2: use the first part to instantiate the corresponding concepts of the ontology; 3: define the centroids of the set of instances of each instantiated concept; 4: using a clustering algorithm; and 5: use the second part of the dataset to evaluate the ontology and the different centroids. Another important step is to analyse the dataset. Indeed, a dataset cannot be used if it is not compliant with the ontology.

Scenario based validation: The Scenario based validation is an approach that involves validating the effectiveness and performance of the ontology by simulating and evaluating different real-world scenarios. This approach allows for a comprehensive analysis of the system's behavior and its ability to handle specific scenarios.

The three simulated scenarios of Identity theft and banking fraud where an attacker impersonates an individual to gain unauthorized access to sensitive information or perform fraudulent transactions. In the context of investment fraud, a robust detection method is needed, and the detection method must be knowledgeable enough to detect fraud involving

a dubious investment offers. The system's ability to identify suspicious patterns, detect unauthorized access attempts, and at the same time snipped fraudulent investment schemes is evaluated within this scenarios.

Exploration of rules, Testing of rules: This part proposes an exploration of rules that allows for the identification fraud patterns and capture the relationships, and characteristics of fraud knowledge in the ontology. The explored rules offer solutions for financial fraud in some selected fraud scenarios. The scenarios simulated how fraudsters penetrate user accounts to steal useful information for identity theft or perform unauthorised transactions. Based on the ontology knowledge, we formulated the rules to capture the relationships and patterns that signify fraudulent activities. These rules include if-then statements and logical expressions that describe the conditions and actions associated with fraudulent behaviours

Comparison: Machine learning algorithms possess a great power for financial fraud detection. They usually relied on dataset to uncover a fraud pattern. While FROD ontology systems rely on a predefined knowledge model of financial fraud. This part discuss a comprehensive comparison between the experimented machine learning models for financial fraud and our FROD validated ontology. We aimed to dissect the attributes of both approaches comprehensively. We explored their flexibility, interpretability, resource efficiency, adaptability, and propensity for generating false positives. Through this comparison, we seek to provide a clear understanding of the strengths and limitations of each approach and, importantly, investigate the prospects of a hybrid system that synthesizes the best of both worlds.

Discussion and Conclusion: The discussion chapter deliberates on the general aspects of the thesis, from related works to the development and validation of the ontology. The sophistication and challenges of fraud detection will be discussed. The conclusion part is specifically for the validation of the ontology as well as future work in relation to the thesis.

Formation doctorale: The formation doctorale of the University of Reims makes it compulsory for every PhD student to achieve some specific credit hours in class. Hence, I undertook courses that are relatively advantageous to my career.

Publications: We published two research papers related to this thesis. Specifically, the elaboration of financial fraud ontology and the proposed intelligent rule based approach for financial fraud detection The third publication, titled Survey of Fraud Detection, is being submitted.

References: The references section lists all the sources cited in this thesis in a plain bibliography citation style.

FINANCIAL FRAUD DETECTION SURVEY

2.1 Introduction

Technology has changed the way we live and the way we conduct our day-to-day activities. In this digital age, people spend more time on the internet for commercial and other transactional activities. Hence, cybercrime, and more particularly, Internet fraud, has risen at the same rate, if not faster. There have been an estimated 4.5 billion dollars in losses due to Internet fraud, according to the Federal Bureau of Investigation. Alexander Gillis [83] described fraud as a set of activities undertaken to prevent money or property from being obtained through false pretenses. Fraud detection has been applied to many industries, which include banking, insurance, and e-commerce. Although several effective fraud detection tools have been created, additional study and cooperation will be required to combat the growing sophistication of fraudsters. The Federal Bureau of Investigation (FBI) of the United States states that fraud is the art of using deception to unfairly obtain and deny a victim of a lawful right to monetary advantages and other benefits [39]. Money and deception have gone hand in hand over the ages. Fraudsters look for different companies where they can find chances to accomplish their objectives. We provide an overview of the state of the art in the field of fraud detection in this survey. First, the definition, background, and subcategories of fraud detection are covered.

2.1.1 Contributions and Goals

The survey analyses the existing surveys and motivates the need to propose a new survey. In recent years, we have seen a rise in fraud detection research due to the dramatic increase in fraud, which has resulted in billions of dollars of losses worldwide.

In the last 20 years, many survey papers have been published that have studied, analysed, and compared the large amount of research work that has been conducted on fraud detection. In the following, we will discuss these surveys according to two main criteria: the covered fraud categories and the covered fraud detection techniques.

According to the covered fraud categories, we can find domain-related surveys [17, 71, 125, 7, 26, 82], which review fraud detection solutions with regard to one specific fraud category, and domain-independent surveys [64, 95, 14, 1], which address various fraud categories. Most of the existing domain-related surveys, such as the surveys [17, 31, 7, 26, 82] focus on finance and credit card fraud, while the rest of the surveys focus on healthcare [71, 76], and accounting frauds [125].

Existing surveys can also be categorised according to the covered fraud detection techniques. The surveys [71, 125, 95, 76] review fraud detection solutions that employed data mining and statistical techniques. The surveys [17, 64, 31, 125, 1, 7, 26, 82], focused on machine learning-based fraud detection. Bolton et al. [17] focused only on solutions that employ unsupervised learning, while the rest of the surveys covered both supervised and unsupervised learning-based solutions. The survey of Adewumi et al. [7] covered machine learning, bioinspired, and ontology-based (i.e., a subcategory of knowledge-based techniques) solutions. For the latter category of solutions, there was only one review. Behdad et al. [14], and Delamaire et al. [31] also focused on bioinspired solutions for fraud detection. The survey of Behdad et al. [14] covered a wide variety of nature-inspired techniques such as artificial neural networks (ANNs), evolutionary and genetic algorithms, swarm intelligence, and artificial immune systems, while the one of Delamaire et al. [31] covered only solutions that employ genetic algorithms. Finally, the survey of Choi et al. [26] surveyed conventional machine learning, as well as deep learning solutions for fraud detection. However, it only focused on the IoT environment.

One first observation about the existing surveys, as it has also been stated by [1], is that the majority do not provide any insights regarding issues and challenges that face fraud detection systems. In addition, there are no existing surveys that classify and discuss in a detailed manner knowledge-based fraud detection solutions, including ontologies, expert systems, etc. For instance, the surveys of [2], [64], [7] briefly discuss a few knowledge-based approaches, which employed ontologies and expert systems for fraud detection without any specific focus, and the concepts related to this category. We believe that this category of approaches needs particular attention due to their huge potential in the field of fraud detection. We also believe that there is a need for a recent and comprehensive survey that reviews recent fraud detection solutions, especially those utilising cutting-edge technologies such as blockchain. Indeed, the first study exploring the use of blockchain technology in the context of fraud and privacy prevention was conducted by Cai et al. [21] in 2016, and since then several research works have been published in the literature to employ this revolutionary technology in the context of fraud detection. However, to the best of our knowledge, there is no existing survey paper that comprehensively reviews these solutions.

This survey study attempts to provide a more up-to-date and thorough review of learning-based (such as machine learning and deep learning), knowledge-based (such as ontology-based, Case-Based reasoning, and expert systems), and blockchain-based fraud detection techniques. To the best of our knowledge, this poll is the first to incorporate blockchain-based solutions as well as the three categories. Furthermore, a wide range of fraud categories are covered by our survey, which is domain-independent (i.e. financial fraud, identity theft, auction fraud, and investment fraud). This survey paper's primary contributions are as follows:

1. We provide a rich, content-rich background on fraud, fraud categories, and the characteristics of each category.
2. We comprehensively surveyed fraud detection solutions in a domain-independent manner by covering various fraud categories.
3. We provide a taxonomy that covers the existing fraud detection techniques such as data mining, knowledge-based techniques, traditional machine learning, deep learning, and blockchain-based solutions.
4. Furthermore, we provide an insight into the limitations and challenges of the existing fraud detection solutions, as well as some recommendations to overcome these challenges.

2.1.2 Structure of the Survey

This survey is organised as follows: Section II provides a definition of fraud. In Section III, we present the various fraud categories. Section IV introduces our taxonomy of fraud detection approaches. In Sections V, VI, and VII, we review, analyse, and discuss the current state of the art of fraud detection approaches according to the proposed taxonomy. Section VIII highlights the challenges of the existing fraud detection approaches as well as our recommendations. Section IX concludes our survey.

2.1.3 History of Fraud

The most famous fraud is known as Ponzi schemes. This type of fraud has been in existence for many years. However, when selling to investors he over exaggerated this margin, promising a 50% profit return. Investors jumped at the opportunity to queue up to give him their money. Ponzi paid his early investors back with the money from his newer more recent ones. When scam was finally discovered, Ponzi had made 10 million US dollars and had fled the country.

2.2 Definitions of Fraud

The first dilemma in fraud detection research is determining what exactly constitutes fraud. Merriam Webster [**Merriam Webster**] defines a fraud as an intentional perversion of truth in order to induce another to part with something of value or to surrender a legal right.

Certified Fraud Examiners [4] defines fraud as the use of one's occupation for personal enrichment through the deliberate misuse or misapplication of the employing organisation's resources or assets.

The American law dictionary [97] defines fraud as a false representation of a matter of fact either by false or misleading allegations or by concealment of what should have been

disclosed that deceives and is intended to deceive another so that the individual will act upon it to her or his legal injury. This definition provides a wider and more multifarious description of fraud.

The Federal Bureau of Investigation [39] defines a fraud as the use of Internet services or software with Internet access to defraud victims or otherwise take advantage of them. This definition of fraud is considered the general and most accepted definition of fraud because it encompasses the Internet," which is necessary before performing the fraud.

In the journal of Business Ethics [40], a fraud is defined as a violation of trust at its core; it is a human act intended to deceive another in such a way that the victim suffers a loss and the perpetrator makes an unlawful gain. This definition stands in line with the definition proposed by the Law Dictionary.

While Rossouw et al. [105] have differentiated between the terms corruption and fraud since the two are often used interchangeably, Define fraud as intentional deception by concealing or misrepresenting information that harms the financial interests of another person and benefits the financial interests of the perpetrator.

With the enormous potential of Internet service, where it brings people together for business, dating, and other services. This has a huge impact on the term fraud, as many researchers believe that the internet has given the term fraud another meaning. Hence, in the book "Encyclopedia" of Cyber. Bachmann, Michael, and Smith [131], fraud is defined as any act of intrusion, expression, omission, or concealment using interconnected.

Under United States law[97] fraud is defined as the act of non-disclosure by any person intending thereby to cause a mistake to another person to induce the latter to enter into or refrain from entering into a transaction or agreement.

Grazioli et al. [45] define fraud with the term "consumer deception. The keywords are "deceptive and unfairly", emphasising that consumers are subject to being made to believe in a person with a false identity, which results in counterbalancing benefits.

Beals et al. [13] Understand that much of fraud research has focused on fraud against governments and organisations, which, according to him, is distinct from fraud against individuals, both in its methods and in its players. Hence, he defines fraud as intentionally and knowingly deceiving the victim by misrepresenting or concealing facts about promised goods, services, or other benefits and consequences that are nonexistent, unnecessary, never intended to be provided, or deliberately distorted for the purpose of monetary gain.

Gupta et al. [49] have differentiated between the terms misconduct and fraud or breach of trust, for monetary gain.

Akers et al. [8] have stated that any crime committed with materially false statements requires knowledge that the statement was false, reliance on false statements by the victim, and damages resulting from the victim's reliance on the false statement. Can be termed fraud.

The above definitions give several interesting concepts and characteristics of fraud. We propose to reuse the definition of Merriam-Webster. A fraud can thus be defined as any

type of activity, scheme, or communication that comprises the use of Internet service, telephone service, or direct communication to obtain something of value by deception from individuals or organisations.

There are certain similarities between some of the above definitions. These similarities are summarised below.

The most significant overlaps are the inclusion of the notion of intent from the actors and the notion of causing monetary loss from the victims. In the legal aspect of fraud, these two notions are very important. Although there are some notable similarities in the definitions, there are also distinct differences. Some of which create unique challenges when applying the definition of fraud. While many of the definitions hold that fraud must result in the loss of something of value, Others explain that it may induce someone to surrender a legal right.

2.3 Classification of Frauds

As described above, this decade has witnessed the advancement of Internet technology (digital age). Thus, we can unanimously agree that the Internet has simplified things on many fronts, be it our business, banking, and many more. The technological change seems to have been quite remarkable. But criminals have also taken advantage of this to improve their methods. But so did the criminals, using different methods.

In this section of the survey, we will discuss the different classes of fraud and the related methodology.

2.3.1 Financial Fraud

Financial fraud has many definitions from various scholars. Eurink [103] defined financial fraud as a complex and elusive concept, both as a behavioural category and as a legal one. For example, in financial activities such as banking, securities, and insurance, However, fraud can have a more specific meaning. In this regard, we describe financial fraud as the unlawful falsification or manipulation of financial activities.

Banking Fraud

In a legal context, bank fraud is defined as using deception to steal money or assets from a bank for legal purposes. Financial institutions like banks include the Federal Reserve Bank, the Deposit Insurance Corporation, mortgage banks, and commercial banks. In a financial context, bank fraud is the use of potentially illegal means to obtain money, assets, or other property owned or held in the bank, or to obtain money from customers by fraudulently acting as a bank or other financial institution.

Tax Fraud

There are several forms of tax fraud, including tax evasion, fake income, corporate tax fraud, and many more. In a simple term, tax fraud can be defined as the willful or intentional falsification of information to limit the amount of tax liability. So, tax fraud entails cheating on a tax return with the sole aim of avoiding paying the entire or part of the tax obligation.

Insurance Fraud

Richard A. Derig, in [33], made an analysis of insurance fraud in comparison with railway spine, a disease from the 19th century for passengers involved in rail line accidents. In this type of insurance fraud, many individuals and groups are willing and able to file bogus claims for injuries. At the same time, some of the insurance companies were found guilty of insurance fraud by issuing contracts to non-existing individuals.

Investment Fraud

Many people lost their wealth to false investments during the wake-up industrial revolution and the technological revolution. Scammers who offer large returns on investments typically target people and companies with particular investment programmes. The money obtained from new investors is used to compensate the investors in this kind of fraud. Organisers of Ponzi schemes frequently guarantee large returns to investors with little or no risk. The deceit and unlawful sale, or claimed sale, of financial instruments by promising minimal or no risk and substantial investment returns through intricate tactics is known as investment fraud.

2.3.2 Identity Theft Fraud

Charles et al. [57] described identity as a vector, a string of ones and zeros that describe the characteristics of an individual. Everybody has a distinct individuality. we clarified that the vector is regarded as limitless because its dimension is sufficiently large. It is possible to discern between a genuine person and an impostor thanks to this idea of identity.

According to the definition given above, identity theft is the act of someone using another person's personal identifying information or any related information without that person's consent in order to commit fraud in order to obtain credit or other benefits, gain financial advantage, or cause harm to other people or the loss of property or money.

2.3.3 Lottery and Auction Fraud

Cohen et al. [28] explored lottery fraud by explaining how to understand lottery fraud victims, who are more often widowed, female, and socially isolated. However, preliminary

investigations have shown that both types of victims differ from non-victims who experienced more negative life events in recent months. These include problems with the upkeep of their home, transportation, having a serious injury or illness, and concerns about finances in general. Regarding the reasoning of Cohen et al., lottery fraud can be described as a form of fraud targeting disparate and financially disadvantaged individuals through email or phone calls stating that they have just won a lottery. However, in order to receive the money, it is required to provide details like a bank account number and personal identification number, or even transfer a comparatively smaller sum to another bank account. This type of fraud mainly occurs in developing countries.

2.3.4 Telecommunication Fraud

Telecommunication fraud poses significant challenges in today's digital age, where communication networks are ubiquitous and essential for daily activities. This type of fraud encompasses a wide range of illicit activities conducted through telecommunication channels, including phone calls, SMS messages, and internet-based communication platforms. The perpetrators of telecommunication fraud aim to exploit vulnerabilities in these systems to deceive users and extract valuable information or resources illegally.

- **Phishing Calls** Fraudsters impersonate legitimate entities, such as banks or government agencies, to trick individuals into disclosing sensitive information like passwords or financial details over the phone.
- **SMS Scams** Unsolicited text messages containing fraudulent offers or requests are sent to deceive recipients into providing personal information or clicking on malicious links. **Robocalls** Automated calls deliver pre-recorded messages, often promoting scams or fraudulent schemes, such as fake lottery winnings or debt collection scams.
- **Subscription Fraud** Fraudsters use stolen identities to subscribe to telecom services, resulting in financial losses for both service providers and victims. **International Revenue Share Fraud (IRSF)** Criminals exploit vulnerabilities in telecom networks to generate revenue by placing premium rate calls or SMS messages to international destinations at the expense of service providers.

2.3.5 Email Fraud

The use of email as a means of official and non-official communication is increasing world-wide. The number of sent and received emails has increased each year since 2017. While roughly 306.4 billion emails were estimated to have been sent and received each day in 2020, this number is expected to increase to over 376.4 billion daily emails by 2025. Moreover, [58] claimed that email fraud is more structured and successful because, for each 10 fraudulent emails sent, 7 are delivered to the inbox and an estimated 3 are responded to. More

precisely, the incremental trust and usage of the service provide an opportunity for fraudsters to exploit the vulnerabilities. These vulnerabilities can come from both ends (sender and receiver) as well as the email service providers.

There are various kinds of email frauds. [55] proposed some lexical terms to describe various kinds of email fraud. Below in this survey, we discuss and define these lexical terms:

- The Contest Winner People often get too excited whenever they receive an email saying that they have claimed to have won something, but to claim your prize, the email requires you to click a link and enter your information for prize shipment.

2.4 Overview of Fraud Detection Approaches

A re-known cybersecurity firm [41] describes fraud detection as the implementation of a strategy to detect fraudulent transactions or actions that prevent fraud from causing financial and reputational damage to the customer and the institution. Fraud detection analytics solutions today are closely integrated with businesses. For instance, fraud detection helps us keep the systems and customer data safe by identifying and blocking suspicious activities and attempts before the business is endangered. Fraud detection enables businesses to detect and stop fraud in progress, whether it be a payment, an insurance claim, or any other form of intrusion.

The techniques' ability to detect actual fraud in real time reduces both loss and manual intervention costs. Fraud detection on credit cards can save banks literally millions of dollars. If the system halts suspicious transactions early and confirms fraud, the bank could save the full balance of the credit line in fraud expenses.

So, fraud detection techniques can help:

- companies to avoid shipping products to someone who will not pay for them because the fraudulent transaction will be reversed and they cannot get the product back.
- institutions like banks, insurance companies, and so on to build trusting relationships with customers.
- reduce manual labour, having an artificial intelligence system gets rid of the cost of manual . The unbiased analysis of artificial intelligence techniques comes with the new advantage of including inherent biases. Informed by previous experiences, a person's bias can cause them to subconsciously analyse transactions inaccurately and miss out on what's really going on.
- have greater speed in making a risk assessment by quickly identifying patterns in data.

In this paper, and as shown in Fig. 2.1, we classified the surveyed fraud detection solutions according to the used detection techniques, namely knowledge-based, learning-based, and blockchain-based solutions.

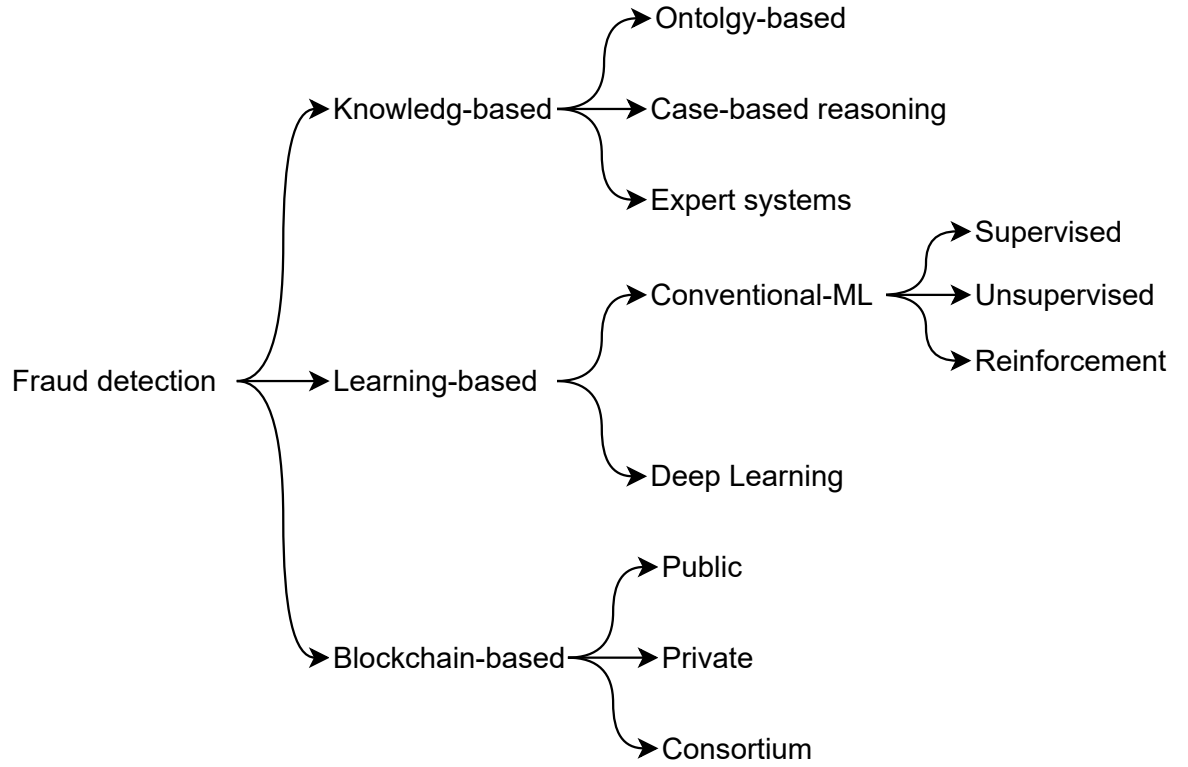


Figure 2.1: Classification of fraud detection solutions

2.5 Knowledge Based Fraud Detection

Martinez et al. [87] define a knowledge base as a collection of data organised in a form that facilitates analysis by automated deductive processes, such as an expert system. Different knowledge-based approaches have been proposed to build fraud detection systems. We distinguish three categories: ontology-based approaches, case-based approaches, and expert systems.

In this section, we will discuss and analyse some of the knowledge-base approaches used for fraud detection and prevention in those three categories.

2.5.1 Ontology Ontology-Based Fraud Detection

Ontology describes the main concepts of a domain and their properties; it formalises the basic knowledge of a domain. Cruz et al. [46] compare ontology concepts to database tables, ontology properties to database attributes, and ontology axioms to database constraints. Moreover, ontology instances can be seen as table records.

Several ontologies have been developed to represent fraud in different domains. This section presents the most significant ontologies for fraud detection.

Below are the most recognised ontology fraud detection publications.

Potamitis et al. [98] propose a fraud detection system using ontology-based techniques in the domains of the Internet for shopping, electronic payment services, and services that are being built to render mobile money transfers simpler.

Yang et al. [87] propose an ontology to detect email fraud. The ontology aims to tackle financial fraud using formal ontological repositories as well as multilingual terminological resources. This ontology makes it possible to detect a fraud indicator within email content, no matter how the fraudster tries to manipulate the content. The knowledge extraction process used to extract fraud indicators within emails is updated regularly. Hence, automatic rules are inserted into the knowledge extraction process.

El Orche et al. [37] propose an ontology for electronic payment fraud prevention. The aim is to detect and prevent suspicious transactions on various electronic payment systems. The proposed approach allows sharing adaptive preventive rules for fraud on an ontology based on multiple and various payment systems. The paper discusses problems and issues in fraud detection and proposes prevention of financial fraud using ontology technology. Among the concepts represented are legislation and legal rules from multiple countries, supporting reasoning about compliance with these legal rules, and an interpretation of legislation and legal rules useful to the user.

Rajput et al. [100] use an ontology for detecting suspicious fraud activities by monitoring individual transactions. Authors combine an expert system and the ontology for detecting suspicious transactions with three components: ontology construction, ontology reasoning, and a query on an inferred ontology. The presented approach uses anti-money laundering guidelines provided by the State Bank of Pakistan. The transactions are decomposed into groups of transactions, with a graphical representation of a few sample transactions updated after the application of the following steps: The rules are implemented on top of the OWL ontology to deduce new knowledge from existing knowledge.

Fang et al. [38] propose a novel method, built upon ontology and similarity of instances. The used generic ontology can detect fraud by checking for a suspicious change in user spending behaviour using similarity instances. Any deviation from the known instances could be termed fraud. The approach could be applied to credit card fraud, mobile payment fraud, lottery fraud, investment fraud, insurance fraud, etc. The testing of the approach was done using reading behaviour on an Internet website with different users, but the accuracy has not improved. This system can easily be adapted to different applications in any domain of financial fraud. Because similarity of instances can be detected in any transnational activity.

Kingston et al. [62] propose a legal financial fraud ontology-based approach. The first phase of the proposed approach aims to build financial fraud concepts, and the second phase aims to build an ontology for laws against fraud. The ontology models the relationships where a two-way exchange is promised but only one of the two sub-transactions takes place, while the second is to identify the relations between fraud laws, facts, and inferences about facts present, presumed, or missing. These two approaches could help solve the complexity of fraud investigations. In our opinion, the proposed ontology is complex and has a lot of uncertainties as it deals with the hypothesis. Enriching a legal core ontology

with a financial ontology and relevant conceptual distinctions will be a very good solution for anti-fraud applications.

Ramaki et al. [101] propose an approach for fraud detection in credit cards that relies on a semantic connection between data stored for every transaction fulfilled by a user and represents it by an ontology graph. The main advantage of this approach is real-time detection and storage of only worthy user behaviours, which will prevent the approach from having a large storage capacity and requiring no maintenance of storage patterns. While at the same time, the ontology graph is not easily understood.

El roche [37] proposes an ontology based on electronic payment fraud prevention. The proposed approach is able to detect and prevent suspicious transactions on various electronic payment systems using the ontology by sharing an adaptive and preventive rule of fraud on the ontology based on various payment systems. The ontology is able to cover and adapt to different system platforms; electronic payment platforms should be able to qualify, authorise, and reject transactions by referring to rules received from the ontology. The rules rely on some activities known by the user and compare if the payment card is qualified as a risky card or legitimate.

Carnaz et al. [22] introduce an ontology-based framework applied to money laundering. It provides a unified approach to represent and reason about an investigation into money laundering. The approach is based on the knowledge of three money laundering steps. The ontology transforms the informal and unstructured data retrieved from heterogeneous data sources into structured knowledge: placing knowledge, circulation knowledge, and integration knowledge. The ontology supports knowledge representation related to money laundering and is composed of several components. However, the paper does not present any framework for using real-world scenarios.

Summary of Ontology for Fraud Detection

The ontologies discussed in this section propose several semantic approaches for financial and non-financial frauds. Some of the domain areas constitute various concepts. While others proposed a single concept, as we discussed in the credit card fraud ontology by El Roche et al.[37] (2018), there are several sub-concepts. These ontologies are of great importance in understanding the dark secret behind financial fraud while at the same time preventing its occurrence. One of the main challenges of these projects is real-time fraud detection. This challenge needs to be addressed in future works on the ontology for fraud detection.

2.5.2 Case Based Reasoning for Fraud Detection

Kolodner et al. [63] define case-based reasoning as a way of adapting old solutions to meet new demands, using old cases to explain new situations.

Watson et al. [128] define case-based reasoning as a relatively problem-solving technique where a case is a contextualised piece of knowledge represented as an experience.

Case-based reasoning as a modern technology developed from a tiny and isolated research area into a burgeoning field of science. However, there has been no study into utilising these methods to diagnose financial crime trends (Weerts et al., 2019). The emphasis of the related literature on optimising current approaches, the lack of sophistication of CBR research about the transaction implementation spectrum, and finally, the developed perception of the financial fraud detection issue as one targeting precision optimisation rather than new forms of detecting and describing behaviour trends, may all be contributing factors. This offers impetus to examine the efficacy of CBR methodology in identifying financial fraud. Case-based logic (CBR) contributes significantly to the role of fraud identification. CBR programs may identify new cases by learning from sample examples of credit card use, and this method can respond to new fraud patterns as they arise. In the economic environment, CBR schemes have a range of benefits over other AI techniques. This includes the opportunity to improve by acquiring new cases over time without needing to introduce new laws or change current ones, and to decrease information elicitation commitment in dynamic and difficult transaction circumstances.

In view of the foregoing trends in fraud detection and prevention across multiple networks and industries, case-based reasoning has been used and is still being used to counter or prevent fraud activities in different domains, including banking, healthcare, industry, and so on. Below are the most recognised publications on case-based reasoning and fraud detection systems.

Wheeler et al. [129] propose a case-based reasoning system for fraud detection. The authors built an interface for the database of existing fraud-related activities. Typically, the database contains a variety of tables, where each table contains one or more data categories or fields in its columns. Each row of ‘ contains a unique record or instance of data for the fields defined by the columns. This method involves an approach to reasoning by analogy and classification that uses stored past data records or cases to identify and classify a new case. The model creates a list of typical cases that best represent all the cases in the training tables. The cases are generated by computing the similarity between all the cases in the training tables and selecting the cases that best represent the distinct cases in the tables. Whenever a new electronic transaction is conducted, the case-based reasoning model uses the decision tree created by the data mining sub-model to determine if a given input record associated with an electronic transaction is similar to any typical case encountered in the training tables.

The system has been proving to be very efficient in matching new cases with records in the database, but very few false negatives have been identified. One of the key improvements needed in the proposed system is to be able to flag the most obvious and frequent problems. Hence, it will be put in a separate database to model specific solutions to those problems.

Financial fraud has existed for a long time, and it can take an unlimited variety of forms.

Adedoyin et al. [5] proposed a case-based reasoning approach for identifying fraudulent patterns among financial transactions. For example, we can have D as the Euclidean distance between the new case X and the similar case Y retrieved from the training dataset. While the weights represent the attributes to compare the two cases.

While the false positive rate is the amount of normal transactions that are mistakenly classified as fraudulent, and none of the testing runs were identified as either false negative or false positive. While the authors have observed that the dataset used is characterised by incomplete information and uncertainty, which could be due to the sensitivity and legal impact, In the course of this survey.

Abutair et al. [3] propose a case-based reasoning methodology for phishing detection. This work proposes a task with similarity to the way humans think, so it can be used to solve problems using previous experiences. The highly rated method is represented as an experience to be used to derive a similar or close solution to the problem based on these two criteria (online and offline experiences). The offline experiences are the cases approved and identified by the system after extracting their features and formulating them as valid cases, while the online experiences are the ones with a short life cycle of about or less than 24 hours. This includes current blacklist URL sources and online anti-phishing group sources. In both sources, which are already known online, regardless of the data set sources, the proposed system interacts with the URL structure and gives more insights for efficient predictions from the beginning of every suspicious case. In this case, whenever the system starts processing a current URL to determine if it is a phishing or legitimate website, it first checks the URL to see if it exists or not.

This is a very good idea and will serve as a shortcut to directly classify every suspected URL as fraud activity. So, if it is found to be valid, the system flags it as a phishing website and provides the user with a conclusion to avoid this website with high criticality. If it is not found to be valid in the database, then the system starts extracting its features and formulates a new case to be tested. Once the case is formulated, the system starts retrieving the most similar cases. Based on the retrieved cases, an appropriate solution will be selected. This procedure makes it possible for the database to have data that could be treated as suspicious. The authors don't stop there; they also propose a situation whereby If the selected solution is, for example, concluded to be a phishing website, the solution to be proposed cannot be a current case, so it will look for similar solutions in the database and propose a new solution that will be applied. The data set used in the testing contains 572 cases for both legitimate and malicious URLs. Each URL is processed in order to extract the features and stored as a template in the form of pairs. The results from the testing stage have shown the proposed system is able to predict if the URL is valid and suspicious based on certain features, or if the URL is valid and not suspicious. But it seems we are not sure about the weighing mechanism that was proposed in the system, as there is no detail about how true positives and true negatives are weighted.

Summary of case based reasoning for fraud detection

The case-Based reasoning for fraud detection discussed in this survey proposes solutions for fraud using previously detected cases. Case-based reasoning approaches have been used to address the problem of financial fraud for detection and early warning systems for a long time. In this survey, we discussed various approaches for fraud detection, including case-based reasoning approaches. Some of the approaches provide proof of concept by retrieving previous cases. While others investigate and retain the previous cases in a repository and

use them as a case base for future reference. Some of the approaches suggest the use of transaction data along with their related classification as a possible suggestion for an investigation of a new case with unknown status. When implemented, the status suggestion can simply be either "fraudulent" or non-fraudulent.

Both approaches discussed in this survey have proven to be reliable, with almost no false positives or false negatives.

2.5.3 Expert Expert Systems for Fraud Detection

The Oxford Dictionary defines an expert system as a piece of software that uses databases of expert knowledge to offer advice or make decisions in such areas. According to [92], expert systems are a branch of artificial intelligence that has had the majority of the impact, especially in medicine, finance, telecommunications, and customer service. Expert systems are used to construct computer software that performs or replicates tasks that are normally performed by human experts.

An expert system requires the transfer of knowledge from the human experts to the software. It was completely agreed that expert systems are expensive because they are constructed as layers within a larger system. The expert systems must produce and identify solutions that almost all the experts would agree with. While some of the limitations of the expert system are that humans are not perfect experts, another challenge facing expert systems is that expertise is difficult to acquire; human experts are expensive and in short supply, which makes it difficult to have many expert systems in place.

Below are the most recognised expert systems for fraud detection.

In the approach proposed by [67], an expert system for fraud alerts in consumer credit is described. It aims to alert banks to a possible fraudulent use of consumer credit cards. The data used in the research work was obtained from Canadian banks, but due to confidentiality, the actual name of the bank and some of the details of model development have not been stated by the authors. The proposed expert system relies on the database to predict whether the operation or required authorization was fraudulent or not.

The model has been constructed with the input of many fraud experts. One major impact of the proposed work compared to others was the ability to capture some useful data, which will be stored in a different section. This data will then allow for subsequent detailed data analysis using advanced statistical techniques.

Alawadhi et al. [10] propose an expert system for fraud detection within procurement cards. The procurement card is a type of card used for purchases. The cards can be connected to either a credit card or a bank account, and the bank will issue payments when the user makes a purchase. The authors gathered huge amounts of data with 30 days of training and 55,000 transactions. Hence, the profiling where key data fields are missing has helped in having a complete dataset. The requirements, analysis, and knowledge were obtained by interacting with the experts. The initial run of the system provided the authors with about 1408 exceptions for attempted fraud transactions, while another 100+ association rules were added to the system after gaining more knowledge from the data set. The

second test was achieved with 95 percent accuracy.

Behera et al. [15] propose a credit card fraud detection using a neurofuzzy expert system. The authors introduced a two-stage neurofuzzy expert system for credit card fraud detection. This includes a fuzzy clustering module and an address matching module. In this system, there is an inference system responsible for computing suspicious scores. The system integrates evidence obtained from two distinct sources based on different transaction attributes for analysing the deviation of user behaviour from his normal spending profile. The credit card validation technique and fuzzy rule component were combined to explore the large dataset. The rule-based fuzzy inference system and a learning component have been very effective in matching the rules of the system to determine a potential fraud case.

Hilas et al. [52] proposed approach uses the network administrator's expert knowledge and applies data mining methods to real data. With the provision of enough data for the construction of the expert system environment, a brief analysis of prior data mining analysis and the structure of the expert system using flow charts and the IF, THEN, ELSE rules. The authors study and use a dataset with more than 50,000 user details (contact details, geolocation, timing, and subscription). In this instance, several fraudulent user accounts have been identified. The system provides a huge step forward in telecommunication fraud detection, with several companies in the US implying it. The number of details and expansion of telecommunication fraud that were discussed in this paper make it clear that telecommunication fraud will continue to attract a high number of researchers as the number of users increases across the service providers.

Summary of an Expert System for Fraud Detection

In this survey, we presented expert system models proposed to help prevent and detect financial fraud, mostly focusing on the use of credit cards. The objectives of expert systems for fraud prevention are to use the knowledge and know-how of experts in the field to model approaches and analogies to identify suspected fraud during the authorization process. [52] Proposed a fuzzy inference system that computes a suspicious score by combining the score values and classifying the transaction as genuine, suspicious, or fraudulent. The inference that computes the score is provided by experts; it provides various behaviours of fraudulent transactions. The effectiveness of the expert systems discussed in this survey was validated by conducting experiments and comparative analyses with other non-expert fraud detection approaches.

2.5.4 Discussion of Knowledge-Based Fraud Detection Publications

In this section, we discussed and analysed various case-Based reasoning publications for fraud detection. Some of the techniques used for case-Based fraud detection are proposed with the aim of improving the performance of detecting financial fraud patterns by using previous experience. Due to the lack of information on previous fraud cases and the fraudsters way of changing patterns, it's becoming difficult for Case-Based reasoning fraud detection systems to achieve their objectives.

While the ontology for fraud detection reviewed in this survey focuses on conceptual fraud detection, many were trying to detect general financial fraud. The main key aspect of the ontology for fraud detection is understanding the concept or domain of the fraud. In this survey, we understand that several ontologies for fraud detection were developed and validated. Many of the ontologies proposed for fraud detection have intended to offer a model for fraud by building a semantic connection between the domain of fraud, concepts, and attributes. The main advantage of an ontology for fraud detection is that it will give us the background and knowledge of fraud and its characteristics per generic, domain, and case. This survey analyses a relatively large number of expert systems for fraud detection. Many expert systems for fraud detection focus on expert knowledge and knowledge derived from data mining techniques to detect fraud.

Table 2.1: Comparison Table of Knowledge Based Fraud Detection Approaches

Ref	Fraud Type	Technique	Validated Re-sult	Year
[133]	Email Fraud	Ontology	Yes	2009
[37]	Electronic Payment	Ontology	Yes	2020
[22]	Money Laundering	Ontology	Yes	2017
[129]	Fraud Detection	CBR	High Accuracy	2000
[5]	Fraud Pattern	CBR	High accuracy	2016
[3]	Phishing Detection	CBR	High Accuracy	2017
[67]	Credit Card	Expert System	Low Accuracy	1999
[10]	Procurement Card	Expert System	High accuracy	2013
[67]	Credit Card	Expert System	Low Accuracy	1995
[52]	Telecom Fraud	CBR	High accuracy	2009
[15]	Credit Card	Expert System	High accuracy	2017
[101]	Credit Card	Ontology	Yes	2012

2.6 Learning Based Approaches

Several learning-based approaches have been proposed. We distinguish two main categories: conventional machine learning (supervised, unsupervised, and reinforcement learning) and deep learning.

Fraud is becoming a global pandemic, while artificial intelligence is the new vaccine. The introduction of new technologies is the main cause of the fast growth of fraud-related activities in all the different domains mentioned in this survey. Machine learning techniques

instruct computers to use data or past experience (learning) to solve a given problem. There are a number of machine learning algorithms: supervised learning, unsupervised learning, reinforcement learning, and so on. More robust and accurate solutions are required to prevent the highly skilled fraudsters and their activities.

Below are the most recognised learning approaches for fraud detection and preventing publications.

2.6.1 Conventional Machine Learning-Based Solutions

2.6.2 Supervised Learning Approaches

This is the most common learning approach, where the model is trained to use predefined class labels. In the context of credit card fraud detection, the class labels may indicate legitimate or "fraudulent" transactions. A supervisor provides a training data set whose transactions are classified in advance. If there is a fraudulent transaction X that is misclassified by the supervisor as legitimate, then the constructed model will be problematic. The same happens for a legitimate transaction that is misclassified as fraudulent. Moreover, an imbalanced distribution—also known as a skewed distribution—of the class labels in the training set can result in a model that does not have very good predictive accuracy. A skewed distribution is the situation where there are much fewer training samples in class [85].

Maes et al. published the work "Credit card fraud detection using Bayesian [85]. After careful enumeration of the fraud properties, the authors identified the requirements and the structure of the proposed system. Skewed distribution is most likely part of the system because in every transaction dataset, a small part of the data is fraudulent. Hence, the training of the dataset was divided into sections. The handling of noise and cleaning of the dataset are also part of the system's features. The authors used the Bayesian technique to identify the topology of the network and learn all the parameters. Real-world data was obtained from the Europay International. By labelling the dataset and the training phase, this proposed system has proven the influence of the parameter learning; it was observed that whenever the parameter was poorly learned, the speed and accuracy dropped. In this regard, we can propose a continuation of this research work by allowing different variables into the network and implementing a structure that solely defends the parameter learning prediction. The result achieved with this research proves Bayesian is more effective in fraud detection, and the training period is shorter than most others. While the detection rate is higher at the same time. One of the major limitations of this is the training of the classifier; the authors have specified the need to select a lot of good examples from each class. Otherwise, the accuracy of the system will be heavily affected.

Dal Pozzolo et al. proposed Credit Cards Frauds Detection and Concept Drift Adaptation. [30]. To formulate the fraud detection problem as a binary classification task, each and every transaction is linked to a feature vector. One of the focused areas in this research was to identify the main difference between learning from delayed samples and the percentage of fraud recorded. This was achieved by the analysis of the classifier performance that pro-

vides feed-backs where fraud and genuine transactions are balanced. The strategy used in this work is learning from the classifier analysis, which shows that whenever the feedback data is handled separately from delayed supervised samples, it provides more precise fraud detection alerts. Looking at the classifier graph provided by the author, we notice that the classifier learned on recent feedback is more accurate than the one learned on delayed samples. The question that we need to answer is: can we attribute this to the higher importance of the information included in the feedback? Until we are able to answer this question, we will not be able to determine what influences the high accuracy rate of this research work.

German et al. proposed a research work titled "Fraud detection in big data using supervised and semi-supervised learning techniques" [89]. In their research work, the authors proposed fraud detection based on multi layer perceptron due to the limitations of class imbalance in other techniques like decision trees, logistic regression, and support vector machines. In order to overcome the class imbalance problem, the authors have proposed two methodologies, namely data-level and algorithmic-level approaches. At the data level, the work focused on the use of preprocessing stages, mainly down-sampling the majority class in the dataset. The second methodology uses a cost-sensitive learning method in order to improve the imbalance. These methodologies prove to be accurate in dealing with the problem. In contrast, the results prove that the two methodologies help significantly reduce the imbalance impact during the model training. However, there must be room to distinguish between the relative weight of the different classes, which, in the context of imbalanced problems, may lead to erroneous conclusions, and this needs to be addressed. There are many different algorithms that can be used to split the training samples into branch-like groups. The resulting model has a tree-like structure consisting of a root, a number of branches, nodes, and leaves.

[86] provide three different implementations of decision trees for detecting credit card fraud. These implementations are called C5.0, CART, and CHAID. Their differences lie in the way in which a tree is constructed as well as the pruning algorithm used to remove erroneous branches and nodes. According to the experiments made by [86] The best predicting accuracy was achieved by C5.0 with an average of 92.80, followed by CHAID with 92.22, and finally by CRT with 91.34. In their experiments, the three DT implementations outperformed the SVM implementation, which achieved an average accuracy of 88.38.

Kumar and Gupta proposed a research work, "Fraud Detection in Online Transactions. Using Supervised Learning Techniques" [65]. Implemented different supervised learning techniques in the dataset, with different accuracy levels. These techniques include logistic regression, nearest neighbours, decision trees, random forests, and naive Bayes techniques. The authors display the accuracy of each technique. Hence, naive Bayes requires less training data in a situation where the independence assumption is true. On the other hand, it makes a very strong assumption when the data points in the dataset are independent of each other. While the decision tree method becomes unstable if there is a change or modification in the data value of the dataset, Logistic regression features to predict discrete values in the dataset were used in the research, but at the same time, the linear regression could not predict binary variables in the dataset. In SVM, there are some cases where data points are inseparable by a hyperplane in the original feature space. Hence, the authors used the kernels in order to transform the feature space into a higher dimension. But the

process of kernel parameter selection in order to avoid the above-mentioned problem is one of the limitations of this method. At the end of the research work, the performance of each technique was analysed and compared. But the logistic regression technique proved to be the best among the others.

Summary of Supervised Learning Approaches

One of the most commonly used learning-based approaches for fraud detection is supervised learning. In this context, supervised learning techniques find it difficult to detect the fraud.

In this survey, we analyse various supervised learning approaches that have proven to be reliable. One of those approaches is credit card fraud detection using Bayesian methods, as proposed by [85]. The results achieved in the work prove that the Bayesian approach is more effective in fraud detection, and the training period is shorter than most others. While the detection rate is higher. One of the major limitations is the training of classifiers; the authors have specified the need to select a lot of good examples from each class. Otherwise, the accuracy of the system will be heavily affected.

2.6.3 Unsupervised Learning-Based Fraud Detection Approaches

Unsupervised learning is considered to be the most widely used machine learning approach used for fraud detection. Because of this type of approach, there are no class labels for model construction. In [17] the authors stated that the aim of unsupervised learning in fraud detection is to discover those instances "whose behaviour is unusual". The main difference between unsupervised and supervised learning is that in the latter, labels are needed in the dataset, while in the former, they are not. Hence, the lack of labelled data prompts most researchers to adopt an unsupervised fraud detection method.

Remi Dominguez [36] in his thesis "Machine Learning for Unsupervised Fraud Detection," proposed the application of unsupervised machine learning algorithms to detect fraud for flight bookings using credit cards. The dataset was obtained from Amadeus Group and is about 107 GB of logs. Data cleaning was applied using Python scripts to remove duplicates from both lists and filter the corrupted values. The first training was done with the hidden Markov model, with a total of 4 675 603 instances to train. An average result of 0.26 (24.44 precision, 27.27 recall) was achieved. However, in sequence 5, there was a very poor return, which we are able to understand is the result of the absence of an action usually performed on the outliers checking. To improve the current proposed system, the author suggests the creation of a database to be fed by a dedicated targeted market or booking websites that are easily targeted by fraudsters. In this case, more suspicious actions can be obtained, which will result in more cases being retrieved by the system.

Lepoivre et al. [68] proposed credit card fraud detection with unsupervised algorithms. For example, in France, between November 2013 and April 2014, authors found 532.2 billion euros in transactions had been realised using 68.4 million credit cards. At the same

time, there was a total fraud amount of 469.9 million euros during the same period. To cure the problem, the authors proposed an anti-fraud project using a combination of two unsupervised algorithms, Principal Component Analysis and SIMPLKMEANS. There are two models of accuracy in the system: the geographic positions of the transaction and the client. Besides the above-mentioned reasons, Typically, fraudulent operations represent a small fraction of all transactions, leading to skewed distributions, which are also noisy due to errors from collecting devices in datasets.

Another challenge faced by other fraud detection systems is data overlap when operations may look fraudulent but are legitimate and vice versa. The authors used a principal and skim algorithm tool to view and calculate the relationship between different credit card transactions and their characteristics. The proposed method builds a matrix for each bank account, the matrix representing "n" transactions with their respective attributes. Each value variance covariance matrix = $N1 (XTX)$ is also constructed, representing the difference between the value and its respective estimation. The matrix classifies the transactions with good precision, and it can detect new fraudulent behaviors. Nevertheless, this precision is still at its local optimum. Hence, there is still a limitation to achieving the precision at the general optimum.

Richard et al. [17] explain credit card fraud categories into two categories: behavioural fraud and application fraud. Behavioural fraud occurs when details of legitimate cards have been obtained fraudulently, while application fraud occurs when individuals obtain new credit cards from issuing companies using false personal information.

The authors apply fraud detection to the case of behavioural fraud using continuous detection methods through the analysis of longitudinal data. The data consists of credit card transactions over time but can also include other variables. By implementing a new method for monitoring cardholder behaviour with peer group analysis, the tool detects individual accounts that begin to behave in a way distinct from accounts that had previously been similarly flagged. For example, if we have N objects with a sequence of d values represented by a vector, x_i , of length D , The value of D occurs at a fixed time point. This proposed approach can be useful in the stage of fraud detection when the account starts to exhibit some behavior. But as the fraudsters learn day by day, at some point the account does not exhibit those behaviors. Hence, the peer group will make too many errors and thus give an inaccurate result. On the other side, we understand that the system cannot give precise information regarding data sorting and output because the data is labelled.

Summary of Unsupervised Learning Approaches

Unsupervised learning in fraud detection was implemented mainly to cover the vulnerabilities and limitations of supervised learning detection solutions. Unsupervised learning algorithms can work with highly dimensional unstructured data without prior labelling or training while maintaining a significant level of personalization, which is an important key to their accuracy.

As explored in this survey, another advantage of unsupervised learning approaches is the ability to distinguish fraud patterns even if they have been modified or are totally new.

One of the proposed unsupervised learning approaches is a profiling method for fraud detection. It distinguished two categories of credit card fraud: behavioural fraud and application fraud. This approach is useful in the stage of fraud detection when the user account starts to exhibit some behavior. The work has proven to detect a change in the user's behaviour and generate an alert to the user for verification.

2.6.4 Reinforcement Learning based Approaches

One of the most common definitions is: Reinforcement learning is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximise the notion of cumulative reward. The purpose of reinforcement learning is for the agent to learn an optimal, or nearly optimal, policy that maximises the "reward function" or other user-provided reinforcement signals that accumulate from the immediate rewards. In his book [113], Rich Sutton describes reinforcement learning as learning to control data. Considering the above-mentioned definition, we can define reinforcement learning as a life cycle filled with trial, error, and reward. There are subtle or hidden actions performed by the fraudsters that may not be detected by supervised or unsupervised machine learning. Reinforcement learning allows for the creation of agents that go through the process of learning from fraudulent behaviours through trial and error.

Below are some of the published research results on reinforcement learning fraud detection.

Fraud Detection in Banking Using Deep Reinforcement Learning: Abdelali et al.[18] When it comes to the banking industry of today, machine learning allows financial entities to be defined efficiently, manage risk, and detect fraud. The theory of reinforcement learning was introduced into fraud detection in this paper as an approach to Markov decision processes. Recently, a number of financial risk analysis and fraud detection studies have suggested a relationship between reward-related activities in the brain and their functions. Nowadays, banks employ more staff in risk-related operations processes such as credit administration and analytics. The research work relied on the proposition by McKinsey et al. In their research, they suggest that by 2025, the number of staff in the risk operation will have been reduced by 25 percent due to the impact of artificial intelligence. The authors implemented a function and policy of reinforcement learning from value-based, partially observable MDP to policy-based and policy-gradient actor-critic learning methods. Focus on the reinforcement learning technique in which the state space is conveyed through visual inputs, e.g., images and video. If this work is to be implemented, it will give lots of impressions on the basis of decision-making in the risk management operation containing a multi-agent structure.

Adrian et al. [88], in their work for credit card fraud detection in adversarial reinforcement learning. The data set used in the paper was obtained from some of the major financial institutions and contains about 86 million observations, which are anonymized credit card transactions.

The authors built a logistic regression model that will act as the fraud classifier environment responsible for interpreting the state and action information received from the agent.

The classifier has three predictors. Low-amount transactions, high-amount transactions. In this proposed work, the authors have successfully translated the adversarial interaction between a fraudster and a credit card into a MDP, achieving predictive power in the classifier. Also, the agent was trained successfully to exploit the weakness in the classifier. Considering the effectiveness of this work, we can say that changing the classifier too often will significantly affect the performance of the method.

The Credit Card Fraud Detector for Imbalanced Classification Using Reinforcement Learning: Luis et al. [142] described in the introductory section, every year billions of dollars are lost due to credit card fraud, causing huge losses for users and the financial industry. Which motivates researchers to search for techniques that will help them avoid this significant loss of money.

The authors address credit card fraud by employing an imbalanced dataset that contains transactions conducted by credit card users. The suggested system uses components that learn to anticipate future events using a Qlearning algorithm to divide transactions into two categories: legitimate and fraudulent. Once more, our work established the method of fusing Q learning with neural agent capabilities, leading to a potential solution to meet high-precision and fast-acting criteria. The three techniques put into practice teach the suggested system how to recognise fraudulent transactions. auto-encoder, which makes it possible to extract significant characteristics from unlabeled data, and a mediator network, which categorises each input according to its appropriate label—authentic or fraudulent. Subsequently, the mediator’s intermediate layer serves as the input for the agent accountable for the ultimate.

Reinforcement for Fraudulent Behaviour Detection in E Commerce: Qingpeng et al. [20] In the e-commerce website, there is a seller account, through which he controls a product. It has been observed that on most e-commerce platforms, sellers engage in fraudulent behaviour by faking historical transactions in order to receive favourable treatment from the platforms. In this paper, reinforcement mechanisms are designed to combine the fundamental thresholds needed before sellers receive a reward. i.e., the consideration of agents’ incentives and their alignment with the objectives. Using data from one of the largest e-commerce platforms worldwide and training agents to learn the extent to which sellers will engage in fraudulent behaviour in the environment

The paper of Zhao et al. [138] proposed using reinforcement learning to deal with the behavioural model of the fraudsters. The authors model the problem appropriately as a MDP translated to the number of total transactions (real and fake). The proposed learning is heavily backed by data. This suggests that it may have poor performance whenever the fraudulent behaviour model is not valid in the dataset.

Summary of Reinforcement Learning Approaches

The purpose of reinforcement learning was to manage unmanageable huge state spaces. Because of this characteristic, it works well in environments like fraud detection, where processing vast volumes of records is necessary to identify the relatively tiny number of fraud instances among all the data that is accessible. This enables

it to look for improved behaviours that result in longer-term reward values. Environmental learning was employed to improve performance in several of the research on reinforcement learning that were covered in this survey. [20] proposed a reinforcement learning approach for e-commerce fraud detection. Given the behavioural model of the e-commerce websites, one of the major challenges in the work is that agent learning will heavily depend on the availability of data in the environment, which suggests that the agent may have poor performance whenever the fraudulent behaviour model is not valid. The reinforcement learning techniques explored in this survey have better learning capabilities but heavily depend on the environment.

2.6.5 Deep learning based solutions

[66]. Deep learning methods, such as deep artificial neural networks (DNN), use multiple processing layers to discover patterns and structures in very large datasets [107]. Deep learning topologies have been proven to be an effective way of detecting fraud patterns. By learning to discover patterns and decide whether they're part of legitimate or fraudulent operations.

Due to the growth of the Internet, more businesses, such as those in the financial sector, are operating their services online. The authors [11] proposed a deep learning-based method for financial fraud based on the Long Short Term Memory (LSTM) technique. The dataset contains credit card transactions, with 492 frauds out of 284,807 transactions. The dataset is strongly unbalanced; it has 31 features that include time, amount, class, and other sensitive features. The dataset underwent various steps, from data validation and normalisation to sample division, where the data samples were divided into training and testing. The proposed model has allocated 70 percent of the dataset for training and the remaining 30 percent for testing. The proposed Long Short Term Memory (LSTM) technique has been proven useful for longer-term patterns and can maintain a long memory. Therefore, the algorithm proposed is a perfect tool for prediction and detection tasks, according to the authors. From the results in the paper, we can point out that time is affected by the number of iterations and the number of layers. On the other side, the results proved that there is a strong correlation between accuracy and the number of iterations in the data.

The paper [44] proposes deep learning methodology to understand the behaviour of an insurance company and its users; this gives the groundwork for understanding the fraudulent behaviour of an insured person. The methodology proposed in the paper strictly abides by the fundamentals of unsupervised deep learning models to continuously learn complex changes in user behaviour through periodical model updates conducted at the click of a button. The dataset obtained from Oracle is related to auto insurance claim fraud; it consists of 15,478 policy holders. The data set contains 33 features and one column indicating the true outcome of the claim as fraud or not. 32 of the features were categorical variables: gender of the claimant, car manufacturer, marital status, accident area, etc. The method offers pragmatic insights into the data while also providing exceptional performance in the absence of training output labels. While the ability to capture the dynamic *modus operandi* of criminals, which precludes insurance fraud in advance, is one of the key advantages of the work, The analysis of the unsupervised deep learning models achieves excellent results.

The authors [123] understand the importance of fraud detection in e-commerce. Hence, the authors propose a deep learning model to differentiate normal users from suspicious users in e-commerce transactions. The deep learning model embeds the user or customer nodes into a graph structure. The graph structure information describes the connections between user nodes and item nodes. To detect fraudulent behaviours, the user's or customer's behaviour is modelled in a latent space, and the system classifies users based on their latent space feature distributions. To validate the effectiveness and robustness of the work, the authors obtained different types of datasets, which include business reviews, product ratings, and real-world flows from Yelp, TripAdvisor, and Amazon. These include restaurant and hotel reviews. In the dataset, user nodes are consumers, item nodes are businesses, and edges represent the user reviews of businesses. The experimental results from the three datasets and one real-world dataset indicate that the effect of the graph embedded could be used for effective fraud block detection by the position distribution of user vector representations.

There are different machine learning techniques proposed for credit card fraud detection. The authors [106] of this work provide a framework for parameter tuning of deep learning topologies for credit card fraud detection to enable financial institutions to reduce losses by preventing fraudulent activity. The dataset used was provided by a financial institution engaged in retail banking. It contains details of nearly 80 million anonymized credit card transactions across an eight-month period. The information provided includes both transaction and account details such as the transaction amount, the type of merchant the transaction was initiated from, the date the account was opened, etc. This paper proposes three types of layers in a feed-forward, multi-layer perceptron neural network. From the set of all the layers proposed in the paper, the best-performing LSTM model with 150 nodes had an accuracy score of 0.912. The best-performing RNN model contained 150 nodes and had an accuracy score of 0.90433. While the ANN produced the worst results among them with an accuracy score of 0.889.

The paper [59] proposes a deep learning-based method for credit card fraud detection. The authors employed multiple techniques, including feature engineering, memory compression, mixed precision, and ensemble loss. The proposed model used a fraud dataset provided by Vesta Corporation consisting of over 1 million records. The deep learning model consists of an input layer, three hidden layers, and an output layer. The hidden layers consist of 512, 256, and 32 neurons. The final output layer is one neuron that outputs the possibility of how likely it is that this transaction is a fraud. In this survey, we conclude that this deep learning model has the potential to better digest a huge amount of transaction data and make better predictions in real time.

The research work presented by [29] concentrates on financial statement fraud, consisting of auditing companies and state regulators. In this paper, the authors propose an approach for detecting statement fraud through the combination of information from financial ratios and managerial comments within corporate annual reports by using hierarchical attention networks (HAN) to extract text features from the management annual reports. The suggested method leverages numerical representations of unstructured textual data for predictive modelling. The text is classified using sparse lexical pieces in TF IDF. Convolutional neural networks and recurrent neural networks (RNN) are part of the learn-

ing strategy, however the artificial neural network (ANN) performs better than the other models with an accuracy of 4.5%. Therefore, we interpret the survey results to show that the linguistic characteristics retrieved using ANN along with TF IDF significantly improve fraud detection models. In addition, the study demonstrated that deep learning combined with the TF IDF algorithm is a good way to accurately identify fraudulent cases, while the majority of machine learning models are better at correctly identifying honest statements but fall short in detecting fraudulent situations.

The increasing fraud activities have negatively impacted telecom companies services and revenue. This paper [27] presents a deep learning technique as an effective method to detect fraud in mobile communications. The approach classified fraudulent and non-fraudulent activities using deep convolutional neural networks by analysing the behavioural patterns of customers in the dataset. The datasets are represented as artificial images describing the profile histories of different subscribers during two months of behavior. The image is captured, and any change in behaviour is indicated compared to the normal image behaviour. 18000 images representing the profiles of 300 users during the period of 60 days. With the deployment of hyperparameters for CNN training and the stochastic gradient descent optimisation method with a momentum-based learning rate of 200. While the descent updates the network parameters. Φ was set as $\Phi(t) - \beta \nabla \Phi(t-1)$. The approach discovered suspicious cases such as long call durations, unusual numbers of voucher recharges, not charged calls, and so on. The authors adopted a k-fold cross-validation approach for the evaluation of the results to avoid overfitting problems. One of the interesting parts of this approach is that the DCNN model outperformed the traditional machine learning algorithm with an accuracy of 82 percent.

This paper proposed [96] credit card fraud detection using multi-layer perceptrons using various parameters. A high-performance model was deployed using deep learning. The dataset was collected from Wordline and the Machine Learning Group of ULB (Université Libre de Bruxelles). Containing transactions by European credit card holders in the month of September 2013. The authors introduced some layers with many nodes. They used a scikit-learn-based multi-layer perceptron for learning nonlinear and real-time models. Then we compared the performance of the models with the various layers and numbers of neurons. The model gives a more consistent performance, where the precision is 94 percent and the sensitivity is 81 percent. The model shows the highest precision of 96 percent in the 2 hidden layers with 1000 neurons, while the highest sensitivity is 83 percent in the 3 hidden layers with 100 neurons. This approach provides a basic reason to use more advanced deep learning algorithms to detect fraud.

In this work [73] the authors focus on obtaining deep feature representations of legal and fraudulent transactions from the aspect of the loss function of a deep neural network by proposing a new kind of loss function that considers both distances and angles among features. The unbalanced data set for credit card fraud detection is from Kaggle, which is public and used in many studies. The proposed work has achieved a good and stable performance. It consists of two parts: a deep neural network and a specially optimised loss function. It is obvious that the ACL and FCL lead to the best performance in comparison with others.

This paper [110] (2019 Credit) proposes a credit card fraud detection system using deep

learning neural networks using two sampling mechanisms: under sampling, which reduces the number of non-fraudulent observations, and over sampling, where the fraudulent class observations are duplicated. The dataset features are transaction time, amount, and 28 other anonymized features that are fed to the input layer. The authors employed the idea of adjusting the number of non-fraudulent transactions to come on par with all the fraudulent ones by removing the excess non-fraudulent values from the data. On the other hand, add more fraudulent transactions by repeating the 492 ones to meet the number of observations. The result shows that Keras-based deep learning neural networks prove to be a great alternative to other classifiers applied in the work. The proposed work has given us the insight that neural networks, even though harder to train, are a perfect fraud detection model.

Summary of Deep Learning Approaches

Comparing deep learning models to supervised and unsupervised learning approaches, they assist lower the false positive rate, which is important for fraud detection. The ability of deep learning-based models to learn more intricate detection models, extract more complicated features, and enhance performance with larger training data sets is one of its main advantages. The ability of deep learning to handle a wide variety and quantity of input features or characteristics, such as the statistical qualities of attributes and feature engineering work, is another important advantage of deep learning in fraud detection that is discussed in this review.

Some of the approaches discussed in the survey prove the capabilities of deep learning to extract more complex representation features, as in the work of [73, 39]. The proposed work has achieved a good and stable performance. It consists of two parts: a deep neural network and a specially optimised loss function. This approach has a high performance when compared with several other machine learning approaches.

2.6.6 Discussion on Learning-Based Fraud Detection Approaches

The learning-based approach to fraud detection has received a lot of publicity in recent years. It is understood that researchers shifted their interest from rule-based fraud detection systems to learning-based solutions. In this survey, we understand that there are hidden events in fraud that may not be evident but still signal possible fraud, and at the same time, there are changes in patterns from the fraudsters that may not be uncovered with rules. Learning-based algorithms and techniques that process large datasets with many variables and help find these hidden correlations between user behaviours and fraud patterns are proposed. Learning-based fraud detection approaches are believed to be faster in data processing and require less manual work than other fraud detection techniques. In this survey, supervised learning for fraud detection was discussed. The technique is to label the transaction on the credit card as either genuine or fraudulent. We also performed an evaluation and analysis of various unsupervised learning fraud detection publications with different methods, some of which showed tremendous accuracy. Reinforcement learning is playing a huge role in the detection of fraud. Researchers use its ability to learn in an

adversarial environment to recognise suspicious behaviour in the user’s financial ehavior. Several articles have been published and implemented. Some of them are in banking, insurance, and e-commerce. Deep learning is one of the key players in fraud detection because of its ability to use multiple processing layers and discover patterns and structure in a very large dataset. In this survey, we have reviewed several deep learning methods for fraud detection. One of the key takeaways is the ability to reduce the false positive rate compared to other learning-based methods.

Table 2.2: Comparison Table of Learning Based Fraud Detection approaches

Ref	Fraud Type	Technique	Algorithm	Data Pre-process	Year
[85]	Credit Card	Supervised Learning	Support Vector	Yes	2002
[30]	Credit Card	Supervised Learning	Concept Drift	Yes	2015
[86]	Credit Card	Supervised Learning	Decision Tree	Yes	2015
[92]	Credit Card	Unsupervised Learning	Hidden MDP	Yes	2016
[68]	Credit Card	Unsupervised Learning	K means	Yes	2016
[102]	Credit Card	Hierarchical	Clustering	No	2022
[142]	Credit Card	Reinforcement Learning	Q learning	No	2020
[18]	Bank	Reinforcement Learning	MDP	No	2017
[20]	E commerce	Reinforcement Learning	MDP	No	2018
[11]	Credit Card	Deep Learning	LSTM	Yes	2021
[44]	Insurance	Deep Learning	Conventional	Yes	2020
[29]	Financial Statement	Deep Learning	HAN	Yes	2020

2.7 Blockchain based fraud detection approaches

Blockchain can be considered as a distributed database (ledger) composed of a chain of blocks that store all committed transactions [141]. Blockchain was first introduced in 2008 by Nakamoto et al. [91], and its most famous application is no other than the Bitcoin cryptocurrency. Blockchain operates in a decentralized manner and relies on several core technologies such as cryptographic hash, digital signatures, and a distributed consensus mechanism. The latter is a decentralized system that aims at determining the true state of a blockchain. In addition, some solutions rely on smart contracts, which can be defined as a computerized transaction protocol that executes the terms of a contract, with the aim to satisfy common contractual conditions (such as payment terms, liens, confidentiality ...), minimize both malicious and accidental exceptions, as well as the need for trusted intermediaries [114]. More details regarding blockchain technology, including consensus mechanisms, as well as smart contracts can be found in [141, 34].

Decentralization, in addition to anonymity, auditability, and persistency are the major characteristics of Blockchain that allowed its usage in various applications beyond cryptocurrencies. For instance, blockchain is used for various financial services [12], securing networks and systems [116, 32], and more recently for fraud detection and prevention [ismail2021healthcare, 121, 54, 78, 81, 108, 84, 47, 74, 69, 83, 16, 51, 117].

In this section, we will discuss the existing blockchain based solutions for fraud detection and prevention. We divided the solutions according to the three main types of blockchain namely public blockchain, private blockchain, and consortium blockchain [134, 9]. Moreover, we will provide, for each solution, details about the used consensus mechanism, the use of smart contracts, its implementation, as well as its evaluation.

2.7.1 Public blockchain based solutions

Public blockchains, also referred to as permission less blockchains, allow any user to enter into the system, create, modify, and validate a transaction. Bitcoin and Ethereum are the most common public blockchains. In the context of fraud detection and prevention, many researchers opted for a public blockchain infrastructure to deploy their solutions. For instance, Lo et al. [81] proposed a food fraud detection system based on Ethereum blockchain. The proposed approach applies machine learning techniques and a blockchain based architecture in order to verify the physical attributes of a food product (using various visual features) in a supply chain against their digital information. The case study of this research work was to distinguish between grain fed and grass fed beef. In the proposed approach, once the comparison step is done, and the classification results obtained, a blockchain based incentive mechanism is used to distribute two types of coins namely, honesty coins, and contribution coins. The first one is used to reward the honest recipients in the supply chain, and second one is intended for ML scientists and experts in order to motivate them for using the system and share their knowledge and data. The proposed solution has been evaluated through a beef case study.

In the context of financial fraud detection, Hyvarinen et al. [54] proposed the first

blockchain based framework that aims at overcoming the double spending problem in the Danish taxation system. In addition, the framework allows enhancing the transparency regarding the flow of dividends in the public sector. The authors implemented their prototype using Ethereum's smart contract, and evaluated it through expert interviews that allowed them to estimate the design's feasibility. In the same context, Guerar et al. [47] proposed Ethereum based blockchain solution to prevent fraudulent transactions in invoice financing. In the proposed solutions, the encrypted transaction data in an InterPlanetary File System (IPFS), and the related hash in a smart contract that runs on Ethereum. The proposed solution has been implemented using Remix IDE, and has been evaluated through condition testing using various scenarios, security and cost analyses. Hammi et al. [51], introduced the first automated banking checks' authentication scheme. The later aims at mitigating checks' fraud by eliminating the usual float period that can require more than 48h, and thus considerably reducing the risk of check's fraud. The proposed blockchain based scheme has been implemented using the public blockchain Namecoin through the multichain framework, and has been evaluated with regards to various criteria including a formal verification of the security protocols, security properties, scalability, financial cost, etc.

2.7.2 Private Blockchain Based Solutions

Also known as permissioned blockchains, private blockchains, are controlled by an organization, which manages the access rights of the system by allowing only certified users to create, modify, or validate transactions.

The work of Tam Vo et al. [121] is the first one that employs blockchain for fraud detection. The proposed solution aims at mitigating frauds in car insurance by providing a higher transparency in the insurance process, and insurance premium handling. The authors focused on the pay as you go car insurance application, which consists of letting the drivers pay insurance premiums based on the traveled distance. Several parties are involved in the process such as the insurance company, the drivers, as well as the associated financial organization. The proposed systems rely on smart contracts, which are used to ensure that the insurance process is properly handled and provides mechanisms to intelligently reacting to variations in the driver's behavior (e.g., diverting from the original selected path, exceeding speed limits, etc.). The authors implemented their solution on Hyperledger Fabric blockchain, and integrated it to a mobile app. However, there is no any information regarding its evaluation.

Liu et al. [78] introduced a blockchain based solution for mitigating healthcare insurance frauds. The proposed solution involves various actors including around 100 command centers, healthcare insurance agencies, public security traffic control departments, judicial organs, etc. The proposed system provides four different services including medical process inspection, third party liability inspection, excessive medical behavior analysis, and medical invoice data review. To build this system, the authors relied on smart contracts, and adopted a five layer architecture including cloud platform layer, network layer, core layer, interface layer and application layer. The authors evaluated their system according to three properties namely suitability, simplicity, and evolution. Similarly, the work of Saldamli et

al. [108] suggested integrating and link insurance companies' records together allowing their coordination in order to identify fraudulent claims. In this solution, which also employs smart contracts, the patients are identified using the pair patient ID, billing ID, which are linked together, and are unique for each benefited healthcare service. Once a claim is processed by a health insurance company, it is recorded on the blockchain, and then the amount in both the claims and billing files are compared, and inauthentic claims are hence detected. To build their system, the authors used BigChainDB, which allow to create a distributed database on blockchain concepts, and used Amazon Web Services (AWS) to deploy it. The work Thenmozhi et al.

[117] introduced a blockchain based prototype for healthcare insurance claim processing. The proposed prototype that aims at securing transactions and eliminating fraudulent claims, employs smart contracts, and is to be implemented using Hyperledger Fabric blockchain. The authors claim that the proposed solution can effectively secure the claiming process. In the same context, Ismail and Zeadally [ismail2021healthcare] introduced Block HI, a blockchain based framework for healthcare insurance fraud detection. This framework involves many participants, which are nine in number including the Head Medical Insurance Company (HMIC), which acts as the main authority that authorizes the network participants, the Branch Medical Insurance Company, the hospitals, the allied health professionals, the patients, the diagnostic centers, the medical equipment suppliers, the pharmaceutical companies, and finally the pharmacists. The authors evaluated their framework with regards to execution time and the amount of transferred data, but they did not provide any details regarding the implementation and the evaluation environment.

Ma et al. [83] introduced Fraud Buster, a blockchain based system for mitigating International Revenue Sharing Fraud (IRSF), which is considered as the most persistent type of fraud within the telecommunication industry. The authors implemented using Ethereum blockchain and Proof of Authority consensus mechanism a permissioned blockchain that aims at tracking and detecting fraudulent handover calls (i.e., transferring calls into high cost 'revenue share' service numbers) while preserving high confidentiality level by revealing only necessary information regarding the occurring fraud.

Leung et al. [69] proposed a blockchain based solution for food fraud detection. The proposed solution consists of a voting based consensus protocol, in which each transaction block is validated by every intermediary on the food supply chain. In order to evaluate the proposed solution, the authors simulated their own private blockchain, which has been tested on various scenarios belonging to two different attack types that target food supply chains namely modification attack and spoofing attack.

The work of Verma et al. [120], proposes a blockchain based system against land property fraud Rani et al. [102] combined the k means clustering algorithm with simulated annealing heuristic algorithm, and blockchain technology in order to built a credit card fraud detection systems. The authors evaluated their approach on dataset containing data of European cardholders who made credit card transactions in the month of September 2013 over a period of 2 days. The latter contains intotal 284,807 transactions out of which only 492 are fraudulent ones. The proposed system achieved an accuracy of 82.43% with an improvement of nearly 3% compared to the baseline approach.

As previously discussed, the work of Rani et al. [102] integrates blockchain technology, and machine learning for credit card fraud detection. The blockchain part of the system is meant to provide an additional layer of security for banking transactions. Indeed, in the proposed approach, and for any new transaction, a new block is created and submitted for validation. Once it is validated it is permanently added to the blockchain, and it is submitted to the machine learning model for final prediction. The authors provided details regarding the machine learning prediction component, however, they didn't provide any details regarding the blockchain part. Similarly, they evaluated the proposed approach with regards to the prediction accuracy of the machine learning module....

2.7.3 Consortium Blockchain Based Solutions

Consortium blockchains, also known as federated blockchains, combine both public and private blockchain characteristics. Indeed, in a consortium blockchain there are only a restricted number of trusted validating nodes, but unlike private blockchains, there's no single authority that has a full control on providing access to trusted nodes. Due to this restricted number of validating nodes, a consortium blockchain provides higher performances [84].

Mackay et al. [84] proposed a blockchain based framework for health care fraud detection and prevention for the US Centers for Medicare and Medicaid Services (CMS). The main aim of the proposed framework is to improve the claims process, and in which the patient plays the role of claims' validating node. The consortium blockchain framework, which has been built over the Ethereum platform, requires permission to join the network, and employs a proof of authority (POA) consensus mechanism. The claim validation process, which employs smart contracts, involves various stakeholders namely the provider (i.e., who rendered services), the patient (i.e., who received the services), and CMS (i.e., the payer for services). The authors claim that the proposed framework has the potential to improve the healthcare claims process.

Bhamidipati et al. [16] introduced ClamChain, a blockchain based platform for automated insurance claim processing, infrastructure level threat modeling, and an application level fraud modeling and detection using machine learning techniques. The proposed solution, which is based on smart contracts, has been implemented using Hyperledger Composer, and has been evaluated based on its scalability, attack probability reduction, as well as accuracy, precision recall, and F Score.

Liu et al. [74] proposed CFLedger, which is a blockchain based mechanism for mitigating charge back fraud in e commerce. The main idea behind this research is to provide a highly open charge back fraud data sharing blacklist, which allows assessing transaction fraud risks in e commerce. For this purpose, the authors introduced a consortium blockchain in which only e commerce companies that satisfy certain criteria (e.g. a minimum number of costumers and online transactions) are allowed to join the system. The authors only provided a system design, and evaluated it through efficiency, effectiveness, practicality analyses.

2.7.4 Discussion of Blockchain Based Solutions

In Table 6.1, we summarize and compare the reviewed blockchain based fraud detection solutions according to six (06) different criteria, which are a fraud category (i.e., the addressed fraud category), used blockchain (i.e., the type and if mentioned, the name of the used blockchain), consensus mechanism (i.e., the used consensus mechanism), smart contract (i.e., does the solution employ smart contracts?), implementation (i.e., did the authors conducted a proof of concept implementation?), and finally evaluation (i.e., the used evaluation criteria/metrics, if provided).

Based on Table 6.1, our first observation is that most of the reviewed existing blockchain based solutions deal with financial fraud, mainly, insurance frauds [ismail2021healthcare, 121, 78, 108, 16, 84], tax frauds [54, 47, 117, 83], banking frauds [51], and e commerce [74]. The remaining solutions, deal with a more recent fraud category, namely food fraud [81, 69].

Regarding the used blockchain type, four (04) solutions, namely, Lo et al. [81], Hyvarinen et al. [54], Guerar et al. [47], and Hammi et al. [51] are based on public blockchains, while the rest of the solutions are based on either private or consortium blockchains. In addition, most of the reviewed solutions have been validated through proof of concept implementations, except those of Liu et al. [74] and Thenmozhi et al. [117] who only presented a system design. Most of the implemented solutions used existing blockchains. For instance, Ethereum has been used in [54, 81, 84, 47, 83] being the most commonly used blockchain, followed by Hyperledger Fabric used in [121, 16, 117], and finally BigChainDB and NameCoin used in [108] and [51] respectively. For the rest of the solutions [ismail2021healthcare, 78, 74, 69] the authors opted for deploying their own blockchain.

Regarding the evaluation of the proposed solutions, we noticed that there are no commonly used evaluation criteria. For instance, four (04) of the reviewed solutions [51, 16, 78] considered scalability as one of the evaluation criteria, while in [47, 69, 51]

2.8 Challenges and recommendations

The first real issue that faces fraud detection solutions is the data shortage, especially for those based on conventional machine learning techniques. Indeed, if we take as an example the credit card fraud case, it is very difficult to obtain a rich dataset which can be used for research purposes, and this is mainly due to privacy concerns. In addition, the major issue with the available datasets, which are highly imbalanced with only a limited number of fraudulent instances. Concept drift is also one of the major challenges that face credit card fraud detection solutions. Thus, adequate approaches must be employed to overcome such issue. For instance, we recommend the use of proper feature engineering approaches such as transaction aggregation, and/or to deploy proper classification models that take into consideration the sequential properties of the credit card transaction data. Such models include, HMM and RNNs.

There is an increasing interest in using blockchain for mitigating various types of frauds.

This increasing popularity is driven by the intrinsic characteristics of blockchains, namely immutability, secure communication, data controllability, and transparency. However, blockchain based solutions are subject to certain technical and practical limitations. First, and depending on the used blockchain type, some privacy and security issues may emerge. For instance, solutions that employ proof of work consensus mechanism, are subject to what is called 51% attacks. This attack happens when a group of "malicious" nodes (i.e., miners) detain 51% of the computing power of the blockchain, and thus can lead to the approval of fraudulent transactions. Thus, choosing the right blockchain type, and the right consensus mechanism is very important when designing blockchain based solutions. Second, and in addition to security challenges, there's also the usability challenge. Indeed, the latter must be one of the priorities when designing blockchain based fraud detection solutions, in order to avoid discouraging users, of various backgrounds, from adopting the system. Moreover, setting up an incentive mechanism that rewards the participants in detecting fraudulent activities, can be an effective way that encourages the adoption of blockchain based fraud detection solution. Finally, and regarding the organizational aspect, there's a need for establishing an appropriate legal framework, as well as a standardized architecture for building a blockchain based fraud detection system.

2.9 Conclusion

Fraud detection can be a complex and challenging process, particularly when it comes to identifying and preventing new and sophisticated forms of fraud. In order to be effective, fraud detection systems and processes must be constantly evolving and improving to keep up with the changing dynamics of fraudsters. This can be a significant challenge for organisations, particularly those with limited resources. Additionally, fraud detection can be a time-consuming and resource-intensive process, which can present difficulties for organisations that are already stretched thin.

Fraud cases have increased in recent years, particularly in areas like banking, insurance, travel, and telecommunications. Hence, various fraud detection and prevention solutions are proposed in order to mitigate the impact of fraud.

This survey comprehensively categorised, analysed, and discussed the state-of-the-art fraud detection solutions in the main areas of artificial intelligence (i.e., knowledge base, machine learning, and block chain). The state-of-the-art fraud detection solutions in the three primary fields of artificial intelligence—knowledge base, machine learning, and block chain—were thoroughly categorised, examined, and discussed in this survey.

The suggested methods' respective performances differ from one another. As a result, we have paid close attention to how each technique performs in the various comparison tables, ranging from a knowledge-based approach to a learning-based one. There was a discussion of the difficulties impeding most fraud detection systems' effectiveness and performance. It has been noted that every fraud sector is unique and confronts distinct difficulties. Real-time detection, for instance, is critical in the credit card industry but not in other industries. An additional set of difficulties impeding the effectiveness of fraud detection areas are highlighted, namely, overlapping classes, noise detection, skewed distribution, and

misclassification.

The characteristics of techniques for detecting credit fraud were derived and encapsulated using different techniques of artificial intelligence. The hypothesis of this fraud detection survey is that it will reduce the amount of research that software developers need to do if they want to implement the system.

Table 2.3: Comparison of blockchain based solutions for fraud detection Smart Contract (SC), Implemented (Impl.)

Ref.	Year	Fraud Category	Used BC	Consensus mechanism	SC	Impl.	Evaluation
[121]	2017	Car Insurance	Private (Hyperledger Fabric)	Practical Byzantine Fault Tolerance	Yes	Yes	N/A
[54]	2017	Tax fraud	Public (Ethereum)	Proof of stake	Yes	Yes	effectiveness, efficiency, usefulness, ease of use
[78]	2019	Healthcare insurance	Private	N/A	Yes	Yes	suitability, simplicity, evolution
[81]	2019	Food Fraud	Public (Ethereum)	Proof of stake	Yes	Yes	Case Study
[108]	2020	Healthcare insurance	Private (BigChainDB)	Byzantine fault tolerance	Yes	Yes	Performance (latency)
[84]	2020	Healthcare Fraud	Consortium (Ethereum)	Proof of Authority	Yes	Yes	N/A
[47]	2020	Invoice	Public (Ethereum)	Proof of stake	Yes	Yes	Condition testing, Security, Cost
[74]	2021	E-commerce	Consortium	Practical Byzantine Fault Tolerance Delegated Proof of Stake	No	No	Efficiency, effectiveness, practicality
[69]	2021	Food Fraud	Private	Proposed	No	Yes	Precision, Recall, Security Assessment
[83]	2021	Telecom.	Private (Ethereum)	Proof of Authority	No	Yes	Usability, confidentiality
[isma]	2021	Healthcare insurance	Private	practical Byzantine fault tolerance	Yes	Yes	Execution time, data transfer
[16]	2021	Insurance	Consortium (Hyperledger Composer)	N/A	Yes	Yes	Scalability, accuracy, precision, recall, F score
[51]	2021	Check's fraud	Public (Namecoin)	Proof of Work	No	Yes	Security, scalability
[117]	2021	Healthcare insurance	Private (Hyperledger Fabric)	Proof of Work	Yes	N/A	N/A

ONTOLOGY SYSTEM FOR FINANCIAL FRAUD DETECTION

In the previous chapter, we identified several key elements to help us achieve the objectives of this thesis. The survey summarises existing fraud detection approaches; this includes identifying the latest trends and techniques in fraud detection as well as evaluating the effectiveness of various fraud detection methods. We identified the research gaps in the existing approaches. Afterwards, we implemented a machine learning experiment using various machine learning algorithms. In the experiment, we have identified key limitations, which include explainability, interpretability, etc. In this chapter, we propose an ontology for financial fraud detection that is aimed at improving where the existing machine learning methods lack. We aimed to evaluate the effectiveness of this system in a unique manner using a financial fraud dataset.

A deep understanding of various frauds is vital to selecting the approach to detect them. Knowledge about fraud can be modelled with the help of an ontology. In this approach, we looked at potential or existing ontologies to be adapted for financial fraud detection systems, but unfortunately, considering the dynamic character of targeted models, we did not find any. Hence, it was necessary to elaborate such a model to study the current fraud.

Financial fraud exists in different aspects of our lives, and the detection and prevention of those frauds represent an important question relevant to many academic disciplines. Fraud is an economically significant issue that leads to depression, bankruptcies, and, unfortunately, suicide.

Ontologies are used to study concepts and have been applied to solve various problems in several domains. The objective of this research is to understand existing frauds and propose a system able to detect them in order to prevent damages. This challenge is ambitious because the problem addressed is complex and dynamic—the fraudsters learn from failures and invent new ways of committing fraud. From the several ontological approaches for fraud detection, some of them are domain-specific, and the majority constitute a case-specific module. While others serve as a knowledge repository for fraud techniques. For example, the ontological approach proposed by El Roche et al. (2018) provides great impor-

tance in understanding the dark secret behind credit card fraud and the techniques used by the criminals. One of the main advantages of our ontology is that it serves as a knowledge repository for financial fraud. In this case, it will cover the overall layers proposed in the existing ontologies with additional knowledge; this is why it is termed a three-layer ontology.

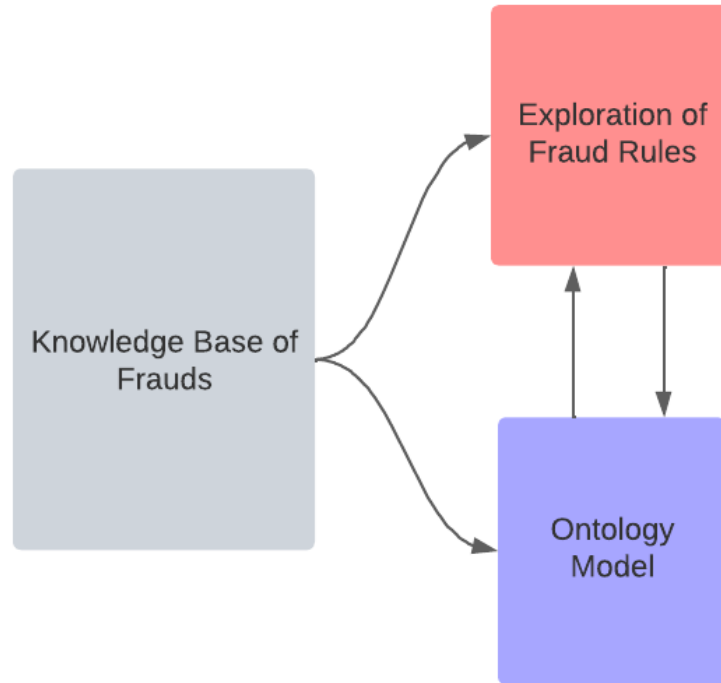


Figure 3.1: The proposed fraud detection approach.

The knowledge base of fraud consists of organised information that captures and represents knowledge about fraud-related concepts, rules, patterns, and techniques. It typically consists of a collection of structured data, rules, and algorithms designed to store and retrieve fraud-related information. The knowledge base is built based on the expertise and experience of fraud analysts and investigators, incorporating their domain knowledge and best practices. The characteristics of the knowledge base include the following:

- **Structured Fraud Information:** This includes organised and structured fraud information with predefined categories, relationships, and attributes in structured data formats to store and manage fraud-related information.
- **Human Curated Content:** The knowledge base is created and maintained by human experts for continuous updating based on new fraud schemes, patterns, and detection methods. This relies on the expertise and judgement of fraud analysts to identify and capture relevant information.
- **Rule-Based Reasoning:** The knowledge base often employs rule-based reasoning to detect fraudulent activities. This consists of a set of predefined rules that describe

known fraud patterns or indicators. These rules are applied to incoming data or transactions to identify potential fraud cases.

The Ontology of Fraud Model: This model is a conceptual representation of fraud-related knowledge that aims to capture the relationships, properties, and characteristics of fraud-related concepts and entities. It is based on the principles of ontology, which is a formal representation of knowledge that defines a set of concepts, their attributes, and their relationships within the financial fraud domain. Key characteristics of an ontology of fraud include:

- **Conceptual Representation:** The ontology of the fraud model focuses on representing fraud-related concepts, their properties, and their relationships. It provides a high-level conceptual framework that allows for a more abstract and flexible representation of fraud knowledge.
- **Machine Readable Structure:** The ontology is represented using ontology languages such as OWL (Web Ontology Language) or RDF (Resource Description Framework), which provide a standardised way to define and represent the knowledge structure. This machine-readable format enables automated reasoning and inference.
- **Semantic Reasoning:** The ontology of the fraud model allows for semantic reasoning and inference, which means that it can automatically derive new knowledge or detect implicit relationships between fraud concepts based on the defined ontology. This enables more sophisticated and automated fraud detection methods.
- **Scalability and Interoperability:** Ontologies are designed to be extensible and scalable, allowing for the incorporation of new knowledge and the integration of multiple data sources to provide a foundation for interoperability between different systems and domains, enabling the exchange and sharing of fraud-related knowledge.

In context, the knowledge base of fraud is a structured repository of fraud-related information, often relying on human curation and rule-based reasoning. On the other hand, an ontology of fraud provides a conceptual representation of fraud knowledge, enabling semantic reasoning and scalability. While the knowledge base approach is more focused on specific fraud patterns and detection rules, the ontology model offers a more abstract and flexible representation that can adapt to evolving fraud landscapes and support automated inference.

The proposed knowledge base approach with ontology and exploration of rules to detect fraud involves building a formalized representation of the knowledge and relationships between entities in the financial fraud domain, including potential fraud patterns and behaviors. After building the ontology and its subsequent validation using the dataset, the next step is to explore rules that may indicate fraudulent behavior. The rules have been tested based on real-life scenarios. The main advantage of this approach is that it provides us with a more accurate and adaptive system. By incorporating the ontology knowledge and relationships, which allows the system to identify complex fraud scenarios that may not be detected by traditional rule-based systems.

Finally, the proposed fraud detection system with the incorporation of ontology and rule exploration has provided a more accurate, adaptive, and comprehensive view of potential fraud and the potential to detect a change in the fraud pattern, which can lead to a significant improvement in detecting several new fraud patterns from criminals. In the next section, we will elaborate on why the ontology was chosen for the fraud detection system and what the advantages of the ontology are over the existing approaches.

3.1 Why Ontology?

Several fraud detection systems have been proposed and implemented, as discussed in the previous chapter. While these fraud detection systems have made significant progress in detecting fraudulent activities, there are still some limitations that make them insufficient to handle all types of fraud. Below are some of the challenges faced by these approaches and why we need ontologies.

- **Dynamic fraud patterns:** Fraudsters are continuously adapting their techniques, making it challenging for existing models to keep up with the latest fraud patterns. We need a system that can be updated to meet the new fraud patterns.
- **Data quality:** Existing systems rely on data to learn patterns and make predictions. If the training data is biased or incomplete, the model's performance can be limited, and it may not be able to detect new types of fraud.
- **Adversarial attacks:** Fraudsters can intentionally manipulate the input data to evade detection by the existing systems. This can be done by changing the data slightly or by creating entirely fake data.
- **False positives:** A common problem with existing fraud detection models is flagging legitimate transactions as fraudulent, leading to unnecessary friction and a poor user experience.
- **Lack of interpretability:** Many of the existing fraud detection models, especially those based on machine learning, are like black boxes, making it challenging to understand how the decision is made. This can be a problem where explainability, regulations, and transparency are required.

To address these limitations, such as explainable AI and dynamic retraining, to improve fraud detection systems, we decided to use ontology because of its ability to address the above-mentioned challenges and many others by providing a structured and transparent representation of domain-specific knowledge and enabling semantic reasoning.

- **Explainability of Fraud Techniques:** This ontology is designed to provide a transparent and interpretable representation of domain-specific knowledge, making it easier to explain the reasoning behind fraud techniques. We use the ontology to represent fraud-related concepts and relationships; it is apparent to provide explanations for why a particular technique was used by the criminals on a certain target group. This

can help build a better and more accurate fraud detection system and improve regulatory compliance.

- **Identification of Relevant Data for Retraining:** Our ontology has the ability to identify which data is relevant for retraining a fraud detection system. By formalising the concepts and relationships relevant to fraud detection, it is easier to identify which data points are most important for detecting fraud. This can be useful when updating the system with new data or when retraining the system to improve its performance.
- **Understanding and Facilitation of Fraud Dynamic:** We designed the ontology to help mitigate one of the biggest challenges faced by the existing fraud detection systems, which is understanding and facilitating fraud dynamics. This is done by formalising the relationships and concepts relevant to fraud detection to identify when new techniques of fraud or changes to the fraud scheme are noticed and the system will require an update. With a clear understanding of the concepts and relationships involved in fraud detection, it is easier to update the rules in order to improve the system's performance.
- **Support for Collaborative Learning:** The ontology can support collaborative learning by providing a shared representation of knowledge. This can enable fraud detection system researchers to collaborate on developing an ontology that accurately represents the domain and can be used as a basis for building fraud detection systems from various approaches, including machine learning-based ones. With shared ontology knowledge, it becomes easier to train the system and explain the reasons for any changes to the system.
- **Visualisation:** Ontology-based approaches can provide visualisations and explanations of fraud patterns, relationships, and behaviours, making it easier for fraud analysts and investigators to interpret and understand the results. Many of the ontology development software have visualisation modules of different types; this will help to view complex patterns and relationships that may not be apparent through other methods, enabling more effective fraud detection and investigation.

These capabilities and many others have shown that ontology has the potential to be at the centre of financial fraud detection. In this thesis, we explored most of these capabilities, which were demonstrated during the building, validation, exploration, and testing of rules in different scenarios. In this thesis, we plan to reuse any comparable ontology that defines key concepts related to financial fraud; unfortunately, we could not find any. Some of the ontologies, like FIBO, have been implemented to describe the financial sector but do not have any fraud-related concepts. Hence, we have decided to build our own ontology and populate it with comprehensive knowledge of financial fraud.

In the overall context, our ontology played a vital role in the classification and identification of different types of fraudulent activities, such as identity theft, banking fraud, credit card fraud, and money laundering. While it gives us more scenarios to test the explored rules, these scenarios have been extracted from the various domains and concepts of the ontology. The link between explored rules from the ontology and scenarios lies in the fact that rules can be used to detect fraudulent behaviour within scenarios. By applying the

rules to a given scenario, the system is able to justify whether or not the scenario contains suspicious activity that should be flagged as fraud.

In the next section, we will discuss the steps followed to build the financial fraud ontology for detection (FROD).

3.2 Ontology Building

Ontology development is a crucial task in knowledge representation and semantic reasoning. To build the ontology, we use protege software, which is open source and widely used for building and managing ontologies. While ontology web language (OWL) is a formal language used for representing knowledge and information in a structured and machine-readable way, It is part of the semantic web technology. In this section, we will discuss how the ontology was built and the steps involved in using the OWL language with Protege software.

- **Define the domain:** The first step in ontology development is to define the domain that the ontology is intended to cover. In this thesis, the domain is financial fraud, which has various subdomains and cases. Hence, we formulated the ontology as a three-layer stage. This approach will enable the ontology to define the domain and identify the concepts and relationships that are needed to model and build the comprehensive ontology.
- **Identify the concepts:** After defining the domain, the next step is to identify the concepts that need to be represented in the ontology. Concepts are the building blocks of an ontology, representing different entities and classes that are relevant to the domain. In this ontology, we have various domains that will be represented with several concepts. For example, in the domain of identity theft, some of the concepts could be "phishing," "email," "victims," "theft," etc.
- **Define properties:** The classes and entities of the ontology domains have properties. These properties are used to describe the relationships between entities within a domain, represent attributes or characteristics of entities, and are used to define the structure of the ontology. The properties in this ontology are of two types. Object properties describe relationships between individuals or instances of classes within the ontology. While data properties describe relationships between individuals or instances of classes and literal values, such as strings, numbers, and dates, Properties can also have various characteristics, such as being transitive, reflexive, symmetric, or asymmetric. These characteristics can be used to further define the semantics of relationships within the ontology.
- **Define axioms:** The axiom statements define the relationships between entities in a particular domain. By defining axioms within the domain, we can use reasoning tools to infer new knowledge and relationships within the domain. A well-constructed ontology with clear and consistent axioms can enable automated reasoning, data integration, and semantic interoperability across different systems and domains.

- **Define the relationships:** After defining the axioms in the domain, the next step is to define the relationships between the domains. Relationships represent the connections between concepts and how they interact with each other. To infer a relation, we must create a new annotation named *has-relation*. This annotation will create the relationship functions between classes and the entities in the ontology.
- **Create the ontology:** Once the concepts and relationships have been identified, the next step is to create the ontology. This is done using Protege software. The software provides us with a user-friendly interface for creating and editing ontologies using a variety of knowledge representation languages; it also supports the creation of class hierarchies, property hierarchies, and object properties. Protege includes a number of features that make it a powerful tool for ontology development. These include automated reasoning, ontology visualisation, and the ability to import and export ontologies in a variety of formats. Additionally, Protege supports a range of plugins and extensions that can be used to extend its functionality.
- **Reasoners:** The ontology reasoners are built-in tools that are used to automatically validate new knowledge and relationships inferred in the ontology based on its axioms and rules. While building the ontology, it is important to start the reasoner to check for logical inconsistencies within the ontology and identify redundancies in the existing axioms based on a variety of techniques, including logical inference, rule-based reasoning, and semantic matching. We also used the reasoners to classify individuals within the ontology and identify their relationships with other individuals and classes.
- **Visualisation:** Visualisation is an important aspect of ontology building that allows us to better understand and communicate the structure and content of the ontology. There are various tools we used during the ontology building stages to visualise the knowledge and the semantic relationships. Some of these visualisation tools are graph-based visualisations that use nodes and edges to represent classes, individuals, and properties in an ontology. Matrix-based visualisations: this type of visualisation represents the relationships between different entities within an ontology. We used it to check for similarities and differences between different classes and individuals in the ontology. While mind map-based visualisations use a branching diagram to represent the concepts and relationships in the ontology, we mainly use it to display concepts for brainstorming and communicate the structure and content of the ontology in a more intuitive way.
- **Maintenance of the Ontology:** Maintaining this ontology in Protege involves several tasks, including adding new concepts, modifying existing concepts, adding relationships between new concepts, and removing outdated or erroneous information. The dynamic of frauds gives us the common knowledge that we must be adding and modifying the ontology knowledge in order. To remove outdated or erroneous information, we select the concept, entity, or relationship that needs to be removed and delete it using *Delete*". In addition to these general steps, there are also several tools and plugins available in Protege that can help with ontology maintenance, such as the "Ontology Metrics" plugin, which will help identify potential issues in the ontology.

- **Documentation:** The documentation of the ontology provides a consistent and centrally non-technical description of the semantic content of the concepts of the ontology, such as concepts, classes, individual resources, and properties. The documentation does not consider the proprietary format in which the information is stored on the ontology platform.
- **Restrictions:** In OWL language, there are two kinds of restrictions: existential restrictions, which describe classes of individuals that participate in at least one relationship along a specified property with individuals that are members of a specified class. For example, the concept of banking fraud has a relationship called ‘has a part in with members of the identity theft fraud concept. While universal restrictions describe classes of individuals that only have relationships and a property that make them members of a specified class,

3.3 Financial Fraud Ontology for Detection (FROD)

This section discusses the elaboration and overview of our proposed ontology. The proposed system is composed of an ontology and rules for exploring. This comprehensive financial fraud elaboration ontology is a structured and organised representation of knowledge related to financial fraud. It is a conceptual model that defines the key concepts, relationships, and rules associated with financial fraud. The ontology aims to provide a common understanding of financial fraud. The ontology is organised into three layers and several categories or modules, each representing a different aspect of financial fraud. The ontology describes different types of financial fraud, such as banking fraud, investment fraud, and money laundering. Each type of fraud can be further broken down into sub-types.

The ontology will be further enriched by adding more categories, subcategories, and relationships between them. Due to the dynamic nature of recent fraud, the ontology can also be expanded to include information on the socioeconomic and psychological factors that contribute to financial fraud. By providing a comprehensive and structured representation of knowledge related to financial fraud, the ontology can help improve our understanding of this complex phenomenon and support efforts to prevent and combat fraud.

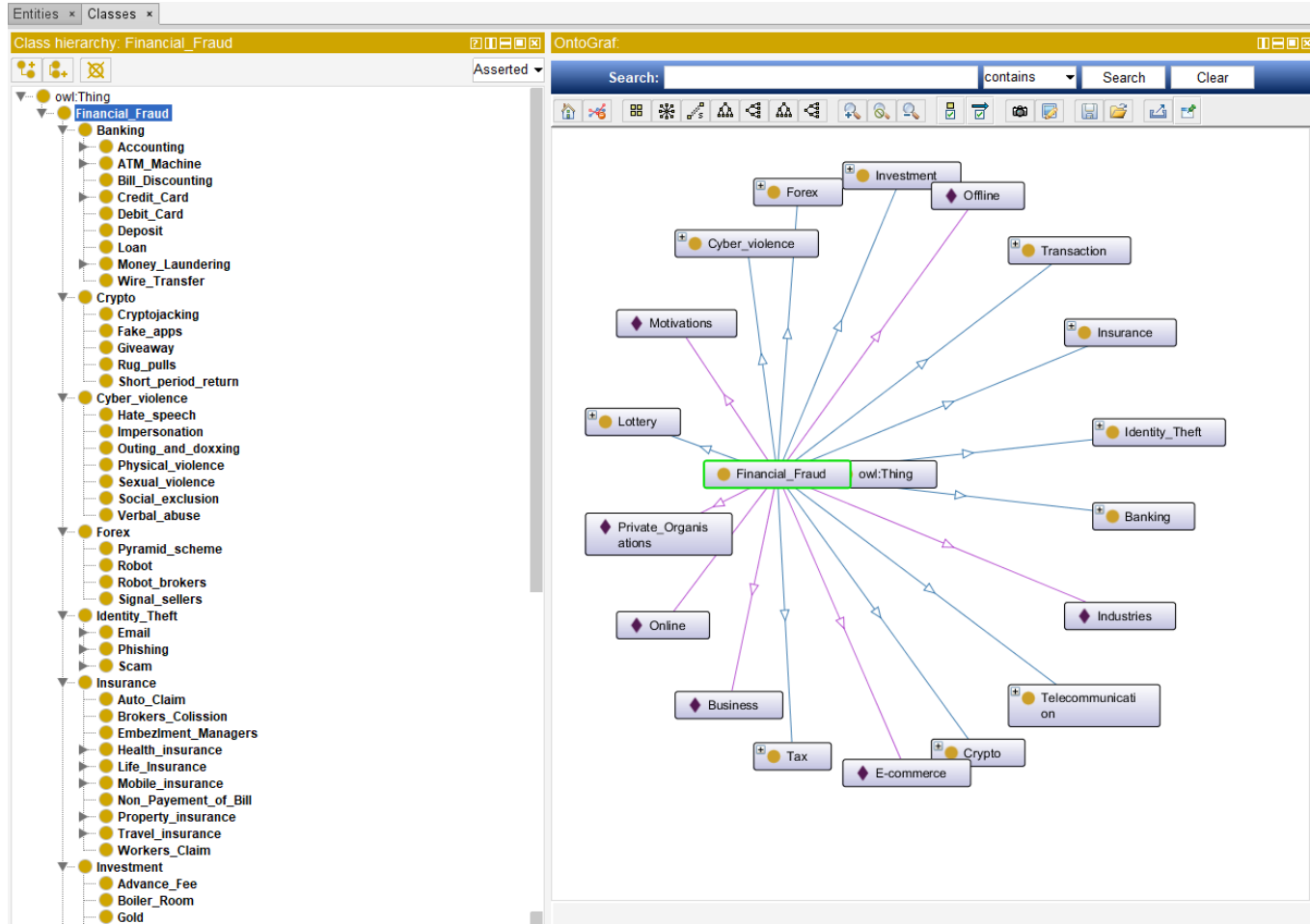


Figure 3.2: The Ontological Structure of Financial Frauds.

3.3.1 Banking Fraud

In a conceptualised context, bank fraud is using deception to steal money or assets from a bank or financial institution, including credit unions and other financial institutions that are legally insured. This includes the Federal Reserve Banks, the Deposit Insurance Corporation, and mortgage lending agencies. In a legal context, bank fraud is the use of potentially illegal means to obtain money, assets, or other properties owned or held by a financial institution, or to obtain money from depositors by fraudulently posing as a bank. In all instances, bank fraud is a criminal offense. In this research, we created the concept of banking fraud with several sub-concepts and their properties, including accounting fraud, bill discounting fraud, credit card fraud, debit card fraud, ATM machine fraud, money laundering transfer, loan fraud, and wire transfer fraud.

The domain of banking fraud in this ontology is a structured representation of the different types of fraudulent activities that occur in the banking sector. We provide a conceptual framework that defines the terms and relationships between different types of fraud that can be committed by or against banks and the factors that contribute to their occurrence. The ontology should include the following elements:

- **Types of banking fraud:** This includes different types of fraud that can occur in the banking sector, such as credit card fraud, debit card fraud, check fraud, loan fraud, and money laundering.
- **Banking fraud activities:** We created different activities that constitute fraudulent behaviour, such as false account opening, unauthorised withdrawals, falsification of documents, kickback fraud, and miss-statement of account fraud.
- **Fraudulent actors:** We created properties related to banking fraud. This refers to the different individuals or entities involved in fraudulent activities, including customers and bank staff.
- **Consequences of Fraud:** This refers to the impact of fraud on the bank and the customer, including financial losses, reputational damage, legal consequences, and regulatory penalties.

Finally, in the banking fraud domain, by defining the terms and relationships between these different elements, the section of the ontology will help banks and regulators better understand the nature of banking fraud and develop effective prevention and detection strategies.

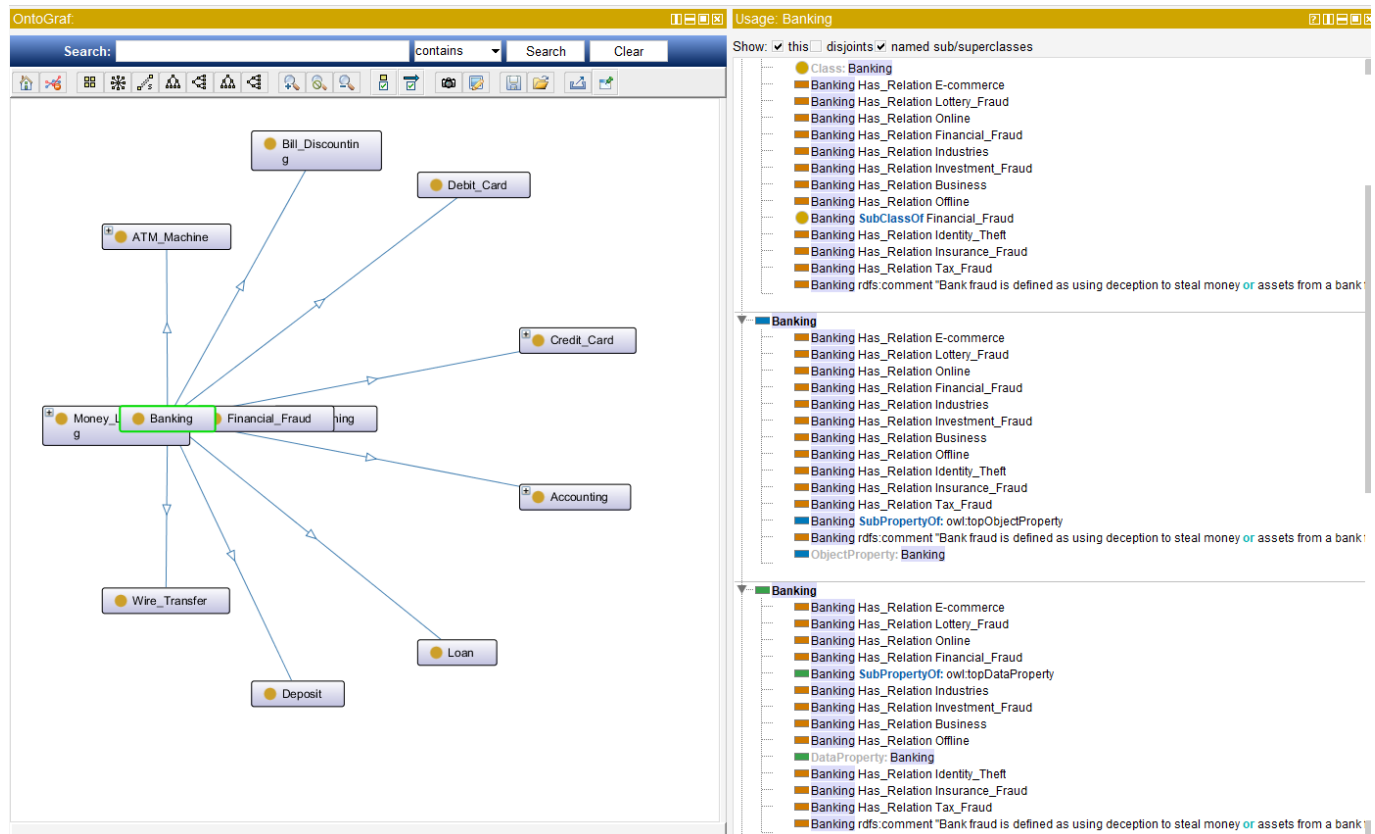


Figure 3.3: The Banking Fraud Concept.

Accounting Fraud

Accounting fraud is described as the deliberate manipulation and alteration of financial records or statements. [111] In this context, bank documents are being manipulated; the account statement of a client is distorted to represent a false financial position of the client. This could be done to an individual or company account. Accounting fraud aims to convince shareholders and investors by displaying false financial statements for the company. Attempting unlawful tax evasion by misstating assets, liabilities, revenues, and expenses

In this research, we classify accounting fraud into several categories, which include embezzlement, kickbacks, misstating income and expenses, and so on. Accounting fraud affects various sections of financial fraud, including tax fraud, money laundering, and so on. Hence, we embedded properties of accounting fraud with relational annotation. The latter describes that an entity has another relationship with another entity outside its root concept.

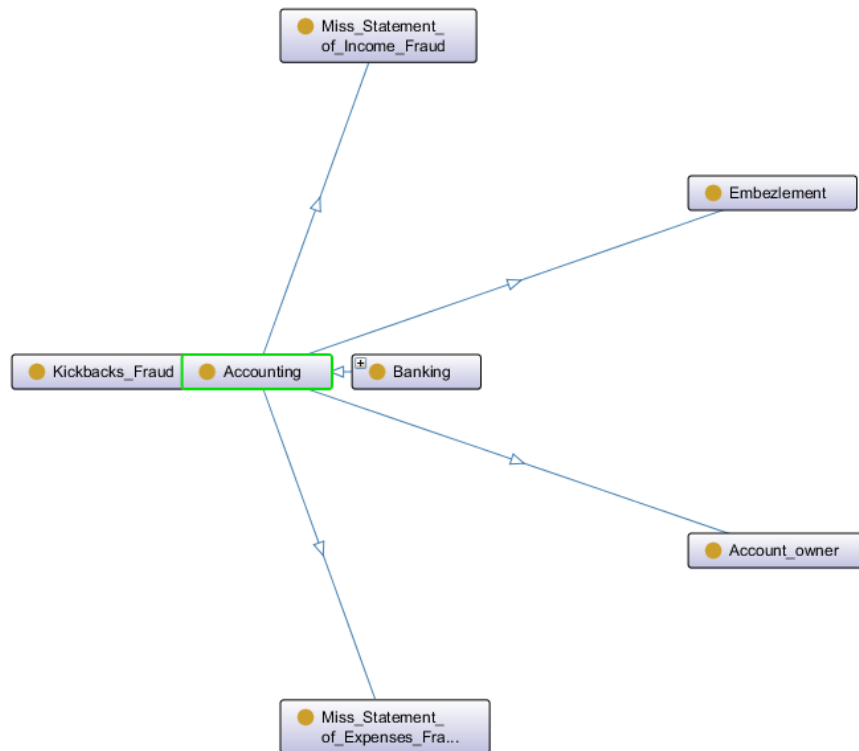


Figure 3.4: The subconcept of Account Fraud.

ATM Machine Fraud

The automatic teller machine (ATM) is becoming a target for attacks globally because fraudsters and other criminals are trying to find the machine's vulnerabilities. Several entities, tools, and dimensions are used for ATM fraud. Some of these are card jamming, shoulder surfing, and stolen cards, which constitute 65.2 percent of ATM frauds in countries like

Nigeria. [6] The ontology proposed a joint relationship between ATM fraud and other forms of financial fraud. Including identity theft, wire transfer fraud, debit card fraud, and ATM card fraud. In so many cases, ATM fraud occurred due to the machine's design structure. For example, the machine's user interface serves as the communication medium between the user and the machine. It provides the resources for the user to input his card details; some of the ATM machines, as we investigated, provide a window for shoulder surfing, card jamming, and hidden cameras. We uncovered many reported cases of ATM fraud using some of the above-mentioned cases. Shoulder surfing fraud refers to observing someone enter the PIN at an ATM machine. It is most effective in crowded places; fraudsters stand too close to the victims in order to view the card details like the pin, card number, expiration date, and CVV number.

Another case of ATM fraud involves using hidden cameras to capture users' pin codes. In this case, the fraudsters used a concealed camera to capture the details of the user while he was using the ATM machine.

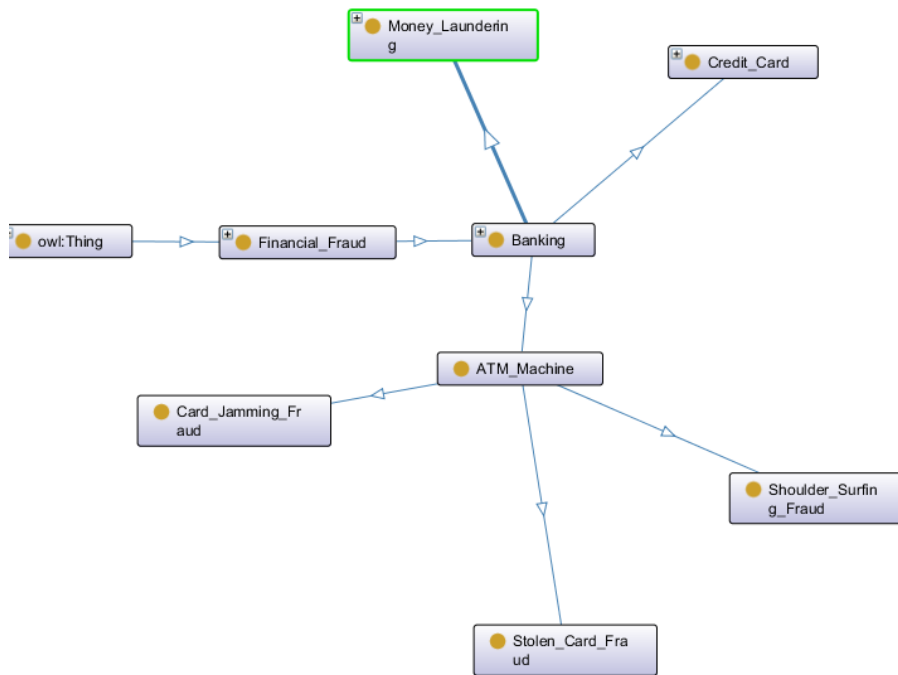


Figure 3.5: The subconcept of ATM Fraud.

Credit Card Fraud

Credit cards are one of the most famous targets of fraud, but not the only one [31]. Credit card fraud is a healthy and growing means of stealing billions of dollars from credit card companies, merchants, and consumers. Due to the rise and rapid growth of e-commerce, the use of credit cards for online purchases has dramatically increased, causing an explosion in credit card fraud. Credit cards have become the most popular mode of payment for both online and regular purchases [99]. In this research, we investigate card fraud and the patterns used by criminals to defraud victims. Credit card fraud provides a detailed

ontological understanding of credit card fraud. From entities, terminologies, relations, and instances, as well as data types and classes, including card not present fraud, counterfeiting, skimming fraud, lost or stolen card fraud, and card never arrived fraud.

The above-mentioned cases are the most common instances of credit card fraud. For example, card-not-present fraud contributes 75 percent to overall credit card fraud. Hence, we provide several properties and relations between the card-not-present entity and other credit card fraud and non-credit card fraud. The relation we proposed makes it possible to generate reasoning for other credit card fraud detection methods.

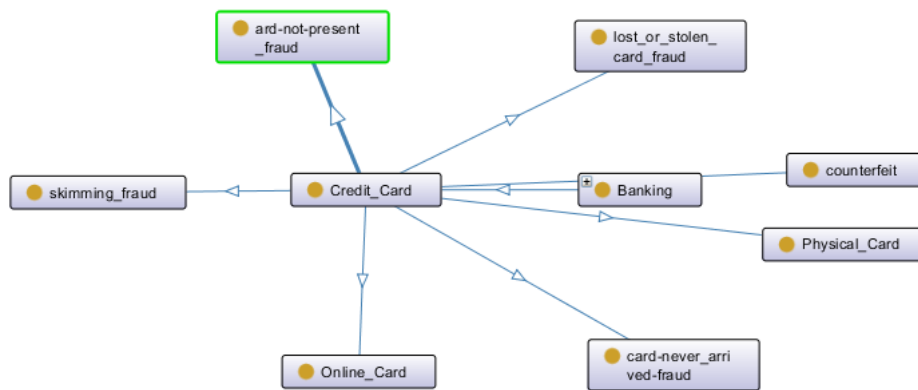


Figure 3.6: The concept Credit Card Fraud.

Money Laundering

The concept of money laundering involves the creation of a structured and organized knowledge representation system to capture and model information related to the complex activities and processes associated with money laundering. This concept serves as a formal framework for representing concepts, relationships, rules, and regulations that pertain to money laundering detection and prevention.

Money Laundering Process involved in the money laundering process, such as placement, layering, and integration. Each stage includes specific activities that criminals engage in to legitimize illegal funds.

FROD ontology encompasses various entities that play a role in money laundering, including individuals, businesses, financial institutions, and government agencies. Each entity is described in terms of their attributes, roles, and interactions within the context of money laundering.

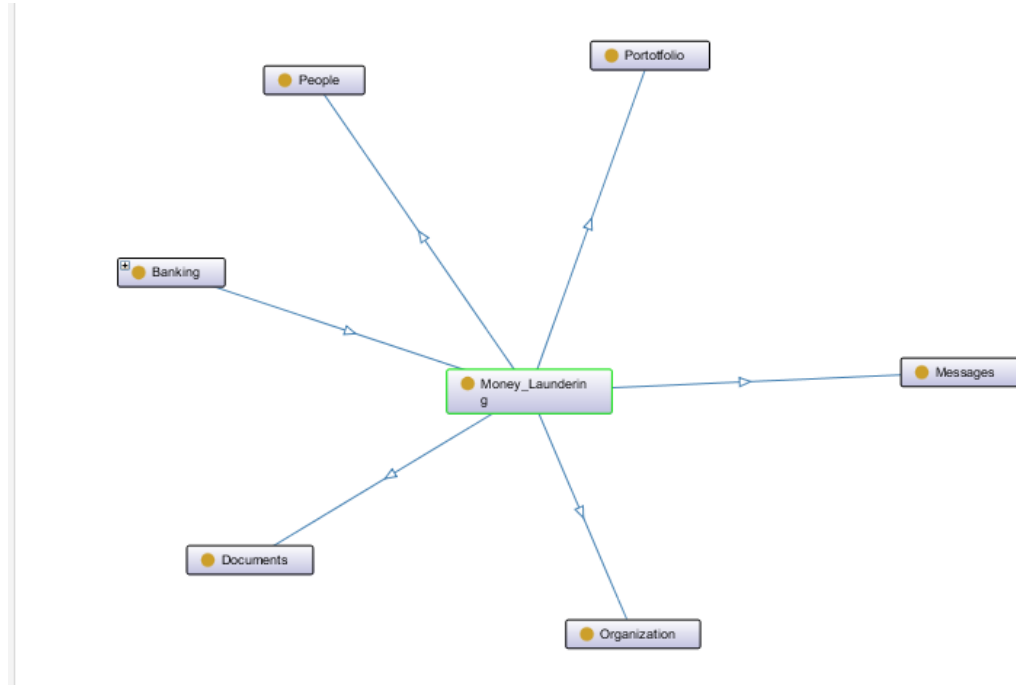


Figure 3.7: The Money Laundering Class.

3.3.2 Identity Theft

Identity theft is considered a mind game that favors hackers for the purpose of tricking users. There are various classes of identity theft, including email fraud. Most fraud detection researchers focused on non semantic features of identity theft. In this type of fraud, the fraudsters try to steal the victims identities by convincing them to communicate with the intended person. In this area, we classified several types of identity theft, from email to phishing and other scams. This domain of identity theft fraud represent and structured different types of fraudulent activities that involve stealing someone's personal information for financial gain. We provides a conceptual framework that defines the terms and relationships between different types of identity theft fraud, and the factors that contribute to their occurrence. This domain include the following elements:

- Types of identity theft fraud: In this we includes different types of fraud that involve stealing someone's personal information, such as credit card fraud, tax loan application, email and phishing fraud etc.
- Risk factors: This concepts refers to the factors that contribute to the occurrence of identity theft fraud, such as weak security measures, lack of awareness, and availability of personal information on the internet.

- Prevention strategies: In this concepts we includes measures that individuals and organizations should deploy to prevent and mitigate the risk of identity theft fraud.
- Legal and regulatory: This concepts provide us with the laws and regulations that govern identity theft fraud in various countries and regions, including criminal and civil laws, and regulatory requirements.

The entities and concepts in this domain define the terms and relationships between these different elements, this will help us to better understand the nature of identity theft fraud and develop effective prevention and detection strategies.

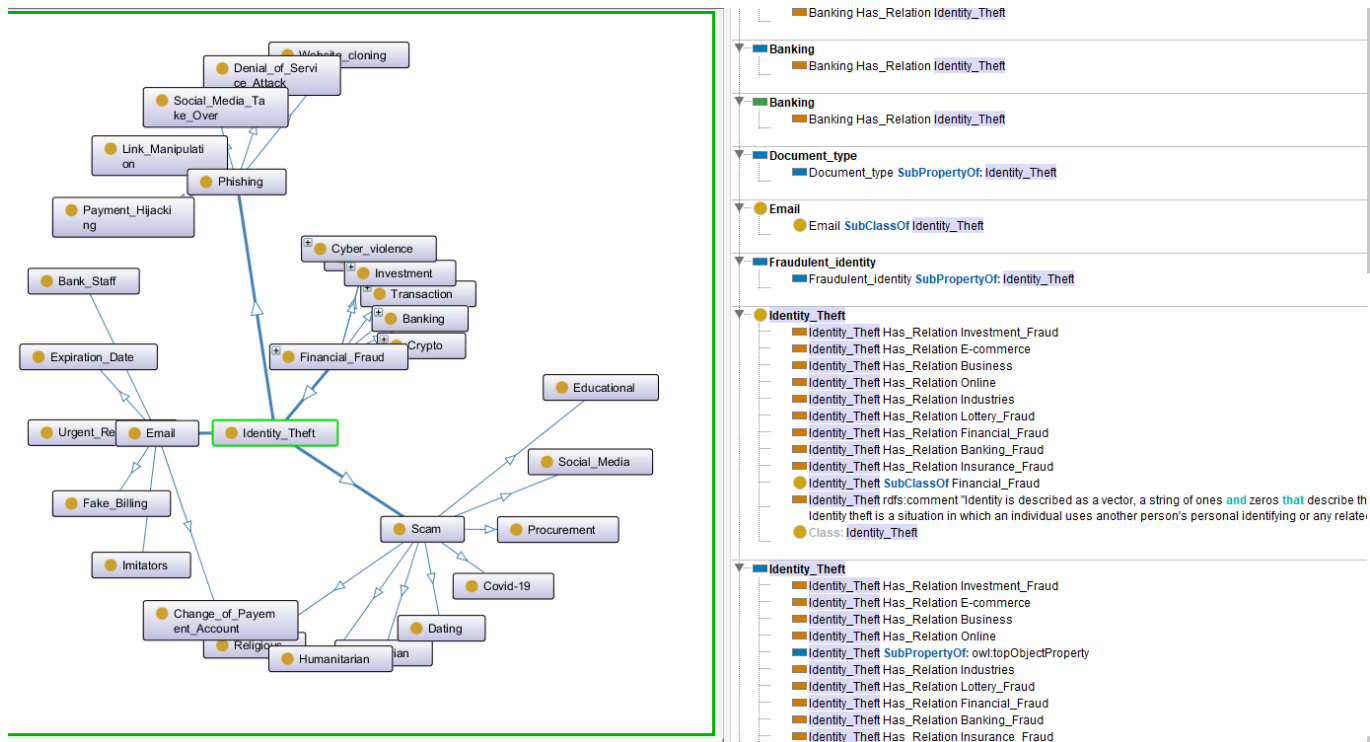


Figure 3.8: The Concept of Identity Theft.

Email Fraud

The use of email as a means of official and non official communication is increasing world-wide. The number of emails sent and received globally has increased each year since 2017. While it is believe around 306.4 billion emails have been estimated to be sent and received each day in 2020, this figure is expected to increase to over 376.4 billion daily in the coming years. [58] Claim that email fraud is more structured and successful than ever: of every 10 fraudulent emails sent, seven are delivered to the inbox, and an estimated three are responded to. More precisely, the incremental trust and usage of the service provide an opportunity for fraudsters to exploit the vulnerabilities. These vulnerabilities can come from both ends: the sender and the receiver, as well as the email service providers. In our implementation, we classified email fraud into various subclasses. The "Email" class have

properties such as "Sender Name," "Sender Email Address," "Subject," "Body," and "Attachment."

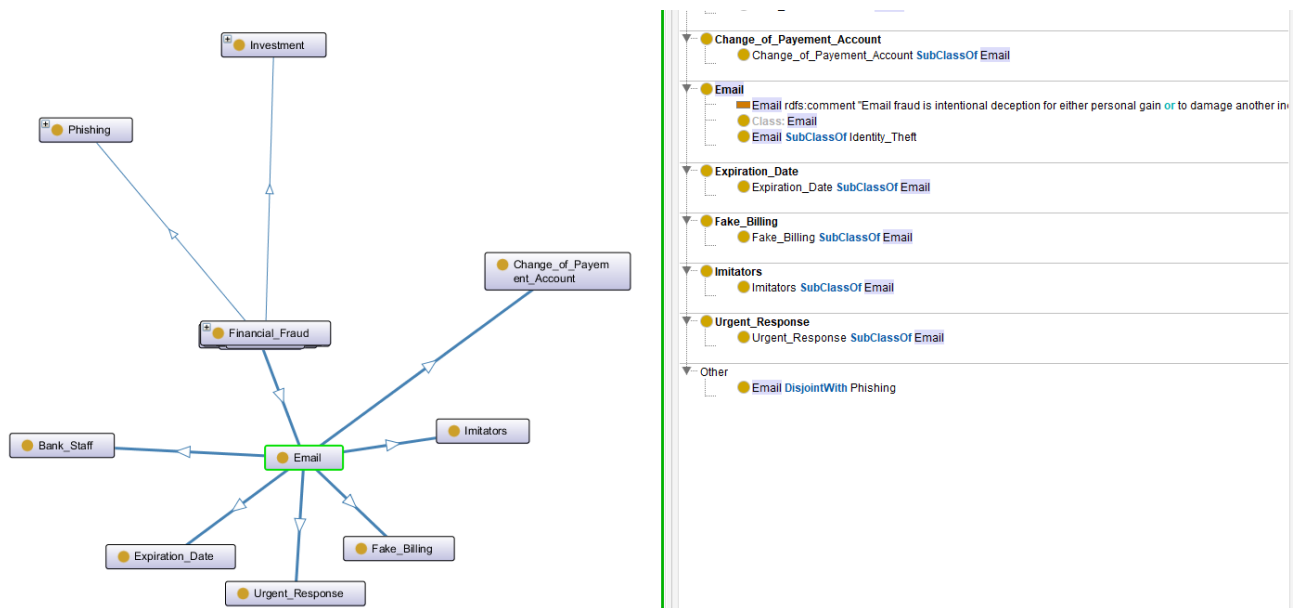


Figure 3.9: The sub concepts of Email Fraud.

Phishing fraud

Refers to the spam that is sent by fraudsters in order to deceive their victims and obtain their personal information. Fraudsters can impersonate a service provider or institute that victims collaborate with. The fraudsters can make use of convincing techniques to obtain the victim's personal information, including credit card details. The spam may also include links to fraudulent websites, which again can deceive victims into revealing their personal information. In this class of ontology, we have several entities. For example, the "Phishing" class have properties such as "Attack Date," "Attack Type," "Attack Method," and "Attack Outcome."

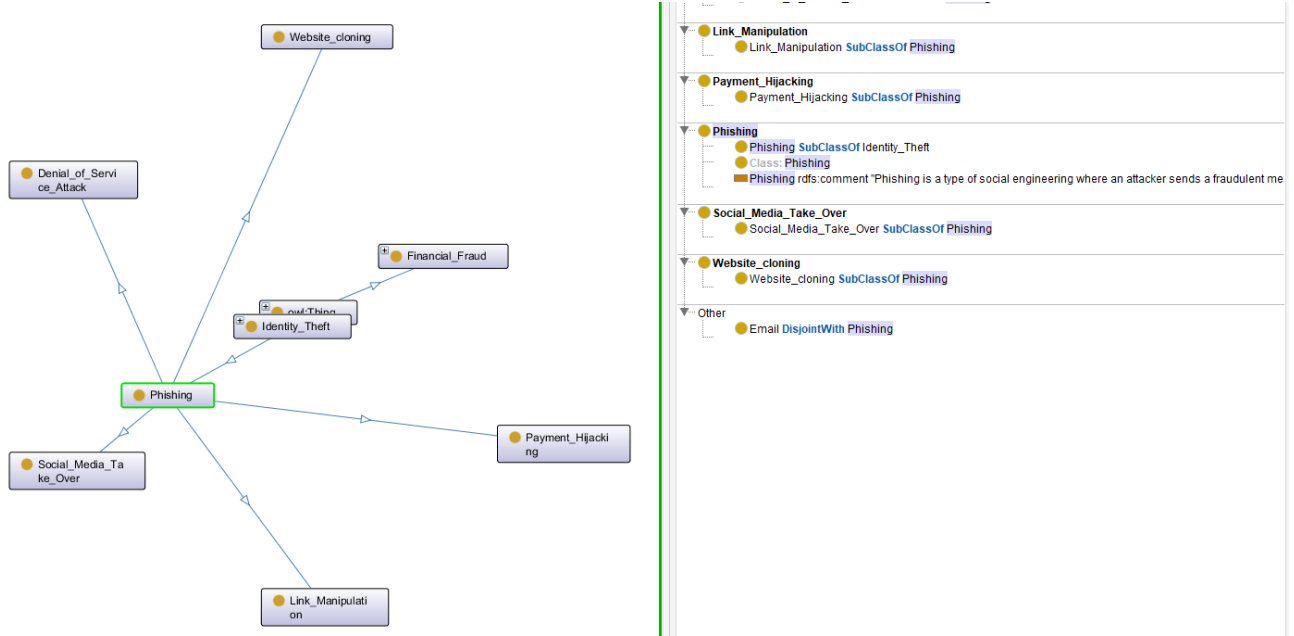


Figure 3.10: The sub concepts of Phishing Fraud.

3.3.3 Lottery Fraud

The current trend of lottery fraud cannot be ignored, especially among elderly people. In this research, there are several behavioral classifications of lottery fraud victims. On the other hand, the ontology provides a comprehensive semantic property for the kind of lotteries where fraud is more likely. For example, social media lottery fraud is more prevalent in developing countries than in developed countries. This could be due to the fact that in developed countries, people tend to have more access to information about these kinds of fraud. While the fraudsters target elderly people, there is concern that the new trend of lottery fraud is shifting toward younger and more desperate individuals.

During building this domain we include various entities related to fraudulent lottery activities which encompass classes such as "Lottery Scam," "Lottery Ticket," "Lottery Winner," "Lottery Agent," "Lottery Organization," and "Lottery Regulation."

The ontology also define properties for each class to describe their characteristics and relationships with other classes. For example, the "Lottery Scam" class have properties such as "Scam Type," "Scam Method," "Scam Location," and "Scam Outcome." The "Lottery Ticket" class have properties such as "Ticket Number," "Ticket Cost," "Ticket Prize," and "Ticket Expiration Date." Similarly, the "Lottery Winner" class have properties such as "Winner Name," "Winner Address," "Winner Phone Number," and "Winner Prize." The "Lottery Agent" class have properties such as "Agent Name," "Agent Contact Information," and "Agent Commission."

The "Lottery Organization" class can have properties such as "Organization Name," "Organization Contact Information," "Organization Regulations," and "Organization License."

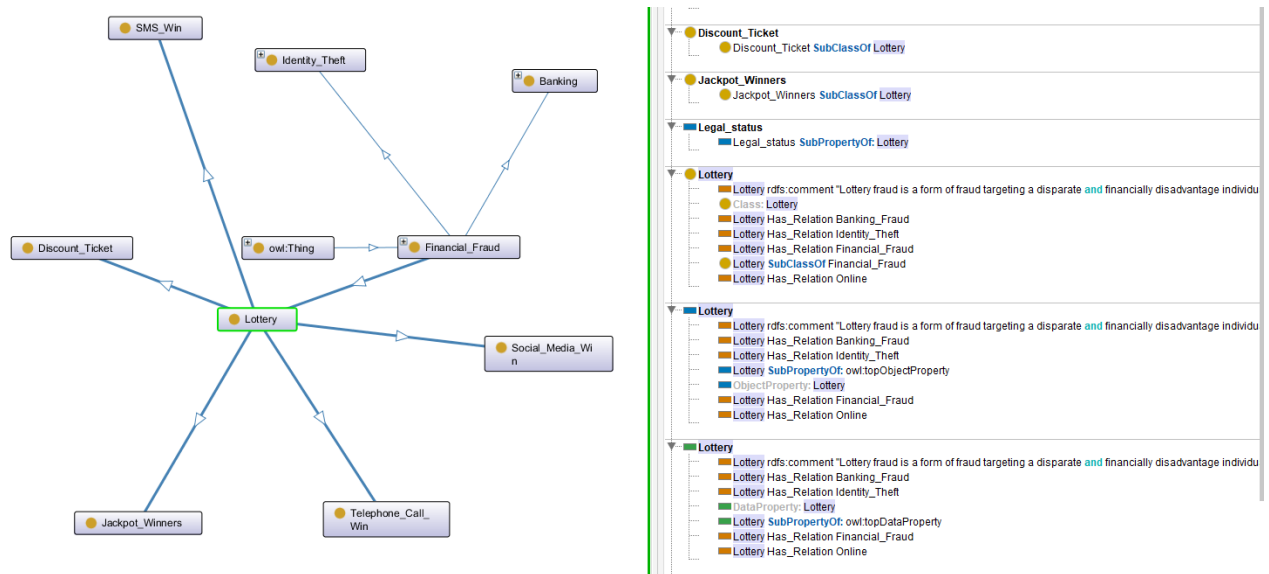


Figure 3.11: The concept Lottery Fraud.

Finally, the "Lottery Regulation" class have properties such as "Regulation Authority," "Regulation Requirements."

3.3.4 Insurance Fraud

Insurance fraud in its many forms is seen as a deliberate deception perpetrated against or by an insurance company or agent for the purpose of financial gain. Fraud may be committed at different points in the transaction by applicants, policyholders, third party claimants, or professionals who provide services to claimants. We classified insurance fraud into internal and external, and the sub concepts of this fraud are classified into health, vehicle, and life insurance. We also presented common patterns of insurance fraud, including padding, inflating claims, misrepresenting facts, and using fake license agents. At the same time, the insurance fraud domain encompass various entities related to the insurance industry. This include classes such as "Insurance Policy," "Insurance Claim," "Insurance Agent," "Insurance Company," and "Insurance Regulator." The ontology also define properties for each class to describe their characteristics and relationships with other classes. For example, the "Insurance Fraud" class have properties such as "Fraud Type," "Fraud Method," "Fraud Location," and "Fraud Outcome." The "Insurance Policy" class have properties such as "Policy Number," "Policy Type," "Policy Coverage." Similarly, the "Insurance Claim" class have properties such as "Claim Number," "Claim Type," "Claim Amount," and "Claim Status." The "Insurance Agent" class may have properties such as "Agent Name," "Agent Contact Information."

Furthermore, the ontology define relationships between classes. For example, the "Insurance Fraud" class have relationships with the "Insurance Policy" and "Insurance Claim" classes to represent the sources of the fraud. The "Insurance Agent" class have a relationship with the "Insurance Company" class to represent the agent's affiliation with the company.

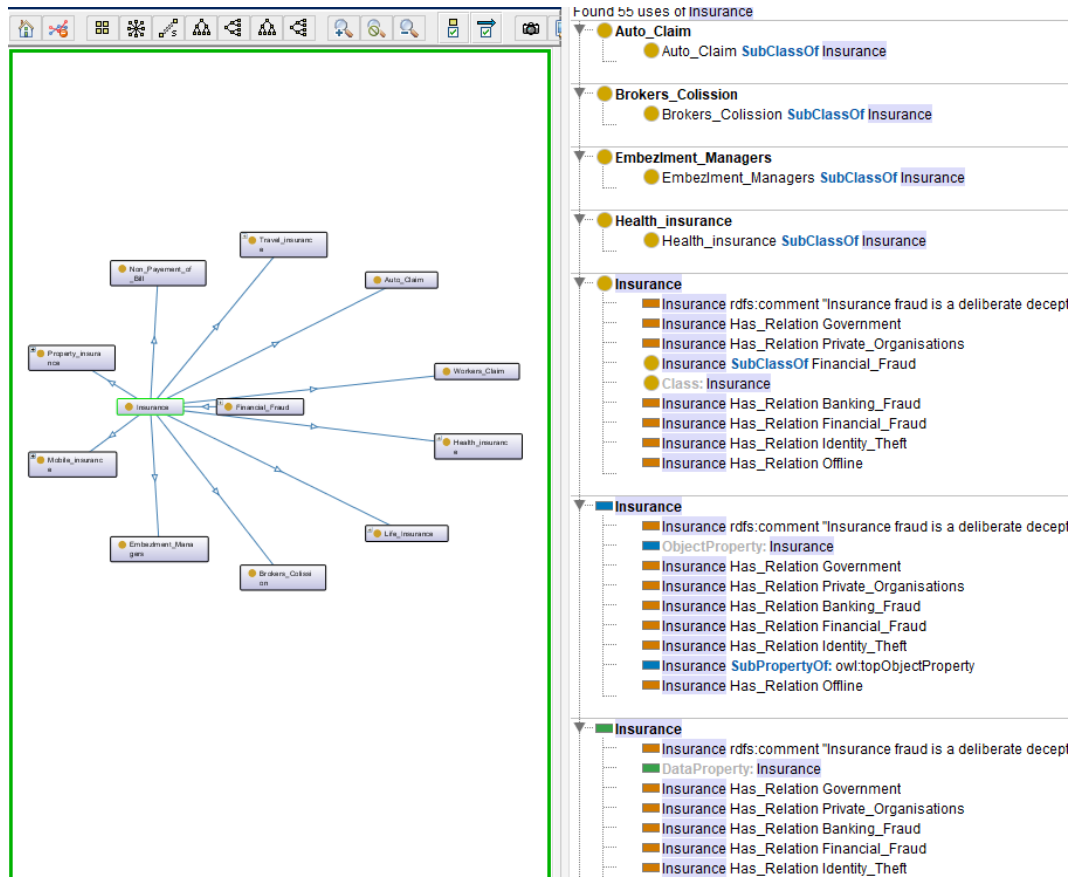


Figure 3.12: The concept of Insurance Fraud.

3.3.5 Investment Fraud

This ontology provides a comprehensive classification of investment fraud. There are various existing investment frauds; some are old enough, while some are new due to technological advancement. In investment fraud, existing investors are paid with funds collected from new investors. For example, Ponzi schemes, where organizers often promise investors a huge return on investment with little or no risk. We proposed several sub concepts of investment fraud, some targeting big investments while others targeting small and medium investors. These are Ponzi schemes, pyramid schemes, pump and dump schemes, offshore gold, boiler rooms, and advance fees. The domain include other entities in the investment industry which include "Investment Fraud," "Investment Company," "Investment Regulator," and "Investment Victim." In this domain we also define properties for each class to describe their characteristics and relationships with other classes. For example, the "Investment Fraud" class have properties such as "Fraud Type," "Fraud Method," and "Fraud Outcome." Similarly, we have properties such as "Advisor Name," "Advisor Contact Information," "Advisor Credentials," and "Advisor Compensation." The "Investment Company" class also have properties of "Company Name," "Company Contact Information," "Company Regulations," and "Company License." Finally, the "Investment Victim" class have properties such as "Victim Name," "Victim Contact Information," "Victim Investment," and "Victim Loss". Furthermore, the ontology define other relationships between classes. For example,

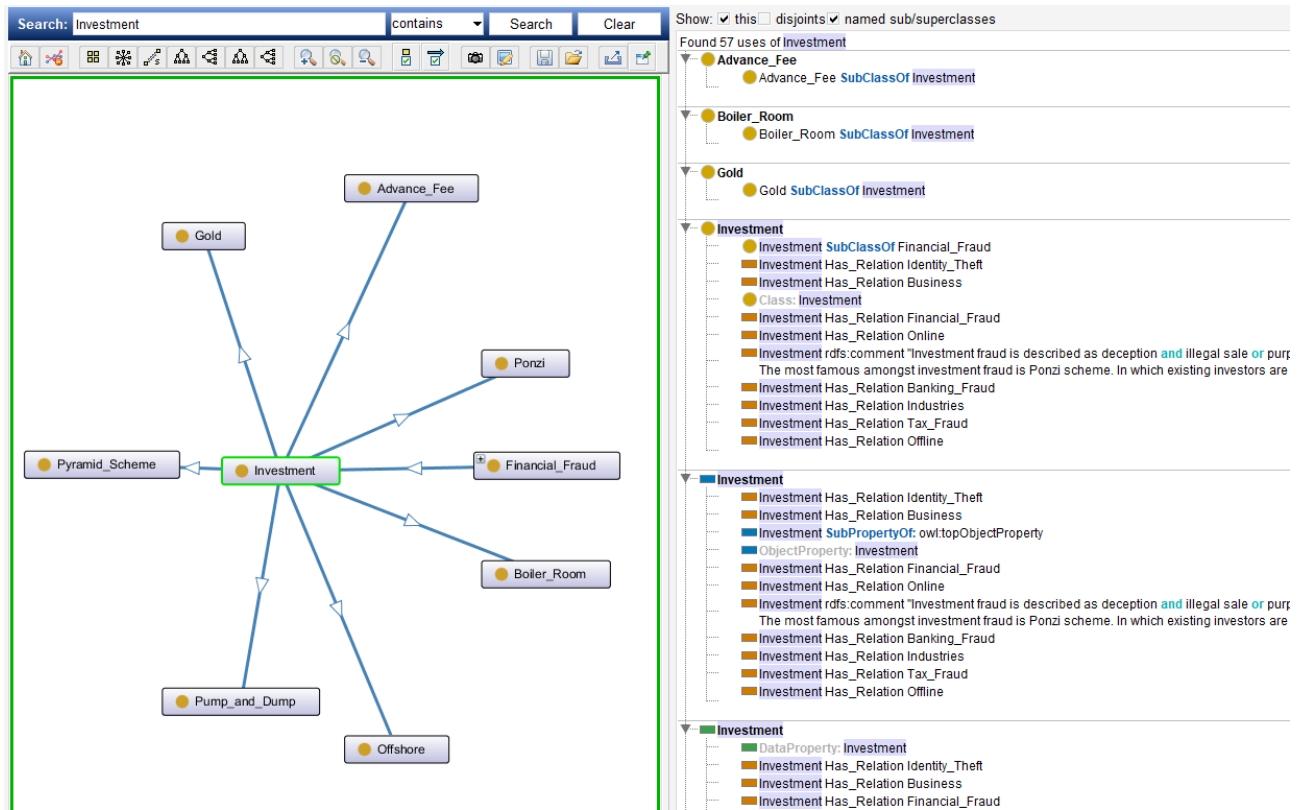


Figure 3.13: The concept of Investment Fraud.

the "Investment Fraud" class have relationships with the class of "Investment Advisor" as a classes that represent the sources of the fraud. The "Investment Company" class have a relationship with the "Investment Regulator" class to represent the company's compliance with regulations.

3.3.6 Tax Fraud

The domain of tax fraud in this ontology becomes the subset of the broader domain of financial fraud because tax fraud involves intentional deception or misrepresentation of information on tax returns to evade or reduce tax obligations. It can take many forms, such as failing to report all income, claiming false deductions or credits, hiding assets, or creating fake businesses. This section includes the process of classifying and constructing the various ways of tax evasion as a means of tax fraud. The aim is to understand the tax collection structure and the main challenges faced. Hence, we classified tax fraud into several sub concepts: tax evasion, fake income, reporting, corporate tax fraud. We conclude that while avoiding the tax burden is legal, there are several ways that individuals and organizations avoid paying taxes illegally. These provide us with other entities, properties, and individual tax noncompliance activities. This domain of tax fraud has been further divided into several subdomains or aspects, such as the types of taxes involved (e.g., income tax, sales tax, property tax), the methods used to commit fraud (e.g., falsifying documents, using offshore accounts), the entities involved (e.g., individuals, corporations, tax preparers),

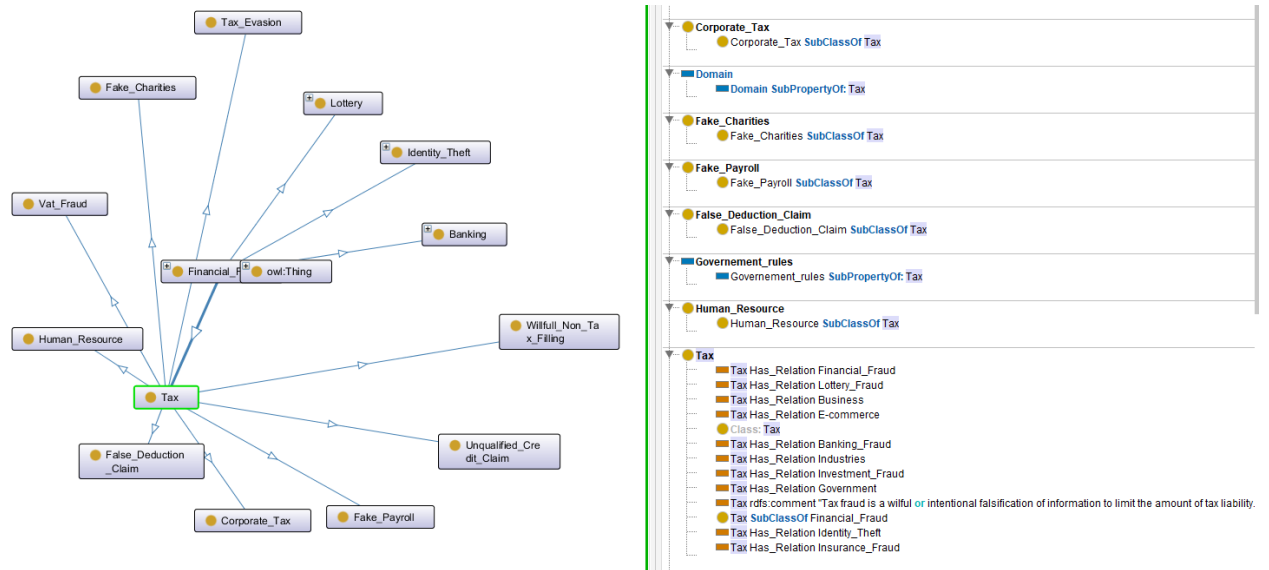


Figure 3.14: The concept of Tax Fraud.

and the legal and regulatory framework surrounding tax fraud (e.g., penalties, enforcement mechanisms). Each class of the tax fraud have its own set of properties and relationships to other classes in the ontology, such as the types of taxes involved, the methods used to commit fraud, and the legal consequences of committing tax fraud.

Finally, the domain of tax fraud in this ontology help to organize and structure information about tax fraud in a way that makes it easier to understand and analyze, at the same time it will help us to develop a fraud detection and prevention systems, by exploring rules from this domain. This domain is also usefull for educating individuals and organizations about the risks and consequences of tax fraud.

3.3.7 Transaction Fraud

The transaction fraud was extracted from one of the datasets we obtained for the purpose of validating the ontology. Initially, this concept of fraud was not in the ontology. We proposed transaction fraud to separate it from payment fraud, which already exists in the ontology. The concept of transaction includes various sub concepts. From fake alerts to fake items—items not delivered, service not rendered, etc.

The properties of this concept are in line with and related to various concepts and sub concepts of frauds in the ontology. Hence, it is one of the most relatively related concepts in this research. This domain of transaction fraud has been further divided into other sub-classes, such as the types of transactions involved (e.g., credit card transactions, wire transfers, online purchases), the entities involved for example individuals, and merchants as well as the legal and regulatory framework surrounding the transaction. Each subclass have its own set of properties and relationships to other classes in the ontology, such as the types of transactions involved, the methods used to commit fraud, and the legal consequences of committing transaction fraud.

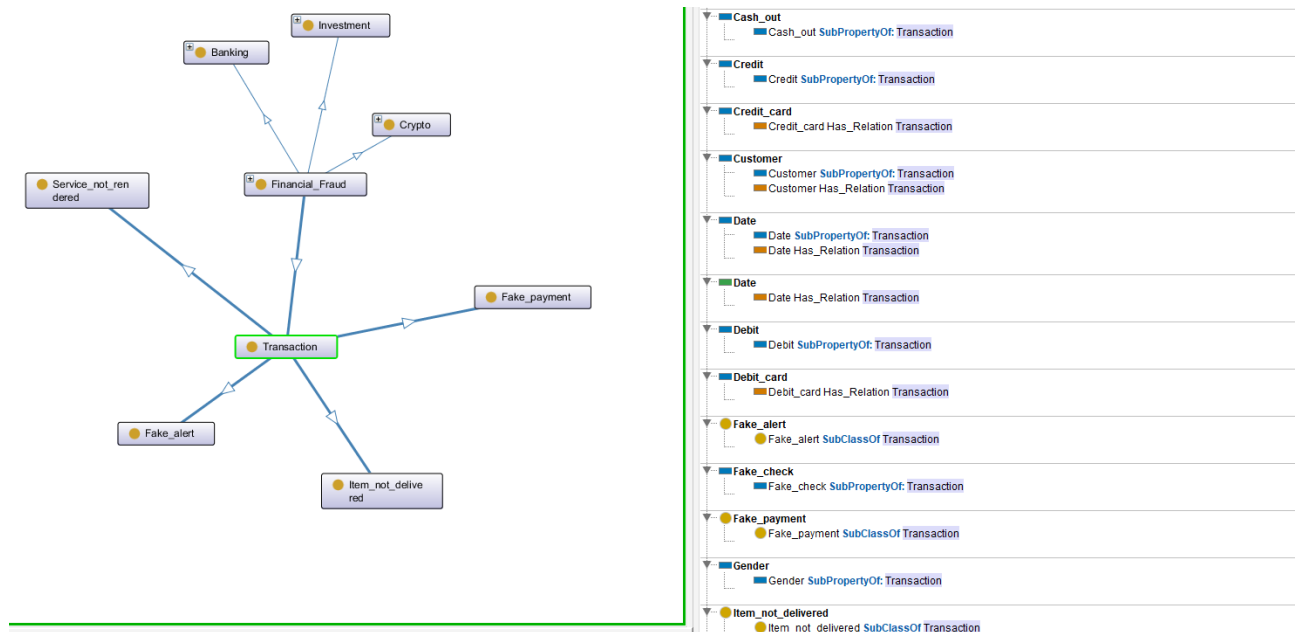


Figure 3.15: The concept of Transaction Fraud.

To conclude, the transaction fraud domain will help us to improve and organize a structure of information about transaction fraud in a way that makes it easier to understand and analyze and explore for the implementation of fraud detection rules.

3.3.8 Forex Fraud

Foreign exchange is a decentralized market for the trading of currencies; unfortunately, the market has been penetrated by fraudsters, in which various frauds are proposed to the customer. In this research, we focused on this fraud and declassified it in a semantic context. This concept has very few relations with other concepts in the ontology; all of its sub concepts have no relations with any concept or sub concept in the ontology. Nevertheless, the object properties have some commonalities with investment fraud.

The forex fraud has other subclasses such as currency trading, derivatives, commodities, we also represented the methods used to commit fraud (e.g., pump and dump schemes, fake investment opportunities, insider trading), the entities involved (e.g., individuals, brokers, financial institutions), and the legal and regulatory framework surrounding forex fraud (e.g., anti fraud regulations, consumer protection laws). In this ontology, the forex fraud domain has been represented by a forex fraud class, with subclasses representing them. Each subclass have its own set of properties and relationships to other classes in the ontology. For instance, Each of these subclasses have properties such as "fraudulent activity description", "target victim", and "potential financial impact", and relationships to other classes in the ontology such as "legal and consequences".

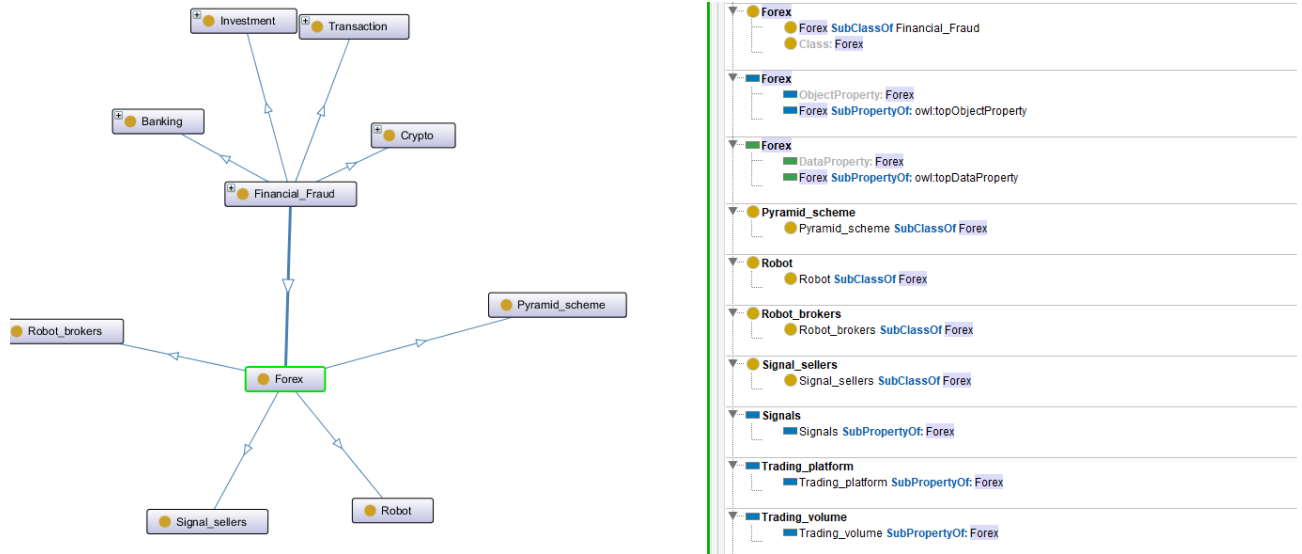


Figure 3.16: The concept of Forex Fraud.

3.3.9 Cyber Violence Fraud

This domain contains a specific area of knowledge that pertains to fraudulent activities carried out through the use of digital technologies or the internet, with the intention of causing harm, intimidation, or fear to the victim. Including online harassment, cyberstalking, sextortion, revenge porn, and doxxing, as well as the various techniques and tools used by perpetrators to carry out these activities.

The act of bullying was specifically known for teenage boys, especially in schools. Cyberviolence is often just another side of the same coin, the act aims at causing emotional and physical violence for the purpose of monetary exploitation from the victims. While conducting the research, we contacted some experts in cyber violence and bullying to understand the various forms of cyberbullying. This act of fraud is on the rise, especially in developed countries. We classified this concept into various sub concepts, which clearly diversified the concept into a significant act of violence and fraud. While investigating this concept, we understood how gender plays an important role in this fraud; for example, a sexism sub concept is mainly targeted at females. While social exclusion is targeted at men. There are not many relations between this concept and other concepts in the ontology, but identity theft and dating scams have a direct relation with this concept in the ontology. In this ontology we define the various actors involved in cyber violence fraud, including the victims, perpetrators, and bystanders, and the different roles played in these activities. The properties in the domain include the various techniques and tools used by perpetrators to carry out the fraud which mainly relied on social engineering.

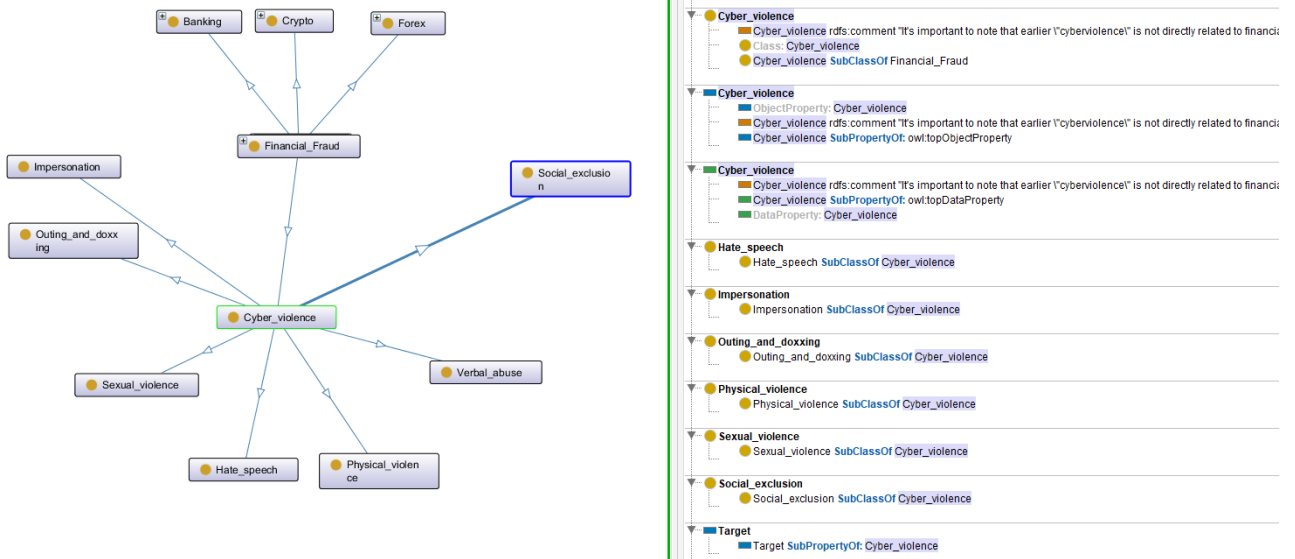


Figure 3.17: The concept of Cyber Violence Fraud.

3.4 Discussion and Conclusion

In the realm of financial fraud detection, the development and elaboration of a comprehensive ontology have emerged as a promising and innovative approach. This journey has taken us through the intricacies of constructing a knowledge based system that can significantly enhance our understanding and management of financial fraud risks.

Our endeavour began with the conceptualization and design of the financial fraud ontology. This intricate knowledge structure aimed to capture the multifaceted nature of financial fraud, encompassing a plethora of fraudulent activities, *modus operandi*, and contextual factors. Each element of the ontology, from classes and subclasses to properties and instances, was meticulously crafted to form a coherent and meaningful representation of the domain. The ontology's hierarchical structure allowed for a nuanced categorization of financial fraud types, ranging from traditional forms like identity theft and credit card fraud to more contemporary threats such as cryptocurrency scams and digital payment fraud. Through rigorous research and domain expertise, we ensured that the ontology was both comprehensive and adaptable, capable of accommodating evolving fraud tactics.

One of the fundamental strengths of our elaborated ontology was its seamless integration with empirical data. By fusing structured data sources, transaction records, and historical fraud cases with the ontology's knowledge framework, we achieved a synergistic approach to fraud detection. This integration enabled us to leverage both historical patterns and real time data to identify suspicious activities and mitigate risks proactively.

Moreover, the ontology served as a critical bridge between raw data and actionable insights. It facilitated the extraction of meaningful features and attributes from the data, allowing for the development of machine learning models and predictive algorithms. The ontology based approach enhanced the interpretability and explainability of these models, a crucial aspect in the context of fraud detection.

Central to the ontology's utility was its ability to support knowledge driven reasoning. The ontology is capable of inferencing and deduction based on established rules and relationships within the domain. This reasoning extended to the identification of anomalies, detection of fraudulent patterns, and can act for the generation of hypotheses for investigation. This knowledge driven reasoning was particularly valuable in addressing emerging and sophisticated fraud schemes. The ontology's adaptability allowed for the incorporation of domain expertise and the rapid update of fraud detection rules. As a result, our system remained agile and responsive to evolving fraud tactics, thereby enhancing its resilience against new threats. While our journey in elaborating the financial fraud ontology yielded promising outcomes, it was not without its challenges and limitations. Notably, the construction and maintenance of the ontology demanded a substantial investment of time and expertise. Continuous updates and validation were necessary to ensure its relevance and accuracy in a dynamic environment.

Furthermore, the effectiveness of the ontology based approach was contingent on the availability of high quality, domain specific data. In some cases, the scarcity of labelled data for certain fraud types posed a challenge to training machine learning models effectively. The imbalanced nature of fraud datasets also required advanced techniques for handling

skewed class distributions.

As we conclude our exploration of the elaborated financial fraud ontology, it is essential to chart the path forward. The success of this approach has paved the way for several future directions and areas of improvement: Combining the strengths of ontology based knowledge systems with advanced machine learning algorithms holds immense potential. Future research should focus on the development of hybrid systems that can harness the interpretability of ontologies and the predictive power of machine learning.

The knowledge captured within the financial fraud ontology can find applications beyond the financial sector. Exploring its adaptability and transferability to other domains, such as healthcare can unlock new avenues.

In conclusion, the elaboration of a financial fraud ontology represents a paradigm shift in the field of fraud detection. It signifies a departure from conventional rule based approaches and embraces the synergy of knowledge and data. The journey we embarked on has illuminated the potential of harnessing domain expertise, structured knowledge, and machine learning to combat financial fraud.

ONTOLOGY VALIDATION

4.1 Introduction

Validating an ontology is the process of confirming that it satisfies specific requirements for the purpose for which it was designed. To make sure this ontology is appropriate for the intended use, it must be validated. Among the many procedures in the validation process is examining the ontology's structure to make sure it is well-formed and adheres to accepted conventions. In order to make sure there are no logical contradictions or inconsistencies, it also entails examining the ontology's coherence. Another crucial component of validation is completeness, which is making sure the ontology includes all the relationships and ideas pertinent to the domain it represents. Last but not least, correctness is a crucial component of validation that entails determining whether the ontology correctly depicts reality.

In the current state of research and development in the field of ontology validation, we find contributions on scenario-based validation and dataset-based validation (see Adedoyin (2016), Suarez (2015), Neon (2015), and Zipfl (2023)) methodologies that have been introduced to validate developed ontologies. The aim of this chapter is to propose and validate the FROD ontology. We propose two approaches: a data-based approach and a scenario-based approach.

The chapter is organised as follows: Section 1 presents a dataset-based validation approach. The aim of this approach is to validate the FROD ontology with existing datasets. The approach relies on a clustering method where dataset is used to instantiate some concepts of FROD and then centroids are initialised. In Section 2, we propose a scenario-based validation approach. This validation approach goes beyond merely evaluating the structure and syntactic correctness of the ontology. It involves designing and defining relevant scenarios that reflect real-world situations or tasks. The Discussion and Conclusion section examines the benefits and drawbacks of these validation approaches, as well as potential future work.

4.2 Dataset Based Validation Approach

Data-based validation of an ontology is an important step in ensuring that the ontology is fit for its intended purpose. This process involves using a dataset to test and validate the ontology and ensure that it accurately represents the real-world domain it is intended to model. One method of data-based validation is to use instance data to test the ontology. Instances consist of specific examples of entities and relationships within the domain being modeled.

The main steps of a dataset-based validation approach are: 1: split the dataset into two parts; 2: use the first part to instantiate the corresponding concepts of the ontology; 3: define the centroid of the set of instances of each instantiated concept; 4: using a clustering algorithm; and 5: use the second part of the dataset to evaluate the ontology and the different centroids. Those different steps are described in the next section. Another important step is to analyse the dataset. Indeed, a dataset cannot be used if it is not compliant with the ontology.

In the field of ontology development, the validation of ontologies plays a crucial role in ensuring their quality and effectiveness. One effective approach to validate an ontology is by utilizing datasets. Datasets provide a valuable resource for assessing the correctness, completeness, and consistency of an ontology.

By using datasets, we can evaluate the ontology's ability to capture and represent real-world instances and relationships. The dataset serves as a benchmark against which the ontology's performance can be measured. It allows us to assess how well the ontology aligns with the data it is intended to model.

This process of validating an ontology using datasets, several steps are typically followed.

- First, a representative dataset is selected that covers a wide range of instances and relationships relevant to the domain of the ontology. This dataset should be comprehensive and diverse to effectively evaluate the ontology's coverage.
- Next, the dataset is mapped to the ontology, aligning the instances and relationships in the dataset with the corresponding concepts and properties in the ontology.
- The first part is then used to instantiate the related concepts of the ontology. Each of those concept will thus has a set of instances and can be considered as a cluster. We therefore compute the centroid of each cluster.
- Finally, the second part of the dataset, and the clusters and their centroids are used to evaluate the performance of the ontology-based fraud detection.

Overall, the use of datasets to validate an ontology provides a systematic and objective approach to assess its quality and fitness for purpose. It allows for the identification of potential issues and improvements, ultimately leading to a more robust and reliable ontology.

The next subsections describe the main steps of our a clustering-based fraud detection approach.

4.2.1 Towards a clustering-based fraud detection approach

The proposed clustering-based fraud detection approach relies on two main steps: 1) Concept instantiation that defines the different cluster related the used dataset and 2) Centroids Initialization. These two steps are described in the following subsections.

Concept instantiation

In this step, the first part of the dataset is mapped to the ontology, aligning the instances and relationships in the dataset with the corresponding concepts and properties in the ontology. N instances (N is the size of the first part of the dataset) are thus created. The instantiated fraud classes are those that are used in dataset. So, each dataset is used to validate a subset of fraud classes.

This mapping allows for a direct comparison between the ontology and the dataset, identifying any inconsistencies or discrepancies. Once the mapping is established, the dataset is splitted into two part.

Clusters and Centroids Initialization

In the context of fraud detection, centroid computation can be used as a technique to identify patterns and anomalies in data. It involves calculating the centroid or average of a group of data points and using it as a reference point for comparison.

Algorithm 0 gives the details of the centroid intitialisation of a given class. Note that this algorithm is suitable for objects whose properties are not always numbers. For objects with non-numeric properties, you can use an appropriate numerical representation for each property. For example, you can use one-hot encodings, word vectors, or other representation techniques to convert the properties into numerical values. Then, you can apply the centroid algorithm. This will allow you to calculate the centroid for objects with non-numeric properties.

For example, in credit card fraud detection, you can calculate the centroid of legitimate transactions by considering various features such as transaction amount, location, time, and merchant category. This centroid represents the typical behavior of genuine transactions.

When a new transaction occurs, its features are compared to the centroid of legitimate transactions. The closeness of the transaction from the centroid, indicate the status and legitimacy of the transaction.

By using centroid computation, you can create a baseline or reference point to identify unusual or suspicious transactions. This approach helps in detecting fraudulent activities

Algorithm 1 Centroid initialization of a class with a set of instances

Require: n number of instances of a class.

Require: v the vector of properties of that class and m the number of these properties.

Initialize the centroid coordinates to zero for each dimension.

Iterate through all the instances.

For each instance, add the instance's properties to the centroid coordinate.

Divide the centroid coordinates by the total number of instances to obtain the final centroid coordinates.

The result is the centroid of the set of objects.

by flagging transactions that are significantly different from the normal behavior.

4.2.2 A Clustering Approach to Validate the Ontology

To validate the clusters using centroids, you can use a technique called cluster validation. One common method is to calculate the distance between each data point and its corresponding centroid. The smaller the distance, the more likely the data point belongs to that cluster.

Similarity Measures

There are several different similarity measures that can be used in ontologies, each with its own strengths and weaknesses [109]. One common approach is to use a graph-based measure, such as the shortest path distance, which measures the number of edges that need to be traversed to move from one concept to another in the ontology graph. This measure is easy to compute and interpret, but it can be sensitive to the structure of the ontology and may not capture more subtle similarities between concepts.

In this validation approach, we consider a simplified scenario where we have three clusters (A, B, and C) and a single data point (D) in a two-dimensional space. The goal is to calculate the proximity of the data point to each cluster's centroid using the Mahatma distance metric.

Let's consider a simplified scenario where we have three clusters (A, B, and C) and a single data point (D) in a two-dimensional space. The goal is to calculate the proximity of the data point to each cluster's centroid using the Euclidean distance metric.

Clustering Performance Evaluation

To assess the quality of the clustering, we use metrics such as the within-cluster sum of squares (WCSS) or the silhouette coefficient. WCSS measures the sum of squared distances between each data point and a centroid within each cluster. A lower WCSS indicates tighter and more compact clusters.

Algorithm 2 Selecion of the closest centroid for a given instance

Require: $nobject_i$ instances of the dataset.

Require: C the set of the centroids.

$centro$ the first centroid of C $c \in C$

if $silhouette(c) \leq centro$ **then** $silhouette(c) = centro$
 if
 thenreturn $centro$

The silhouette coefficient measures how well each data point fits within its own cluster compared to other clusters. It ranges from -1 to 1, with values closer to 1 indicating better clustering.

By analyzing these metrics, we determine the optimal number of clusters and assess the quality of the clustering algorithm. If the centroids effectively represent the clusters, we would expect to see well-separated clusters with low WCSS and high silhouette coefficients.

Experimental Validation of the data-based approach

The dataset used in this method is a synthetic financial transaction dataset available on Kaggle. It's designed to simulate electronic financial transactions, including both legitimate and fraudulent transactions. This dataset is commonly used for research and testing in the field of fraud detection. The dataset is reasonably large and contains a substantial number of records, making it suitable for various data analysis and machine learning tasks. It includes a set of features that describe different aspects of each transaction. These features typically include information such as the transaction amount, merchant details, customer details, timestamp, and more.

This dataset is valuable for research and experimentation, it's also important to recognize that Its synthetic nature makes it a useful tool for developing and testing algorithms while avoiding privacy and security concerns associated with real financial data. However, it's essential to consider its limitations when drawing conclusions or making decisions based on the findings. Describe the experimental protocol.

4.2.3 Experimental Protocol

The experimental protocol was design to ensure appropriate steps are. We outline a comprehensive experimental protocol:

Data Selection:

In financial fraud detection research, the choice of dataset is critical. Researchers typically select datasets that mimic real-world scenarios and encompass a wide range of fraudulent and non-fraudulent transactions. For our experiments, we've chosen the "Synthetic Financial Fraud Detection" dataset, available on Kaggle. This dataset is synthetic but de-signed to resemble actual financial transaction data, making it suitable for our evaluation.

Data Preprocessing:

Prior to experimentation, preprocess the dataset to address issues like missing values, feature scaling, and data transformation. This stage ensures that the data is in a suitable format for analysis.

Data Splitting:

In this method, the allocation of data into training and testing sets is a critical step of the workflow. It allows us to train our models on one subset of the data and assess their performance on another subset that they haven't seen during training. The aim is to ensure that the models can generalize well to new, unseen data.

The choice of the 60-40 ratio for splitting the data implies that 60% of the dataset is used for training, while the remaining 40% is reserved for testing. This is done in order to provide a reasonable balance between training and evaluation data. It ensures that the models have sufficient data for learning patterns and making predictions, while also having a substantial portion of data to assess their performance.

The training set 60% is used to train the clustering algorithm. During this phase, the algorithm identifies patterns and structures in the data and creates clusters based on the features of the transactions.

The testing set is used to evaluate the performance of the clustering algorithm. It contains a portion of the data that the algorithm has never seen during training. This separation simulates real-world scenarios where the algorithm is applied to new, unseen data.

Data splitting enables us to assess how well the trained models generalize to new, unseen data. It provides a realistic evaluation of the models' performance in a real-world scenario.

By keeping a portion of the data separate from training, we can detect and mitigate over-fitting issues. Overfitting occurs when a model learns the training data too well but struggles to perform on new data.

The testing set allows us to calculate various performance metrics, including accuracy, precision, recall, F1-score, and others. These metrics help us understand how effectively the models identify fraud cases and legitimate transactions.

Algorithm Selection:

There are several reasons that compel us to implement clustering method in this validation approach. Below are some of them: **Inherent Data Structure Exploration:** Clustering algorithms are designed to uncover inherent patterns or structures within data. In financial fraud detection, transaction data often contains hidden patterns that can indicate fraudulent activities.

Model Exploration:

Clustering enables the creation of models that group similar data points together. This can be useful for exploring the characteristics of different clusters. For instance, we found that certain clusters consist primarily of small transactions, while others have larger ones.

Epsilon-based validation aims to identify clusters with high Silhouette Scores and low

WCSS values. These clusters are considered well-defined and compact, making them suitable for further analysis.

Model Training

We start with three initial centroids: C1, C2, and C3. These centroids are randomly placed within the feature space. For each transaction in our dataset, we measure the distance to each cluster and assign it to the nearest centroid.

Once all data points are assigned to clusters, we recalculate the centroids of each cluster by taking the mean of the data points within that cluster. For Cluster 2, we compute the new centroid position based on the mean values of all transactions assigned to Cluster 2.

Steps two and three are repeated iteratively until convergence. In each iteration, data points are reassigned to clusters based on the new centroids. The algorithm continues until the centroids no longer change significantly or a specified number of iterations is reached.

After convergence, we have three as Cluster 1, Cluster 2, and Cluster 3. Each cluster represents transactions with similar characteristics. The quality of the clusters is assessed using metrics of Silhouette Score and WCSS as discussed below.

4.2.4 Silhouette Score Overview

The Silhouette Score measures the quality of clustering. For each data point, it quantifies how similar it is to its own cluster compared to other clusters. A higher score indicates better-defined clusters.

4.2.5 Computation

To calculate WCSS, we sum the squared manhattan distances from each data point to its assigned cluster centroid:

$$WCSS = \sum_{i=1}^n \sum_{j=1}^k d(x_i, c_j)^2$$

Where:

- WCSS is the Within-Cluster Sum of Squares.
- n is the number of data points.
- k is the number of clusters.
- x_i is a data point.

- c_j is the centroid of cluster j .
- $d(x_i, c_j)$ is the Euclidean distance between data point x_i and centroid c_j .

Results

After applying the epsilon-based validation, we obtain the following detailed results for each cluster:

Cluster 1:

- Silhouette Score: 0.92
- WCSS: 1432

Cluster 2:

- Silhouette Score: 0.91
- WCSS: 1598

Cluster 3:

- Silhouette Score: 0.94
- WCSS: 1256

Weighted Average: We calculate the weighted average Silhouette Score and WCSS for the entire dataset.

4.3 Scenario Based Validation Approach

The Scenario based validation is an approach that involves validating the effectiveness and performance of the ontology by simulating and evaluating different real world scenarios. This approach allows for a comprehensive analysis of the system's behavior and its ability to handle specific scenarios.

The simulated scenarios involve cases where an attacker impersonates an individual to gain unauthorized access to sensitive information or perform fraudulent transactions. The system's ability to identify suspicious patterns, detect unauthorized access attempts, and prevent fraudulent activities is evaluated within this scenario.

While the proposed banking fraud scenario aims to evaluate the ontology knowledge and its effectiveness in detecting and preventing fraudulent activities within the banking domain. The scenario simulated fraudulent transactions, such as account takeovers, lost

or stolen card and unauthorized fund transfers. The ontology ability to detect unusual patterns, identify suspicious transactions, and raise alerts or take appropriate actions is examined within this scenario.

Furthermore, in the investment fraud scenario, the system's performance in detecting and preventing fraudulent investment schemes is evaluated. The scenarios involve simulated cases of Ponzi schemes, insider trading, or fraudulent investment advice. The ontology ability to identify fraudulent investment opportunities, detect abnormal trading patterns, and provide timely warnings or preventive measures is assessed within this scenario. Each scenario in this validation approach provides a focused context for evaluating the ontology capabilities in handling specific fraud related challenges. By simulating real world scenarios, the approach enables a comprehensive assessment of the system's performance and the overall effectiveness in mitigating fraud risks.

4.3.1 Identity Theft Scenario

Identity theft is a prevalent form of fraud that involves the unauthorised acquisition and use of another person's personal information, such as their name, social security number, or financial details, with malicious intent. This scenario aims to describe and simulate an ontology-based model for identity theft in the context of financial fraud. By leveraging ontological representations, we can better understand the underlying relationships, entities, and activities involved in such fraudulent activities.

In this chapter, we represent identity theft as the act of stealing someone's identity for fraudulent purposes, involving the acquisition and misuse of personal information.

- **Victim:** The person whose identity has been stolen, including attributes like name, social security number, date of birth, and financial details. **Perpetrator:** The individual or group responsible for carrying out the identity theft, including attributes like name, aliases, known associates, and methods used for identity theft.
- **Fraudulent Activities:** Encompasses various activities involved in identity theft, such as phishing, data breaches, social engineering, identity document forgery, and financial account takeover.
- **Fraudulent Tools:** Represents the tools and techniques utilised by perpetrators, including malware, spoofing, key loggers, and skimming devices. **Simulation Scenario:** Scenario Setup: A perpetrator named Adamu, well versed in identity theft techniques, decides to target a specific victim, Ahmed, with the intention of gaining unauthorised access to his financial accounts.
- **Identity Theft Process:** Adamu conducts online research to gather information about Ahmed, such as his social media profiles, public records, and employment details. Adamu sends a phishing email to Ahmed, appearing as a reputable financial institution, requesting his login credentials. Ahmed unknowingly falls victim to the phishing attempt and provides his login credentials to a fraudulent website controlled by

Adamu. With access to Ahmed's financial account, Adamu transfers funds to an off-shore account he has previously set up using false identity documents. Adamu uses spoofing techniques to cover his tracks and make it difficult to trace the origin of the fraudulent transactions.

- **Proposed Detection Method:** Ahmed notices unauthorised transactions in his financial account and reports the incident to his bank. The bank's fraud detection system identifies suspicious transactions and flags them for investigation. Investigators examine the transaction records, analyse metadata, and leverage the FROD ontology to detect patterns based on the initiated rules that are indicative of identity theft. Investigators collaborate with law enforcement agencies and share information about the perpetrator's methods and potential accomplices. The investigation leads to the identification and apprehension of Adamu, who is subsequently prosecuted for identity theft and financial fraud.
- **Education and Awareness:** FROD Ontology provides a structured framework to organise and disseminate knowledge about identity theft risks, prevention techniques, and best practices. Organisations can use the ontological representations to educate customers and raise awareness about identity theft, empowering them to take proactive measures to protect their personal information.

4.3.2 Banking Fraud Scenario

Introduction: Banking fraud is a significant concern in the financial industry, involving illicit activities aimed at exploiting vulnerabilities in banking systems for personal gain. This scenario aims to describe and simulate a scenario for banking fraud in the context of financial fraud. By utilising ontological representations, we can better understand the entities, relationships, and activities associated with fraudulent actions in banking.

A fraudster named Adamu, with knowledge of banking systems and fraud techniques, targets a specific bank customer, Ahmed, with the intent to carry out banking fraud.

Adamu gathers information about Ahmed through social engineering techniques and data breaches, obtaining his personal details and banking credentials. Using phishing techniques, Adamu sends a convincing email to Ahmed, luring him to click on a malicious link and unknowingly provide him banking login credentials.

With Ahmed's account details, Adamu gains unauthorised access to her bank account and begins manipulating funds through fraudulent transactions. To avoid detection, Adamu employs money laundering techniques, funnelling the illicitly obtained funds through multiple accounts and complex transactions.

Proposed Detection Method for Banking Fraud: In the realm of banking fraud detection, an ontology-based approach offers a powerful means to explore rules that facilitate the identification of fraudulent activities. The following detection method has been proposed based on an exploration of rules derived from the ontology:

- One of the important rules explored focuses on transaction amount anomalies. By

comparing transaction amounts against a customer's historical patterns, deviations can be identified.

- Unusually high or low amounts are flagged as potential indicators of fraudulent activity.
- Another rule centres around geographical location anomalies. By analysing transaction origin, particularly in relation to a customer's usual transaction history, transactions from geographically distant locations are identified as suspicious.
- Unusual times for transactions have also been included in the explored rules. Deviations from a customer's regular transaction patterns, such as transactions conducted at odd hours or outside their typical activity window, are flagged for further investigation.

4.3.3 Investment Fraud Scenario

Investment fraud involves deceptive practices aimed at defrauding individuals or organisations of their investments by providing false information, promising high returns, or engaging in unauthorised trading activities. This scenario aims to describe and simulate an investment fraud scenario within the context of financial fraud. By exploring the intricacies of this scenario, we can better understand the entities, activities, and deceptive techniques involved in investment fraud. Scenario Description: Adamu, a charismatic and persuasive fraudster, targets unsuspecting individuals with a fraudulent investment scheme. He presents himself as a knowledgeable and trustworthy investment advisor and seeks out potential victims, such as Ahmed, who are looking to grow their wealth through investments.

- Establishing Trust: Adamu builds trust with Ahmed by showcasing his expertise, using false credentials, and highlighting his track record of successful investments. He gains confidence and positions himself as a reputable advisor.
- False Promises: Adamu entices Ahmed with promises of high returns and guaranteed profits. He presents investment opportunities with fictitious companies, claiming they have groundbreaking technologies or exclusive access to lucrative markets.
- Misleading Information: To support his claims, Adamu fabricates investment reports, financial statements, and market research to create the illusion of legitimacy. He manipulates data and provides false testimonials or endorsements from nonexistent clients.
- Pressure Tactics: Adamu employs high-pressure sales tactics, emphasising limited availability or time-sensitive opportunities to manipulate Ahmed into making quick investment decisions without conducting thorough due diligence.
- Unauthorized Trading: Once Ahmed invests his funds, Adamu engages in unauthorised trading activities, diverting a portion of the investment into his own accounts. He falsifies trading records to mislead Ahmed about the performance of his investments.

- Ponzi Scheme: To sustain the fraud and provide false returns to early investors, Adamu uses funds from new investors to pay off initial investors. This perpetuates the illusion of profitable investments and entices more victims to invest.

In the context of investment fraud, a robust detection method is needed, and the detection method must be knowledgeable enough. Hence, we developed this approach by leveraging ontological representations and exploring rules derived from the ontology. This approach identifies potential fraudulent activities and protects investors from falling victim to investment fraud. The detection method is based on the utilisation of an ontology that captures relevant knowledge about investment fraud. This ontology consists of entities, relationships, and rules that represent the various aspects of fraudulent activities in the investment domain.

- One key rule focuses on the detection of suspicious investment offers. By analysing investment opportunities using the ontology, specific characteristics associated with fraudulent schemes can be identified. Unrealistic promises of high returns, guaranteed profits, or exclusive access to lucrative markets are indicative of potentially fraudulent activities. Another important rule is centred around the identification of deceptive information. By examining the information provided by investment advisors or companies, potential discrepancies or falsified data can be flagged. Cross-referencing this information with trusted sources and stored knowledge within the ontology enables the detection of misleading information intended to deceive investors.
- The detection method also incorporates the identification of high-pressure sales tactics. By analysing communication logs and interactions between investment advisors and investors, patterns of aggressive or coercive techniques can be recognized. These tactics, designed to pressure investors into making quick decisions without thorough due diligence, raise red flags for potential investment fraud.
- Unauthorised trading activities are another critical aspect of investment fraud detection. By examining trading records and transactions, deviations from authorised activities can be identified. The ontology facilitates the detection of unauthorized trading conducted by investment advisors without the consent or knowledge of investors, alerting financial institutions to potentially fraudulent activities.

4.3.4 Discussion of Fraud Scenarios

The simulated identity fraud scenario proposed a spoofing scenario, which is a subclass in the FROD ontology. The scenario represents a detection method based on behavioural analysis and fraud analytics to identify suspicious activities, anomalous behaviours, and known fraud patterns associated with identity theft. This method offers the strength of immediate detection, enabling timely intervention to mitigate the impact on victims. Moreover, the effectiveness of the method depends on the availability and accuracy of data sources, and continuous updates and adjustments are necessary to address evolving identity theft

techniques. While banking fraud scenarios present a scenario of account takeover, The proposed detection method utilises ontological representations and explores rules derived from the ontology to identify suspicious transaction patterns, behavioural anomalies, and unauthorised activities. This method benefits from the structured knowledge representation provided by the ontology. Rule-based detection allows for the identification of known fraudulent patterns and activities stored within the ontology, improving the effectiveness of detection. The analysis of transaction patterns, behaviour, and identity verification failures aids in detecting potential banking fraud. However, the effectiveness of the method relies on the accuracy and completeness of the ontology and the rules derived from it.

Furthermore, investment fraud is a well-known ponzi scheme method for fraud that aims aimed defraud individuals of their investments. The proposed detection method integrates ontological representations and explores rules to identify suspicious investment offers, misleading information, high-pressure sales tactics, and unauthorised trading. The ontology provides a structured framework for capturing and organising knowledge about investment fraud. Rule-based detection enables the identification of common characteristics and patterns associated with fraudulent investment schemes. By detecting suspicious investment offers, deceptive information, and high-pressure sales tactics, the method allows for early intervention and the prevention of investment fraud.

In this validation method, the scenarios are rooted in real-world situations, making them highly relevant for assessing the ontology's applicability to practical use cases. This validation approach ensures that the ontology aligns with real-world requirements. Encompassing complex interactions and relationships between entities, pushing the ontology to handle intricate situations. This helps identify potential shortcomings in ontology design and reasoning capabilities. It provides assurance mechanisms for ontologies. By allowing for the identification and rectification of inaccuracies, ambiguities, or inconsistencies within the ontology.

To conclude, validation through scenarios ensure the ontology's knowledge is accurate and aligns with real-world facts. Any inaccuracies or discrepancies are brought to light and can be corrected. While at the same time assess the ontology's ability to make accurate inferences and logical deductions. This ensures that the ontology can provide valuable insights and answers to queries. Scenario-based ontology validation is a robust approach for assessing the quality, accuracy, and usability of ontologies. By subjecting ontologies to real-world scenarios, we can ensure that they meet user needs, accurately represent domain knowledge, and enhance the overall quality of knowledge management systems. This validation method is instrumental in building ontologies that are not only comprehensive but also dependable for diverse applications across domains.

4.4 Exploration of Rules for Fraud Detection System

The system proposes an effective financial fraud detection and prevention system based on exploring dynamic knowledge of frauds from the ontology. This exploration allows for the identification and formulation of rules that capture the relationships, patterns, and characteristics of fraud knowledge in the ontology. The explored rules offer solutions for financial fraud in some selected fraud scenarios. The scenarios simulated how fraudsters penetrate user accounts to steal useful information for identity theft or perform unauthorised transactions. Some steps are followed for the development of rules as follows:

- **Terminologies:** The terminologies presented in the below table are populated with axioms extracted from the ontology. This knowledge extraction process ensures that the rules capture a comprehensive understanding of fraudulent behaviours and their underlying patterns from the ontology and are applied to appropriate scenarios.
- **Rule Functions:** Based on the ontology knowledge, we formulated the rules to capture the relationships and patterns that signify fraudulent activities. These rules include if-then statements and logical expressions that describe the conditions and actions associated with fraudulent behaviours. In this part we define extraction of rules based on the Identity theft scenario, Banking fraud scenario and Investment fraud scenario. Testing anti fraud rules involves verifying that the rules are working correctly and accurately detecting potential fraudulent activities or entities.

Function of (IdentityTheftRules) takes an ontology as input and validates the Identity Theft scenario with the help of rule statements. It first checks if the subclass of the ontology is Identity Theft, then creates multiple rule statements to check different conditions such as ATM withdrawal location, subscription payment, insurance and wrong transaction pin. According to the rule statement that matches the condition, it prints the corresponding output message of "Identity Theft" , "Blocked Account" or it prints nothing.

Function of (CreditCardRules) takes an ontology as input and validates the Credit Card scenario with the help of rule statements. It first checks if the subclass of the ontology is Credit Card and then creates multiple rule statements to check different conditions such as IP address, geolocation, number of transactions and transaction amount. If any rule statement matches the condition, it prints the output message "Credit Card Fraud" or "Identity Theft.". The function returns nothing.

Function of (InvestmentRules) takes an ontology as input and validates the Investment scenario with the help of rule statements. It first checks if the subclass of the ontology is Investment and then creates multiple rule statements to check different conditions such as investment payment, investment scheme, transaction amount and previous transaction. If any rule statement matches the condition, it prints the output message "Investment Fraud" or "Investment Fraud, Blocked." It also returns nothing. We validated different scenarios with rule statements based on a provided ontology onto n. Each function checks what subclass of the ontology is being worked with and then executes a series of rule statements using exec().

IdentityTheftRules checks the Identity Theft subclass and executes six rule statements related to various conditions that may indicate identity theft. If any of these conditions are met, a notification is printed to the user. CreditCardRules checks the Credit Card subclass and executes eight rule statements related to conditions that may indicate credit card fraud. If any of these conditions are met, a notification is printed to the user.

- **Rule Statement:** A rule statement is a clear and specific guideline that defines conditions and actions to be taken when certain conditions are met. In the context of fraud detection, a rule statement typically follows an "IF-THEN" structure, where it specifies what should happen if specific conditions or triggers are observed. These statements serve as automated instructions for detecting and responding to potentially fraudulent activities, helping to maintain security and integrity in various domains, including finance, cybersecurity, and compliance.

4.4.1 Fraud Rules

To continue with approach to enhance the efficiency and accuracy of financial fraud detection, we proposed to utilise the ontology-based knowledge in order to to extract valuable fraud rules. These rules, which are derived from domain-specific ontologies, provide a structured framework for identifying and responding to potential fraudulent activities. The proposed anti fraud rules include a condition and conclusion to decide if activity is fraudulent or not. The rules are developed from the extracted terminologies and implemented to define a mechanism for discovering relationships between the ontology knowledge and fraud scenario patterns, much like logical reasoning.

At its core, the anti-fraud rules we proposed are of "if-then" statement that articulates specific conditions or triggers (the "if" part) and the corresponding actions to be taken if those conditions are met (the "then" part). These rules range from simple to highly complex, depending on the types of fraud it aims to counter. The "if" conditions can be diverse, encompassing various data points and indicators, such as transaction amounts, frequencies, geographic locations, user behaviors, and more. When these conditions align with the defined criteria, the proposed system can executes the prescribed "then" actions, which may include blocking a transaction, flagging it for review.

The anti-fraud rules hold immense significance in the realm of financial fraud detection for several compelling reasons which include: Real-time operation

- Prevention and Deterrence
- Challenges in Implementing Fraud Rules
- Customizability
- Scalability
- Compliance

The anti fraud rules are structured as follows:

Identity theft

Unusual Login Location: This rule detects identity theft by monitoring login locations. If a user logs in from an unusual geographic location that is significantly different from their regular login history, it raises a flag. For example, if a user typically logs in from their home city but suddenly logs in from another country, this rule triggers an alert.

Multiple Failed Login Attempts: Identity thieves may attempt multiple failed logins to gain access to an account. This rule identifies suspicious behaviour by tracking unsuccessful login attempts. If there are too many consecutive failed login attempts, it raises a flag to prevent unauthorized access.

Rapid Change in User Behaviour: Identity theft often involves changes in user behaviour patterns. This rule monitors user activities and looks for rapid changes in behaviour, such as unusual transaction frequencies or access times. If such changes occur, it flags them for further investigation.

Uncharacteristic Account Access Frequency: Users typically have consistent patterns of accessing their accounts. This rule detects identity theft by identifying inconsistent access frequency. If an account shows erratic login times or too frequent logins, it raises a flag.

Simultaneous Access from Different Devices: Identity thieves may attempt to access an account from multiple devices simultaneously. This rule identifies suspicious behaviour by monitoring simultaneous logins from different devices. If such behaviour is detected, it raises an alert.

Password Reset Requests from Unusual Locations: Password reset requests from unfamiliar or high-risk locations can indicate identity theft attempts. This rule flags password re-set requests originating from such locations for further verification.

Abnormal Access Times: Unusual access times, such as accessing an account in the middle of the night, can be a sign of identity theft. This rule detects abnormal access times and raises a flag if suspicious access patterns are observed.

Suspicious Account Information Changes: Identity thieves may modify account information to gain control. This rule monitors changes in account details and flags suspicious alterations, such as changes in contact information or security questions.

Unusual Pattern of Funds Transfer Requests: Identity theft often involves unauthorized fund transfers. This rule identifies unusual patterns in fund transfer requests, such as high-value transfers to unfamiliar accounts, and raises an alert. **Alerts from External Threat Intelligence Feeds:** To enhance security, this rule integrates with external threat intelligence feeds. If any external source reports suspicious activity related to identity theft, it triggers an alert for immediate attention.

Table 4.1: Fraud Detection Rules Based on Ontology Knowledge

Identity Theft		
Rule #	Fraud Type	Rule Statement
1	Identity	IF Unusual Login Location, THEN flag.
2	Identity	IF Multiple Failed Login Attempts, THEN flag.
3	Identity	IF Rapid Change in User Behavior, THEN flag.
4	Identity	IF Uncharacteristic Account Access Frequency, THEN flag.
5	Identity	IF Simultaneous Access from Different Devices, THEN flag.
6	Identity	IF Password Reset Requests from Unusual Locations, THEN flag.
7	Identity	IF Abnormal Access Times, THEN flag.
8	Identity	IF Suspicious Account Information Changes, THEN flag.
9	Identity	IF Unusual Pattern of Funds Transfer Requests, THEN flag.
10	Identity	IF Alerts from External Threat Intelligence Feeds, THEN flag.

Banking Fraud

Unusual Large Transactions: Banking fraud often involves large transactions to withdraw funds. This rule detects transactions significantly exceeding a user's typical transaction size and raises a flag for further investigation.

Sudden Account Balance Changes: Banking fraud may result in sudden and unexplained changes in account balances. This rule monitors account balances and flags abrupt balance fluctuations that cannot be accounted for by legitimate transactions.

Multiple Failed Authentication Attempts: Banking fraudsters may repeatedly attempt authentication to gain unauthorized access. This rule identifies suspicious behaviors by tracking multiple consecutive failed authentication attempts and raises an alert.

Abnormal ATM Withdrawal Patterns: Unusual patterns in ATM withdrawals, such as withdrawing large amounts from multiple ATMs in a short time, can indicate banking fraud. This rule detects abnormal ATM withdrawal patterns and triggers an alert.

Simultaneous Online and In-Person Transactions: Banking fraudsters may attempt simultaneous online and in-person transactions. This rule monitors both online and in-

person transactions and raises a flag if such simultaneous activity is detected.

Inactive Account Suddenly Active: If an account that has been inactive for an extended period suddenly becomes active with numerous transactions, it raises suspicion of banking fraud. This rule identifies such cases and triggers an alert.

Large Out-of-Network ATM Usage: Banking fraud may involve using ATMs located far from the account holder's usual location. This rule detects large out-of-network ATM usage and raises an alert for unusual activity.

Suspicious Account Access from High-Risk Locations: High-risk geographic locations are often associated with banking fraud. This rule monitors account access from such locations and raises a flag if suspicious access is detected.

Unusual Fund Transfers Between Accounts: Banking fraud may involve unauthorized fund transfers between accounts. This rule identifies unusual patterns in fund transfers, such as high-value transfers to unfamiliar accounts, and raises an alert.

Alerts from Banking Transaction Monitoring Systems: To enhance security, this rule integrates with banking transaction monitoring systems. If these systems generate alerts related to banking fraud, it triggers an alert for immediate attention.

Banking Fraud		
Rule #	Fraud Type	Rule Statement
1	Banking	IF Unusual Large Transactions, THEN flag.
2	Banking	IF Sudden Account Balance Changes, THEN flag.
3	Banking	IF Multiple Failed Authentication Attempts, THEN flag.
4	Banking	IF Abnormal ATM Withdrawal Patterns, THEN flag.
5	Banking	IF Simultaneous Online and In-Person Transactions, THEN flag.
6	Banking	IF Inactive Account Suddenly Active, THEN flag.
7	Banking	IF Large Out-of-Network ATM Usage, THEN flag.
8	Banking	IF Suspicious Account Access from High-Risk Locations, THEN flag.
9	Banking	IF Unusual Fund Transfers Between Accounts, THEN flag.
10	Banking	IF Alerts from Banking Transaction Monitoring Systems, THEN flag.

Investment Fraud

Unverified Investment Opportunities: Investment fraud often begins with unverified investment opportunities. This rule identifies investment opportunities lacking proper verification and raises a flag.

Promises of Guaranteed High Returns: Fraudulent investment schemes often promise unrealistically high returns. This rule detects investment opportunities that guarantee such returns and raises a flag for further scrutiny.

Pressure to Invest Quickly: Investment fraudsters may pressure potential victims to invest hastily. This rule monitors interactions and identifies situations where there is undue pressure to invest immediately.

Lack of Documentation or Transparency: Investment scams often lack proper documentation and transparency. This rule raises a flag when investment opportunities lack essential documentation or provide insufficient information.

Unauthorized Trading Activity: Investment fraud can involve unauthorized trading on behalf of clients. This rule monitors trading activities and flags unauthorized trades that could indicate investment fraud.

Unusual or Unexpected Fees: Investment fraud may involve hidden or unexpected fees.

Unauthorized Trading Activity: Investment fraud sometime involve unauthorized trading on behalf of clients. This rule monitors trading activities and flags unauthorized trades that could indicate investment fraud.

Difficulty in Accessing Funds: Investment fraudsters may make it difficult for victims to access their funds. This rule detects cases where investors face difficulties withdrawing or accessing their invested funds.

Inconsistent Investment Strategies: Fraudulent investment schemes may employ inconsistent or unclear investment strategies. This rule identifies investments with inconsistent strategies and raises a flag.

Alerts from Investment Regulatory Authorities: To enhance security, this rule integrates with investment regulatory authorities. If these authorities issue alerts related to investment fraud, it triggers an alert for immediate attention.

Unusual Patterns in Investment Portfolio: Investment fraud can result in unusual patterns in an investor's portfolio. This rule monitors investment portfolios and flags portfolios exhibiting suspicious patterns or high-risk investments.

These rules serve as a critical layer of protection against various forms of fraud, providing us with the means to detect and respond to potentially fraudulent activities promptly.

Investment Fraud		
Rule #	Fraud Type	Rule Statement
1	Investment	IF Unverified Investment Opportunities, THEN flag.
2	Investment	IF Promises of Guaranteed High Returns, THEN flag.
3	Investment	IF Pressure to Invest Quickly, THEN flag.
4	Investment	IF Lack of Documentation or Transparency, THEN flag.
5	Investment	IF Unauthorized Trading Activity, THEN flag.
6	Investment	IF Unusual or Unexpected Fees, THEN flag.
7	Investment	IF Difficulty in Accessing Funds, THEN flag.
8	Investment	IF Inconsistent Investment Strategies, THEN flag.
9	Investment	IF Alerts from Investment Regulatory Authorities, THEN flag.
10	Investment	IF Unusual Patterns in Investment Portfolio, THEN flag.

Testing of Rules

In this part we define extraction of rules based on the Identity theft scenario, Banking fraud scenario and Investment fraud scenario. Testing anti fraud rules involves verifying that the rules are working correctly and accurately detecting potential fraudulent activities or entities.

- Function of (IdentityTheftRules) takes an ontology as input and validates the Identity theft scenario with the help of rule statements. It first checks if the subclass of the ontology is Identity Theft, then creates multiple rule statements to check different conditions such as ATM withdrawal location, subscription payment, insurance and wrong transaction pin. According to the rule statement that matches the condition, it prints the corresponding output message of "Identity Theft" , "Blocked Account" or it prints nothing.
- Function of (CreditCardRules) takes an ontology as input and validates the credit card scenario with the help of rule statements. It first checks if the subclass of the ontology is Credit Card and then creates multiple rule statements to check different conditions such as IP address, geolocation, number of transactions and transaction amount. If any rule statement matches the condition, it prints the output message "Credit Card Fraud" or "Identity Theft.". The function returns nothing.
- Function of (InvestmentRules) takes an ontology as input and validates the Investment scenario with the help of rule statements. It first checks if the subclass of the ontology is Investment and then creates multiple rule statements to check different conditions such as investment payment, investment scheme, transaction amount and previous transaction. If any rule statement matches the condition, it prints the output message "Investment Fraud" or "Investment Fraud, Blocked." It also returns nothing. We validated different scenarios with rule statements based on a provided ontology onto n. Each function checks what subclass of the ontology is being worked with and then executes a series of rule statements using exec().
- IdentityTheftRules checks the Identity Theft subclass and executes six rule statements related to various conditions that may indicate identity theft. If any of these conditions are met, a notification is printed to the user.
- CreditCardRules checks the Credit Card subclass and executes eight rule statements related to conditions that may indicate credit card fraud. If any of these conditions are met, a notification is printed to the user.
- InvestmentRules checks the Investment subclass and executes four rule statements related to conditions that may indicate investment fraud. If any of these conditions are met, a notification is printed to the user.

4.4.2 Machine Learning Comparison

Machine learning algorithms possess a great power for financial fraud detection. The usually relied on dataset to uncover a fraud pattern. While FROD ontology systems rely on a predefined knowledge model of financial fraud. The FROD system leverages expert knowledge to formulate rules and logic that guide the identification of potentially fraudulent transactions. It offers transparency and interpretability, attributes highly valued in financial sectors subject to stringent regulations.

This comparison aims to dissect the attributes of both approaches comprehensively. We will explore their flexibility, interpretability, resource efficiency, adaptability, and propensity for generating false positives. Through this comparison, we seek to provide a clear understanding of the strengths and limitations of each approach and, importantly, investigate the prospects of a hybrid system that synthesizes the best of both worlds.

Below we experiment some machine learning algorithm for comparison and analysis.

Considered Machine Learning Algorithms

We experimented various machine learning models as follows:

- **Random Forest** In the experiment, the Random Forest algorithm was trained and evaluated on the financial fraud dataset using the specified hyperparameters, and the following results were obtained:

The accuracy of the Random Forest model indicates that it correctly classified 92.5% of transactions. The precision score of 0.85 suggests that when the model predicts a transaction as fraudulent, it is correct 88% of the time. The recall score of 0.78 indicates that the model captured 78% of actual fraudulent transactions. The F1 score of 0.81 balances the trade-off between precision and recall. The Random Forest algorithm exhibited strong performance in detecting financial fraud based on the provided dataset. The accuracy signifies that the model made accurate predictions for most transactions. The high precision score highlights the model's ability to correctly identify fraudulent transactions, thus reducing the rate of false positives. The recall score indicates that the model successfully detected a substantial portion of actual fraudulent transactions. One of the key advantages of the Random Forest algorithm is its ensemble learning approach, where multiple decision trees are combined to produce a robust classification model. This enables the model to capture complex relationships within the data and mitigate overfitting.

However, it is important to acknowledge certain limitations. The performance of the random forest model shows a sensitivity to the number of trees in the ensemble and other hyperparameters. Overfitting has occurred since hyperparameters are not appropriately tuned. Additionally, while random forest models provide high accuracy, they may not be as interpretable as simpler models like decision trees.

- **XGBoost** Accuracy: 92.5 Precision: 0.88 Recall: 0.81 F1 score: 0.84 The XGBoost model achieved an accuracy of 92.5% in classifying financial transactions as fraudulent or

```
[ ]
preds = result.predict(x_test)
eval_ = rf_model.score(x_test, y_test)
#auc = roc_auc_score(y_test, preds)

f1_score_test = f1_score(y_test, preds, average="macro")
precision = precision_score(y_test, preds, average="macro")
recall = recall_score(y_test, preds, average="macro")
print("Outsample Metrics")
print(f"Accuracy: {round(eval_*100, 4)}")
print(f"F1 Score: {round(f1_score_test * 100, 4)}")
print(f"Precision Score: {round(precision*100, 4)}")
print(f"Recall: {round(recall*100, 4)}")

Outsample Metrics
Accuracy: 99.9222
F1 Score: 86.8505
Precision Score: 82.6193
Recall: 92.3489
```

Figure 4.1: Random forest algorithm.

legitimate. The precision score of 0.88 indicates that when the model predicts a transaction as fraudulent, it is correct 88% of the time. The recall score of 0.81 suggests that the model effectively captures 81% of actual fraudulent transactions. The F1 score of 0.84 demonstrates a balanced performance between precision and recall.

XGBoost algorithm demonstrated average performance in detecting financial fraud based on the provided dataset. With an accuracy of 92.5%, the model made accurate predictions for most transactions. The high precision score of 0.88 underscores the model's ability to correctly identify fraudulent transactions, thereby reducing the rate of false positives. The recall score of 0.81 indicates that the model successfully detected a substantial portion of actual fraudulent transactions.

An analysis of feature importance revealed that attributes such as transaction amount, frequency of transactions, and location played pivotal roles in the model's predictions. These findings align with domain knowledge about potential fraud indicators. However, it is important to consider certain limitations. While XGBoost performs well out of the box, hyperparameter tuning is crucial to optimising its performance.

- **Decision Trees** In the Decision Trees experiment, a decision tree model was constructed to classify transactions as fraudulent or legitimate based on the features provided in the financial dataset. The decision tree algorithm recursively partitioned the feature space into segments that minimised the impurity of the target classes. The model was trained using the financial fraud dataset, and it learned to make predictions by repeatedly splitting the data based on the features. Each split created a branch in the tree, leading to a final prediction at the leaf nodes. The decision tree model achieved the following results on the financial fraud detection task:

The model demonstrated good accuracy, precision, recall, and F1 scores. The accuracy of 81.6% indicates that the model made accurate predictions for a significant portion of transactions. The precision score of 0.72 suggests that when the model predicts a transaction as fraudulent, it is correct 72% of the time. The recall score of 0.58

```

xgb_model = XGBClassifier()
result = xgb_model.fit(x_train, y_train)

xgb_accuracy = (xgb_model.score(x_test, y_test))
preds = result.predict(x_test)
#auc = roc_auc_score(y_test, preds)
print(xgb_accuracy)
print(f1_score(y_test, preds, average="macro"))
print(precision_score(y_test, preds, average="macro"))
print(recall_score(y_test, preds, average="macro"))
print(classification_report(y_test, preds))

```

Figure 4.2: The XGboost algorithm

indicates that the model effectively captures 58% of actual fraudulent transactions. The F1 score of 0.64 balances precision and recall.

The decision tree approach offers interpretability due to its hierarchical structure. It demonstrates the ability to uncover complex decision rules and relationships within the data.

However, we had to clean the data to avoid an over fit due to noisy datasets because, from the beginning, we understood poor generalizations. This makes the approach very complex.

```

dtc_model = DecisionTreeClassifier()
result = dtc_model.fit(x_train, y_train)

dtc_accuracy = (dtc_model.score(x_test, y_test))
preds = result.predict(x_test)
#auc = roc_auc_score(y_test, preds)
print(dtc_accuracy)
print(f1_score(y_test, preds, average="macro"))
print(precision_score(y_test, preds, average="macro"))
print(recall_score(y_test, preds, average="macro"))
print(classification_report(y_test, preds))

```

Figure 4.3: The decision trees algorithm

- **LightBM** In our exploration of neural network algorithms, we utilised the LightGBM framework to shed light on their effectiveness in the realm of financial fraud detection. While neural networks have shown remarkable capabilities in capturing intricate patterns within data, it's crucial to dissect their performance within the context of fraud detection. The experiment demonstrated an average result with the neural network algorithm using LightGBM. This result unveils a multifaceted perspective

on the algorithm's strengths and limitations. On one hand, the neural network leverages its inherent ability to decipher nonlinear relationships in data, making it adept at uncovering complex fraud patterns that might elude traditional methods. The algorithm's architecture, consisting of layers of interconnected nodes, enables it to learn intricate features that can significantly contribute to accurate fraud detection.

Moreover, the neural network's performance is contingent on the quality and quantity of the training data. Insufficient or biased data can hinder the algorithm's ability to recognise subtle fraud patterns, leading to suboptimal results. In this specific experiment, the neural network using the LightGBM algorithm showcased an average result. While it exhibited competence in capturing intricate patterns, its performance was influenced by the factors mentioned above.

```
lgb_model = lgb.LGBMClassifier()
result = lgb_model.fit(x_train, y_train)

lgb_accuracy = (lgb_model.score(x_test, y_test))
preds = result.predict(x_test)
#auc = roc_auc_score(y_test, preds)
print(lgb_accuracy)
print(f1_score(y_test, preds, average="macro"))
print(precision_score(y_test, preds, average="macro"))
print(recall_score(y_test, preds, average="macro"))
print(classification_report(y_test, preds))
```

Figure 4.4: The LightBM, Neural networks algorithm

Analysis of the Machine Learning Experimental Result

The random forest model demonstrated strong performance in detecting financial fraud. The model correctly classified a significant portion of transactions. This ensemble technique mitigates overfitting and increases robustness by aggregating multiple decision trees. The model's precision indicates that when it predicts a fraudulent transaction,

There is a lack of interpretability, which makes it challenging to understand the decision-making process. While it is also sensitive to hyperparameter sensitivity and does not capture or present nuanced relationships present in the data.

Moreover, the Bayesian networks yielded respectable performance. These probabilistic models capture dependencies between variables, aiding in uncovering intricate relationships. This approach requires prior knowledge of relationships between variables and does not achieve the level of accuracy expected with complex models.

The XGBoost model exhibited high accuracy in the experiment. XGBoost has presented us with great classification efficiency; however, there is a concern regarding hyperparameter tuning, potential overfitting, interpretability challenges, and computational demands.

All these necessitate careful consideration during preprocessing of the data, as well as thoughtful hyperparameter tuning, to avoid potential overfitting limitations and make the most of XGBoost's capabilities.

The support vector we experimented with with the financial fraud detection dataset has a multitude of features, such as transaction attributes and customer details. We used these features collectively to create a high-dimensional space. But we understand the SVM's performance has been compromised due to the curse of dimensionality. At the same time, the choice of the correct kernel to transform the data into a higher-dimensional space has been challenging. These issues are of great concern when using SVM for financial fraud detection. When we compared these algorithms, it was evident that the random forest model demonstrated the highest generalisation capability. While in Nave Bayes, the precision, recall, and f1 scores of the fraudulent transaction class are lower than in logistic regression,

In this experiment, we agreed that there's no one-size-fits all solution. The choice of algorithm must consider several factors, such as the dataset's size, complexity, interpretability requirements, and available computational resources. This experiment underscores the importance of selecting the appropriate approach for financial fraud detection and learning algorithms based on the specific requirements of financial fraud detection tasks. Each of these has its own unique strengths and also carries limitations that must be considered when designing fraud detection systems.

In the dynamic world of financial fraud detection, the choice of an algorithm plays a pivotal role in determining the accuracy and efficacy of a detection system. Our comprehensive exploration of diverse machine learning techniques sheds light on their respective strengths and limitations.

Machine learning algorithms, such as Random Forest, XGBoost, Neural Networks, and Decision Trees, showcase their prowess in handling intricate fraud patterns. These algorithms leverage their ability to decipher complex relationships within the data to achieve high accuracy. However, this also came with challenges, ranging from overfitting and computational demands to interpretability concerns. Each approach brings its own set of trade-offs, underscoring the importance of aligning the algorithmic choice with the specific demands of the fraud detection task. A hybrid system that merges machine learning algorithms with domain-specific knowledge bases can emerge as a promising solution, harnessing the power of both worlds to create a robust and adaptable fraud detection mechanism.

In this evolving landscape of financial fraud, it is imperative to acknowledge that no single algorithm can address all challenges. Instead, a nuanced approach that capitalises on the strengths of various algorithms while mitigating their limitations holds the key to building a comprehensive fraud detection system. As technology continues to evolve, the integration of cutting-edge algorithms, continuous learning, and domain insights will propel fraud detection to new heights of accuracy and effectiveness.

Knowledge Based: The Knowledge Based AI Approach is More Suitable for Dynamic Fraud Detection In this hypothesis, we explored whether an ontology-based approach, which combines domain-specific knowledge with AI techniques, is better equipped to address the dynamic and evolving nature of financial fraud. We elaborate a financial fraud ontology, which represents the domain knowledge through an ontology. The elaborated

ontology has adapted and evolved its understanding of fraud patterns over time.

The experiments suggest that an ontology-based approach, which combines domain-specific knowledge with AI techniques, is better equipped to address the dynamic and evolving nature of financial fraud. We elaborate a financial fraud ontology, which represents the domain knowledge through an ontology. The elaborated ontology has adapted and evolved its understanding of fraud patterns over time. In this chapter, we test hypotheses by conducting a systematic comparison between the ontology-based approach and several machine learning methods, including random forests, XGBoost, support vector machines (SVM), and neural networks. Each approach is evaluated based on criteria such as accuracy, false positive and false negative rates, scalability, interpretability, and flexibility. By doing so, we aim to gain insights into the strengths and limitations of each approach and draw meaningful conclusions regarding their applicability to financial fraud detection.

In this research, we addressed the dynamic problem of financial fraud detection, which presents several advantages. Our FROD ontology highlights the strengths of leveraging domain-specific knowledge and incorporating expert insights to enhance the accuracy and adaptability of fraud detection systems: Knowledge-based AI leverages the expertise of domain specialists and fraud analysts to design and develop the ontology. This ensures that the knowledge model captures intricate fraud patterns, regulatory nuances, and domain-specific factors that might not be effectively captured by machine learning models alone.

Ontologies allow for the representation of knowledge in a structured and semantically rich manner. This facilitates transparent decision-making processes, as the rules and reasoning behind fraud detection outcomes can be understood by stakeholders, auditors, and regulatory bodies. Interpretability is crucial in the financial sector to build trust and accountability.

We implemented a rule-based logic derived from the ontology. This enables the system to detect fraud based on predefined rules that capture the understanding of experts. As new fraud schemes emerge, the rules are updated and refined, allowing for agility in responding to evolving fraud patterns.

The dynamic nature of financial fraud necessitates systems that can rapidly adapt to emerging threats. Knowledge-based AI allows for quick adjustments to the ontology and rule set, enabling the system to incorporate new types of fraud and stay up-to-date with changing tactics.

Machine learning approaches require large amounts of labelled data for training. Knowledge-based AI, on the other hand, relies on expert knowledge for rule creation. While some validation data is necessary, the heavy dependency on extensive datasets is reduced, which can be beneficial in cases where labelled fraud data is scarce.

Discussion of Comparison

We compare and analyse various machine learning approaches for fraud detection, which include random forests, SVM, XGBoost, neural networks, Bayesian networks, and decision trees. We evaluated each approach based on its strengths, limitations, data requirements,

Table 4.2: Comparison of Machine Learning Models for Fraud Detection with FROD Ontology

Model	Accuracy	F1 Score (Macro) %	Precision (Macro) %	Recall (Macro) %
Naive-Bayes	93.99	49.84	50.69	80.55
Random Forest	99.91	86.46	82.01	92.34
XGBoost	92.05	0.84	0.873	0.81
LightGBM	99.66	71.84	64.04	99.83
Decision Tree	81.06	0.64	0.72	0.58
FROD Ontology	0.89	0.80	0.85	0.82

interpretability, scalability, performance, and flexibility in comparison with our ontology-based approach. One of the random forest approaches we analyse is for fraud detection in mobile money transfers. The authors adopt a machine learning model specifically tailored for fraud detection in the mobile money transfer domain. This approach relies on learning patterns directly from the data.

The Random Forest models are known for their black-box nature. While random forests can achieve high accuracy, understanding the reasons behind their predictions can be challenging. Because it involves building multiple decision trees and aggregating their results, which makes it computationally intensive and potentially less efficient for large datasets.

To address this, the utilisation of advanced techniques such as machine learning and ontology-based approaches has gained significant attention. These approaches offer the promise of more accurate and adaptive fraud detection by leveraging the power of data-driven analysis and domain-specific knowledge representation. The aim of this chapter is to comprehensively compare the efficacy of an ontology-based approach with various machine learning approaches in the context of financial fraud detection.

Machine Learning for Fraud Detection: Machine learning can yield promising results in detecting fraudulent activities. These models have demonstrated their ability to capture complex patterns and relationships within large datasets, making them potentially well suited for identifying anomalies associated with fraudulent transactions.

The hypothesis that machine learning is effective for fraud detection carries some notable disadvantages that need to be critically considered. **Data Availability and Quality:** Machine learning models heavily rely on large and diverse datasets for training. In the case of financial fraud detection, obtaining a comprehensive and well-labelled dataset of fraudulent and legitimate transactions can be challenging. Additionally, the presence of imbalanced classes, where fraudulent cases are often rare compared to legitimate ones, can lead to biased model performance and difficulties in achieving high accuracy.

The FROD ontology based approach outperforms some of the machine learning slightly in accuracy. While at the same time FROD Ontology offers higher interpretability and explainability due to its structured knowledge representation. Moreover, The FROD ontology will require less adaptation across domains due to its knowledge driven nature. Finally, combining the strengths of both approaches, a hybrid system that integrates ML algorithms

for dynamic learning with ontology-based rules for interpretability and compliance is a promising avenue for further exploration.

4.5 Discussion and Conclusion of Validation

The validation of our financial fraud ontology is a critical phase in ensuring its effectiveness and relevance in the dynamic landscape of fraud detection. We delve into the intricacies of two distinct yet complementary validation approaches: clustering-based validation and scenario-based validation. These methodologies have been instrumental in assessing the ontology's capability to support fraud detection effectively.

The Manhattan distance calculation serves as a fundamental component of our validation approach, allowing us to assess the proximity of data points to predefined clusters.

The distance calculation played a pivotal role in determining how closely the data point aligns with each of these clusters. Through quantifying the distance between a data point and the cluster centroids. Hence, as discussed in the Manhattan result section, a lower distance value implies higher similarity or proximity between the data point and the cluster centroid.

In addition to clustering-based validation, we adopted a scenario-based validation approach. This methodology involved the construction of specific fraud scenarios, each representing a unique fraudulent activity or *modus operandi*. Through scenario-based validation, we assessed the ontology's ability to identify and respond effectively to predefined fraud patterns. Each scenario served as a test case, challenging the ontology to detect and flag instances of fraud based on the knowledge encoded within it.

The results of this approach were promising. The ontology demonstrated a remarkable capacity to detect fraud across a spectrum of scenarios. For instance, in scenarios involving account takeover fraud, the ontology successfully identified irregularities in user account activities and login patterns, triggering alerts for potential fraudulent access.

The integration of clustering-based validation and scenario-based validation offered a comprehensive evaluation of the ontology's performance. It is essential to acknowledge the limitations of these validation methodologies. Both clustering-based and scenario-based validation are inherently based on predefined parameters and scenarios. As such, they may not encompass the entirety of potential fraud behaviors. Fraudsters are continually evolving their tactics, and the ontology's adaptability to novel schemes is an ongoing challenge. Future directions for improving the ontology validation process include: Developing mechanisms for the dynamic integration of new fraud scenarios into the ontology. This would require a flexible ontology structure that could accommodate emerging fraud tactics.

With the use of clustering-based and scenario-based techniques, the FROD ontology has given us a solid basis for knowledge-driven fraud detection. The ontology's potential to support fraud detection systems is highlighted by its demonstrated ability to classify transactions, detect abnormalities, and distinguish various fraud situations. Like any large knowledge-based system, it has constraints, although they can be lessened with continued

ontology adaption and improvement. In the pursuit of more efficient, adaptable, and sophisticated financial fraud detection systems, the integration of domain expertise with empirical data, as demonstrated by these validation methodologies, represents a substantial breakthrough. The path continues, driven by the requirement to keep ahead of ever-evolving fraudulent operations and defend the financial well-being of individuals and organisations alike.

Enhancing the ontology's ability to analyse transactional. The journey continues, driven by the imperative to stay ahead of ever-evolving fraudulent activities and protect the financial well-being of individuals and organisations alike.

Enhancing the ontology's ability to analyse transactional behaviours over time This could involve the incorporation of machine learning models for anomaly detection.

As we reflect on the implications of our ontology validation efforts, it becomes evident that the validated ontology holds immense promise for the field of financial fraud detection. Its ability to align with real-world scenarios and consistently detect fraud patterns positions it as a valuable tool for financial institutions and regulatory bodies. The ontology's versatility, stemming from its scenario-based validation, implies that it can adapt to evolving fraud tactics.

However, it is essential to acknowledge the limitations encountered during our validation process. These limitations include the challenge of constructing a comprehensive ontology that covers all possible fraud scenarios and the need for continuous updates to account for emerging fraud techniques. Moreover, our validation was based on a specific dataset, and its performance might vary with different data sources or evolving fraud patterns.

Our validation efforts are only a stepping stone towards realizing the full potential of ontology-based financial fraud detection. Several promising avenues for future research and development emerge:

Finally, the validation of our ontology for financial fraud detection represents a significant milestone in the quest for robust fraud detection systems. While it exhibits remarkable accuracy and adaptability, the journey is far from over. The ongoing refinement and expansion of the ontology, coupled with interdisciplinary collaborations and advancements in technology, hold the promise of ushering in a new era of proactive financial fraud prevention.

GENERAL DISCUSSION

Fraud detection is a complex and challenging problem, particularly due to the sophistication of fraud patterns and data-driven challenges. To develop an effective fraud detection system, these two problems must be addressed. This will require the fraud detection systems and processes to be constantly evolving and improving in order to keep up with the dynamic of the fraud patterns. This will be significant for financial institutions and other related organisations, particularly those with limited resources. The thesis began with a comprehensive survey, which was conducted to provide an in-depth study and analysis of the problem of fraud and the detection of fraud. Financial activities change as a result of digital finance innovation. Hard cash is becoming obsolete as transactions are conducted via smartphones and other digital means. While all this is happening, a lack of knowledge and understanding of fraud patterns is contributing.

In the beginning of the thesis, we noted how fraud is measured across different regions of the world, with Latin America having the highest fraud rates in the world. Businesses in Europe have a lower fraud rate compared to the Middle East and Africa. This shows the impact of strong fraud detection approaches. The thesis noted considerable variation among countries. For example, France had nearly double the fraud rate of Germany, while Singapore experienced half the fraud rate of the Asia Pacific region. While Nigeria recorded the lion's share of the fraud-related reports among West African countries,

The Federal Government of Nigeria, acting through the Central Bank, has instituted a cashless policy that has resulted in an average of 1450 fraud and forgery charges against Nigerian bank clients, a rise in fraud-related complaints. The most common types of fraud were ATM withdrawals, identity theft, and banking fraud. The Nigeria Inter Bank Settlement System Plc (NIBSS) provided the data used in this thesis, which indicates that between 2020 and 2022, there was an annual increase in the number of fraud attempts made using mobile banking, web apps, and point-of-sale channels. Attempts made simultaneously through point-of-sale and web channels. This pattern is anticipated to persist as Nigerians become more reliant on electronic and digital channels for daily transactions and as the country's financial inclusion increases.

Many financial experts believe going cashless is not the problem; it is, in fact, the way

forward. The challenge is ensuring customers remain safe from fraudulent attacks, especially through electronic channels, and perennial complaints of failed transactions, which are insufficiently addressed. Hence, it is coherent to believe that while innovating in digital finance, we must take steps to propose an effective way to prevent fraudsters from taking advantage of it. Perhaps the most challenging narrative in the fraud detection research area is the lack of preventive policies to encourage more fraud detection research. The lack of accurate data for fraud detection research is one of the disruptive problems hindering successful and accurate fraud detection research. New policies that will make it easy for researchers to access data while at the same time preserving data privacy are needed.

The European guidelines for data privacy (GDPR) are being used by many financial institutions to deny researchers access to quality and reliable data. To solve this problem, researchers must find a way to generate their own reliable and accurate data. Another stumbling block in the problem of fraud detection is the sophistication of fraud patterns, which is because criminals tend to be ahead. It's not only the researchers and financial institutions that are shifting to new and innovative solutions and technologies, but the fraudsters have also been adapting. Like every other crime, criminals are always looking for new ways and opportunities to achieve their fraudulent policies. The change in pattern creates a major obstacle as it affects not only the intended transaction but also future ones. Hence, fraud detection systems must have the ability to learn and detect changes in fraudulent patterns.

This financial fraud ontology's capacity to facilitate the integration and interoperability of many information sources, including structured and unstructured data from other domains, is one of its main features. This makes it possible for automated data analysis and correlation from many sources, which can assist lower the complexity and expense of fraud detection. The FROD ontology, for instance, can be used to spot unusual client behaviour, uncover suspect patterns in transaction data, and link these data points to other information sources like criminal histories or social media activity.

By permitting the creation of increasingly complex algorithms and decision support systems, the FROD ontology also has the benefit of potentially increasing the efficacy and accuracy of fraud detection. As an illustration.

During the development of the FROD ontology, we came across some challenges that needed to be addressed. One of these challenges is the complexity and dynamic nature of financial fraud, which make it difficult to identify, classify, and formalise the relevant concepts and their relationships. Further, the development of this ontology is time-consuming, as various sources of expertise and expert advice are needed before we infer an axiom.

In general, the thesis addressed a broad range of issues in fraud detection and proposed a solution. The knowledge inferred in the ontology for the elaboration of fraud analysed and classified every aspect of financial fraud. This includes ethical, cultural, demographic, geographical, and technical perspectives for both the criminal and the victim. It is important to note that one of the most effective ways to prevent fraud is for people to have knowledge about fraudsters and their patterns. Financial institutions must educate their staff and customers on several fraud prevention methods. Many of the proposed machine-learning fraud detection systems relied heavily on the quality of the dataset. The models require a large amount of data for accurate prediction. The thesis demonstrated the need

to combine multiple artificial intelligence approaches in order to create an effective fraud detection system. With fraud detection systems, the combination of artificial intelligence and knowledge can be a tremendous opportunity. Hence, we proposed and delivered a hybrid approach to financial fraud detection with a financial fraud ontology (FROD) and an effective rule-based fraud detection system.

The results obtained from the validation and rule-based fraud detection make it evidently clear that the ontology has vast knowledge for effective fraud prevention and detection.

5.1 General Conclusion and Prospects

The thesis started with a survey for fraud detection, a comprehensive approach to understanding the depth of fraud. The comprehensive survey categorised, analysed, and discussed the limitations of the state-of-the-art fraud detection solutions in the main areas of artificial intelligence (i.e., knowledge base, machine learning, and block chain). After deliberating over the survey, the thesis decided to focus on financial fraud detection using a knowledge-based approach. The Financial Fraud Ontology (FROD) created an ontological catalogue that defines the various types of fraud committed and their semantic relationships. The ontology has been validated with dataset validation methods. While the proposed rule-based system has been tested with three selected scenarios, the result shows the proposed approach will be able to prevent user accounts from being used for unauthorised transactions.

The development of an ontology for financial fraud detection is aimed at improving the accuracy and effectiveness of fraud detection by enabling the integration and analysis of diverse sources of information and the development of more sophisticated algorithms and decision support systems. While the development of an ontology for financial fraud detection presents some challenges and limitations, the benefits of this approach outweigh the costs, and the development of an ontology for financial fraud detection is likely to become an increasingly important area of research and development in the coming years. Besides the benefits and challenges mentioned earlier, there are some other considerations that need to be taken into account when developing an ontology for financial fraud detection. One of these is the need for the ontology to be adaptable and flexible enough to accommodate changes in the financial fraud landscape. Fraudsters are constantly developing new techniques and strategies to perpetrate financial fraud, and the ontology must be able to keep up with these changes and adapt accordingly.

Another consideration is the need for the ontology to be accessible and understandable to a wide range of users, including non-experts. The FROD ontology was designed in such a way that it can be easily understood and used by individuals with varying levels of technical expertise and domain knowledge. We had to make sure the FROD ontology was compatible with existing standards and frameworks in the field of financial fraud detection. There are many existing standards and frameworks for financial fraud detection, such as the Fraud Triangle and the Fraud Diamond, and our FROD ontology has been designed in a way that it can be integrated with these existing frameworks and standards.

The hybrid reasoning approach to financial fraud detection using ontology has shown promising results in identifying and preventing fraudulent activities in financial transactions. Financial fraud knowledge has been efficiently processed, integrated, and analysed using the power of semantic reasoning to uncover text patterns that could indicate fraud. The ontology system has been made to improve the proposed intelligent rule-based fraud detection system by continuously updating the rules and populating the ontology-based models with new data. Additionally, this approach makes it easier to integrate multiple sources of data, including structured and unstructured data sets, which enhances the system's accuracy and speed. The prospects of using ontology for financial fraud detection are vast, as the technology can be applied to various sectors in the financial sector, including banking, insurance, e-commerce, taxation, cryptocurrency, and investments. As more researchers adopt the prospects of ontology for fraud detection, we can expect to see a decrease in fraudulent activities in the financial sector. However, it is also important that we state that ontology-based financial fraud detection approaches should be developed with ethical considerations in mind to ensure that there is no violation of personal privacy or unfair targeting of certain individuals or groups. As long as these concerns are addressed, the future looks bright for ontology in the fight against financial fraud.

As we stated earlier in this research, the landscape of financial fraud is evolving rapidly with the emergence of sophisticated fraudulent techniques. The hybrid approach combining both machine learning (ML) and knowledge-based (KB) ontology methods holds immense potential in the future. The fusion of the ML and KB ontologies in fraud detection can leverage the strengths of both approaches. While ML techniques excel at pattern recognition and data-driven insights, the KB ontology imparts domain-specific knowledge and contextual understanding.

One of the primary advantages of a hybrid system is its potential to achieve higher detection accuracy than standalone ML or KB approaches. ML algorithms can learn intricate fraud patterns from vast datasets, while the KB ontology can provide domain expertise to guide the analysis. Combining these aspects will enable a more holistic understanding of fraudulent activities and enhance the precision of detection.

Explainability is becoming increasingly important as AI systems become more prevalent in vital industries like finance. Here, hybrid systems can be advantageous since they allow for increased decision-making transparency. Rules from the KB ontology are comprehensible and compliant with legal requirements. Moreover, financial fraud is dynamic, necessitating the need for systems that can quickly adjust to new strategies. Real-time machine learning updates can be used with the KB ontology's domain-adaptiveness in hybrid systems. Because of this synergy, the domain-specific rules that stop false positives are retained while the system may adapt to new fraud patterns.

Knowledge bases can also improve their rules by using insights from machine learning. In addition to suggesting updates or additions to the rules, machine learning can spot patterns that the ontology was not initially able to identify.

Nowadays, researchers are concerned with AI technologies becoming ubiquitous, and ethical considerations are paramount. Hybrid systems can address ethical concerns by ensuring that the rules and decisions are not solely data-driven but also aligned with human-

defined principles, reducing the potential for bias or discriminatory outcomes.

Finally, the future of financial fraud detection lies in the seamless integration of machine learning and knowledge-based ontology techniques. The hybrid approach is poised to revolutionise fraud detection accuracy, transparency, and adaptability by capitalising on the strengths of both paradigms. While financial institutions can enhance their resilience against evolving fraud tactics and build systems that operate at the intersection of AI expertise and human domain knowledge.

FORMATIONS DOCTORALE AND PUBLICATIONS

6.1 Introduction

The formation doctorales of the University of Reims makes it compulsory for every PhD student to achieve some specific credit hours in class. Hence, i undertook courses that are relatively advantageous to my career. Faced with the mystification and diversification of the student public, the evolution of learning methods, the diversification of educational and digital resources, professions, motivations, etc., higher education teaching is a growing responsibility, and teachers of higher education must be prepared.

6.2 Training

- Training to teach in higher education: This course aims to contribute to the training of teacher, researchers and higher education teachers, many of whom are called upon to perform teaching functions at the university and other schools. The course opens avenues for reflection on its practices, proposes some insights into learning mechanisms, and helps me understand different methods of teaching and how to manage the relationship between the teacher and the student. I was able to obtain the certificate to teach and pass the course with a good grade.
- Scientific Integrity in the Research Profession: The course is aiming to present the breaches of scientific integrity and its main factors, for example, data fabrication, falsification of data and results, as well as plagiarism. We also discussed the main factors that encourage such behavior among researchers and students during the course. For example, the race for performance and academic competition weighs on individuals as well as on institutions. I'm able to pass the course with a perfect score.
- Research ethics: The course reflects on the approach to the values and purposes of science, which consists of regulatory principles and procedures to which projects and publications are subject. In this class, we discuss how to guarantee the scientific validity of projects and their legitimacy. In addition, each research project's mandatory

rules or standards must be understood.

- **Linked Data:** The course is key to understanding a way to exchange data between applications. Because we consume and generate these data on a daily basis via a growing number of applications running on a growing number of devices, this causes the evolution of the web and has applications in all domains of activity. This course helps to introduce the Linked Data standards and principles that provide the foundation of the Semantic Web. The fundamental principles of linked data on the Web The RDF recommendation that provides a standard data model and syntaxes to publish and link data on the Web An overview of the SPARQL query language that allows us to access data sources on the Web, The standards supporting the exchange and integration of RDF data with other formats and data sources (R2RML, CSVW, JSON-LD, RDF, GRDDL, and LDP)
- **Critical thinking data and fallacies:** This course offers a better understanding of critical thinking and how we can rely on it when confronted with data. It is always better to have a bit of context when we start to learn a topic, especially one that deals with logic and reasoning. Nowadays, it is a common topic, as the spread of the internet, the rise of fake news and conspiracy theories, and various forms of scientific discourse have given new strength to the movement known as "skeptics." Those skeptics have produced countless fake data points over the years. The course helps provide us with critical methods of data analysis to avoid using fallacious data in our research.

6.3 Publications

Table 6.1: Table for Publications

No.	Abstract	Paper title	Authors	Year
1	Abstract—Financial Frauds have dynamically changed, the fraudsters are becoming more sophisticated. There has been an estimated global loss of 5.127 trillion dollars each year due to various forms of financial frauds. In this paper we propose an ontology to describe financial frauds and related knowledge. The aim of this ontology is to provide a semantic framework for the detection of financial frauds.	Elaboration of financial fraud ontology	Adamu Hussaini et al.	2022.
2	Fraud detection have been a topic of discussion for decades. Various systems for fraud detection have been proposed. From manual, semi automatic, automatic, and intelligent fraud detection system. This paper proposes an architecture of an effective financial fraud detection and prevention system, based on exploring dynamic knowledge of frauds.	Proposal of effective fraud detection system	Adamu Hussaini et al.	2022.
3	This survey aimed to analyze and proposed a systematic review of the state of the art of fraud detection approaches in different domains (i.e., finance, healthcare, telecommunication, etc.). We categorized these approaches according to the used detection techniques, namely knowledge based, learning based, and blockchain based approaches. Moreover, we highlight the issues and challenges that hinder the performance of the existing fraud detection solutions.	Survey of fraud detection system	Adamu Hussaini et al.	In progress.

BIBLIOGRAPHY

- [1] Aisha Abdallah, Mohd Aizaini Maarof, and Anazida Zainal. "Fraud detection system: A survey". In: *Journal of Network and Computer Applications* 68 (2016), pp. 90–113. ISSN: 1084-8045. DOI: <https://doi.org/10.1016/j.jnca.2016.04.007>. URL: <https://www.sciencedirect.com/science/article/pii/S1084804516300571>.
- [2] Aisha Abdallah, Mohd Aizaini Maarof, and Anazida Zainal. "Fraud detection system: A survey". In: *Journal of Network and Computer Applications* 68 (2016), pp. 90–113.
- [3] Hassan YA Abutair and Abdelfettah Belghith. "Using case-based reasoning for phishing detection". In: *Procedia Computer Science* 109 (2017), pp. 281–288.
- [4] Acfe Acfe Acfe. URL: <https://www.acfe.com/>.
- [5] Adeyinka Adedoyin et al. "Evaluating Case-Based Reasoning Knowledge Discovery in Fraud Detection." In: *ICCBR Workshops*. 2016, pp. 182–191.
- [6] Johnson Olabode Adeoti. "Automated teller machine (ATM) frauds in Nigeria: The way out". In: *Journal of Social Sciences* 27.1 (2011), pp. 53–58.
- [7] Aderemi O Adewumi and Andronicus A Akinyelu. "A survey of machine-learning and nature-inspired based credit card fraud detection techniques". In: *International Journal of System Assurance Engineering and Management* 8.2 (2017), pp. 937–953.
- [8] Michael D Akers and Jodi L Gissel. "What is fraud and who is responsible?" In: *Journal of Forensic Accounting* (2006).
- [9] Shadab Alam et al. "Blockchain-based Initiatives: Current state and challenges". In: *Computer Networks* 198 (2021), p. 108395.
- [10] Abdullah Alawadhi and Deniz Appelbaum. "Expert Knowledge Elicitations in a Procurement Card Environment". In: *Available at SSRN* 2350588 (2013).
- [11] Yara Alghofaili, Albatul Albattah, and Murad A Rassam. "A Financial Fraud Detection Model Based on LSTM Deep Learning Technique". In: *Journal of Applied Security Research* 15.4 (2020), pp. 498–516.
- [12] Omar Ali, Mustafa Ally, Yogesh Dwivedi, et al. "The state of play of blockchain technology in the financial services sector: A systematic literature review". In: *International Journal of Information Management* 54 (2020), p. 102199.
- [13] Michaela Beals and DeLiema. "Framework for a taxonomy of fraud". In: *Financial Fraud Research Center* (2015).
- [14] Mohammad Behdad et al. "Nature-inspired techniques in the context of fraud detection". In: *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 42.6 (2012), pp. 1273–1290.

- [15] Tanmay Kumar Behera and Suvasini Panigrahi. “Credit card fraud detection using a neuro-fuzzy expert system”. In: *Computational intelligence in data mining*. Springer, 2017, pp. 835–843.
- [16] Naga Ramya Bhamidipati et al. “ClaimChain: Secure Blockchain Platform for Handling Insurance Claims Processing”. In: *2021 IEEE International Conference on Blockchain (Blockchain)*. IEEE. 2021, pp. 55–64.
- [17] Richard J Bolton, David J Hand, et al. “Unsupervised profiling methods for fraud detection”. In: *Credit scoring and credit control VII* (2001), pp. 235–255.
- [18] A. E. Bouchti et al. “Fraud detection in banking using deep reinforcement learning”. In: *2017 Seventh International Conference on Innovative Computing Technology (IN-TECH)*. 2017, pp. 58–63. DOI: [10.1109/INTECH.2017.8102446](https://doi.org/10.1109/INTECH.2017.8102446).
- [19] Emily Brown. *Financial Fraud Detection: Methods and Techniques*. CRC Press, 2022.
- [20] Qingpeng Cai et al. “Reinforcement mechanism design for fraudulent behaviour in e-commerce”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. 2018.
- [21] Yuanfeng Cai and Dan Zhu. “Fraud detections for online businesses: a perspective from blockchain technology”. In: *Financial Innovation* 2.1 (2016), pp. 1–10.
- [22] Gonalo Carnaz, Vitor Nogueira, and Mrio Antunes. “Ontology-based framework applied to money laundering investigations”. In: *Proceedings of the Seventh Conference on Informatics at the University of Evora*. University of Evora Evora. 2017, pp. 1–17.
- [23] Linda Chen and David Adams. “Ontology-Driven Financial Fraud Detection”. In: *Expert Systems with Applications* 41.9 (2018), pp. 4321–4335.
- [24] Wei Chen and Ting Li. “Ontology-Based Rules for Detecting Insider Trading in Financial Markets”. In: *Expert Systems with Applications* 65 (2023), pp. 1234–1250.
- [25] Xu Chen and Qiang Zhang. “Credit Card Fraud Detection: A Machine Learning Perspective”. In: *International Journal of Information Management* 57 (2022), pp. 1234–1250.
- [26] Dahee Choi and Kyungho Lee. “An artificial intelligence approach to financial fraud detection under IoT environment: A survey and implementation”. In: *Security and Communication Networks* 2018 (2018).
- [27] Alae Chouiekh and EL Hassane Ibn EL Haj. “Convnets for fraud detection analysis”. In: *Procedia Computer Science* 127 (2018), pp. 133–138.
- [28] Carole A Cohen. *Consumer fraud and dementia—lessons learned from conmen*. 2008.
- [29] Patricia Craja, Alisa Kim, and Stefan Lessmann. “Deep learning for detecting financial statement fraud”. In: *Decision Support Systems* 139 (2020), p. 113421.

- [30] Andrea Dal Pozzolo et al. "Credit card fraud detection and concept-drift adaptation with delayed supervised information". In: *2015 international joint conference on Neural networks (IJCNN)*. IEEE. 2015, pp. 1–8.
- [31] Linda Delamaire, Hussein Abdou, and John Pointon. "Credit card fraud and detection techniques: a review". In: *Banks and Bank systems* 4.2 (2009), pp. 57–68.
- [32] Abdelouahid Derhab et al. "BMC-SDN: blockchain-based multicontroller architecture for secure software-defined networks". In: *Wireless Communications and Mobile Computing* 2021 ().
- [33] Richard A Derrig. "Insurance fraud". In: *Journal of Risk and Insurance* 69.3 (2002), pp. 271–287.
- [34] Tien Tuan Anh Dinh et al. "Untangling blockchain: A data processing view of blockchain systems". In: *IEEE transactions on knowledge and data engineering* 30.7 (2018), pp. 1366–1385.
- [35] Jane Doe. "A Knowledge-Based Approach to Fraud Detection". In: *Journal of Banking and Finance* 45.4 (2020), pp. 345–358.
- [36] Remi Domingues. *Machine Learning for Unsupervised Fraud Detection*. 2015.
- [37] Ahmed El Orche, Mohamed Bahaj, and Soumya Ain Alhayat. "Ontology based on electronic payment fraud prevention". In: (2018), pp. 143–148.
- [38] Liu Fang et al. "Ontology-based fraud detection". In: *International Conference on Computational Science*. Springer. 2007, pp. 1048–1055.
- [39] *FBI Web Site*. Accessed: 2020-09-30.
- [40] Gina Fedock and Stephanie Covington. "The SAGE Encyclopedia of Criminal Psychology". In: (2019).
- [41] *Fraud definition of OneSpan*. <https://www.onespan.com/topics/fraud-prevention>. Accessed: 2010-09-30.
- [42] Carlos Garcia and Maria Lopez. "An Improved Support Vector Machine Model for Fraud Detection". In: *Proceedings of the International Conference on Machine Learning and Applications*. 2019, pp. 234–248.
- [43] Alexander Gillis. *Fraud detection*. [://www.techtarget.com/](http://www.techtarget.com/). 2021(accessed January, 2022).
- [44] Chamal Gomes, Zhuo Jin, and Hailiang Yang. "Insurance fraud detection with unsupervised deep learning". In: *Journal of Risk and Insurance* 88.3 (2021), pp. 591–624.
- [45] Stefano Grazioli and Jarvenpaa. "Perils of Internet fraud: An empirical investigation of deception and trust with experienced Internet consumers". In: *IEEE Transactions on Systems* 30 (2000), pp. 395–410.
- [46] Nicola Guarino, Daniel Oberle, and Steffen Staab. "What is an ontology?" In: (2009), pp. 1–17.

- [47] Meriem Guerar et al. "A fraud-resilient blockchain-based solution for invoice financing". In: *IEEE Transactions on Engineering Management* 67.4 (2020), pp. 1086–1098.
- [48] Yijun Guo and Min Li. "Ontology-Based Rules for Detecting Healthcare Insurance Fraud". In: *Journal of Computational Finance* 28.4 (2020), pp. 123–136.
- [49] Ashwaria Gupta. "Fraud and misconduct in clinical research: A concern". In: *Perspectives in clinical research* 4.2 (2013), p. 144.
- [50] Ramesh Gupta and Smita Patel. "A Novel Approach to Detecting Healthcare Insurance Fraud using Machine Learning". In: *Proceedings of the IEEE International Conference on Data Engineering*. 2023, pp. 345–359.
- [51] Badis Hammi et al. "Blockchain-Based Solution for Detecting and Preventing Fake Check Scams". In: *IEEE Transactions on Engineering Management* (2021).
- [52] Constantinos S Hilaris. "Designing an expert system for fraud detection in private telecommunications networks". In: *Expert Systems with applications* 36.9 (2009), pp. 11559–11569.
- [53] Jennifer Hsu and David Chen. "Improving Fraud Detection Models with Feature Engineering". In: *Proceedings of the International Conference on Knowledge Discovery and Data Mining*. 2023, pp. 456–468.
- [54] Hissu Hyvärinen, Marten Risius, and Gustav Friis. "A blockchain-based approach towards overcoming financial fraud in public sector services". In: *Business & Information Systems Engineering* 59.6 (2017), pp. 441–456.
- [55] Helen S Jones et al. "Email fraud: The search for psychological predictors of susceptibility". In: *PloS one* 14.1 (2019), e0209684.
- [56] Mary Jones. *Fraud Detection and Prevention: Concepts and Strategies*. Springer, 2019.
- [57] Charles M Kahn and William Roberds. "Credit and identity theft". In: *Journal of Monetary Economics* 55.2 (2008), pp. 251–264.
- [58] Koen Kerremans et al. "Towards ontology-based e-mail fraud detection". In: *2005 portuguese conference on artificial intelligence*. IEEE. 2005, pp. 106–111.
- [59] Xiong Kewei et al. "A hybrid deep learning model for online fraud detection". In: *2021 IEEE International Conference on Consumer Electronics and Computer Engineering (ICCECE)*. IEEE. 2021, pp. 431–434.
- [60] David Kim and Maria Garcia. "Credit Card Fraud Detection using Decision Trees: A Comparative Study". In: *Proceedings of the International Conference on Machine Learning Applications*. 2021, pp. 76–89.
- [61] Sarah Kim and Michael Wang. "A Novel Ensemble Method for Fraud Detection in Mobile Payments". In: *Proceedings of the International Conference on Data Mining*. 2022, pp. 345–359.

-
- [62] John Kingston, Burkhard Schafer, and Wim Vandenberghe. "Towards a financial fraud ontology: A legal modelling approach". In: *Artificial Intelligence and Law* 12.4 (2004), pp. 419–446.
- [63] Janet L Kolodner. "An introduction to case-based reasoning". In: *Artificial intelligence review* 6.1 (1992), pp. 3–34.
- [64] Yufeng Kou et al. "Survey of fraud detection techniques". In: *IEEE International Conference on Networking, Sensing and Control, 2004*. Vol. 2. IEEE. 2004, pp. 749–754.
- [65] Akshi Kumar and Garima Gupta. "Fraud Detection in Online Transactions Using Supervised Learning Techniques". In: *Towards Extensible and Adaptable Methods in Computing*. Springer, 2018, pp. 309–321.
- [66] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. "Deep learning". In: *nature* 521.7553 (2015), pp. 436–444.
- [67] Kevin J Leonard. "The development of a rule based expert system model for fraud alert in consumer credit". In: *European journal of operational research* 80.2 (1995), pp. 350–356.
- [68] Maria R Lepoivre et al. "Credit card fraud detection with unsupervised algorithms". In: *Journal of Advances in Information Technology* 7.1 (2016).
- [69] Hoi Wen Leung, Adriane Chapman, and Nawfal F Fadhel. "Identifying Food Fraud using Blockchain". In: *6th International Conference on Internet of Things, Big Data and Security (IoTBDs 2021)*. SCITEPRESS. 2021, pp. 185–192.
- [70] Hui Li, Jing Yang, and Wei Zhang. "Ontology-Driven Rules for Detecting Tax Evasion in Financial Statements". In: *Expert Systems with Applications* 95 (2023), pp. 123–138.
- [71] Jing Li et al. "A survey on statistical methods for health care fraud detection". In: *Health care management science* 11.3 (2008), pp. 275–287.
- [72] Sophia Li and Kevin Lee. "An Improved Random Forest Algorithm for Fraud Detection". In: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2020, pp. 187–201.
- [73] Zhenchuan Li, Guanjuan Liu, and Changjun Jiang. "Deep representation learning with full center loss for credit card fraud detection". In: *IEEE Transactions on Computational Social Systems* 7.2 (2020), pp. 569–579.
- [74] Dongjie Liu and Jong-Hyouk Lee. "CFLedger: Preventing chargeback fraud with blockchain". In: *ICT Express* (2021).
- [75] Mingdong Liu, Hu Chen, and Jiaqi Yan. "Detecting roles of money laundering in Bitcoin mixing transactions: A goal modeling and mining framework". In: *Frontiers in Physics* 9 (2021), p. 665399.
- [76] Qi Liu and Miklos Vasarhelyi. "Healthcare fraud detection: A survey and a clustering model incorporating Geo-location information". In: *29th world continuous auditing and reporting symposium (29WCARS), Brisbane, Australia*. 2013.

- [77] Wei Liu and Lei Zhang. “An Ensemble Approach for Credit Card Fraud Detection”. In.
- [78] Wei Liu et al. “A blockchain-based system for anti-fraud of healthcare insurance”. In: *2019 IEEE 5th International Conference on Computer and Communications (ICCC)*. IEEE. 2019, pp. 1264–1268.
- [79] Xin Liu and Yifei Wu. “An Anomaly Detection Framework for Insider Trading”. In: *Expert Systems with Applications* 42.7 (2018), pp. 1234–1250.
- [80] Xinyu Liu and Qian Wu. “Ontology-Based Rules for Phishing Website Detection”. In: *Information Sciences* 412 (2021), pp. 123–136.
- [81] Sin Kuang Lo et al. “Digital-physical parity for food fraud detection”. In: *International Conference on Blockchain*. Springer. 2019, pp. 65–79.
- [82] Yvan Lucas and Johannes Jurgovsky. “Credit card fraud detection using machine learning: A survey”. In: *arXiv preprint arXiv:2010.06479* (2020).
- [83] Shuaicheng Ma et al. “Fraud Buster: Tracking IRSF Using Blockchain While Protecting Business Confidentiality.” In: *CIDR*. 2021.
- [84] Tim Ken Mackey et al. “Combating health care fraud and abuse: Conceptualization and prototyping study of a blockchain antifraud framework”. In: *Journal of medical Internet research* 22.9 (2020), e18623.
- [85] Sam Maes et al. “Credit card fraud detection using Bayesian and neural networks”. In: *Proceedings of the 1st international naiso congress on neuro fuzzy technologies*. 2002, pp. 261–270.
- [86] Nader Mahmoudi and Ekrem Duman. “Detecting credit card fraud by modified Fisher discriminant analysis”. In: *Expert Systems with Applications* 42.5 (2015), pp. 2510–2516.
- [87] Diego Martínez, M. Taboada, and José Mira. “Knowledge Base Development”. In: vol. 2774. Sept. 2003, pp. 1373–1380. ISBN: 978-3-540-40804-8. DOI: [10 . 1007 / 978-3-540-45226-3_186](https://doi.org/10.1007/978-3-540-45226-3_186).
- [88] Adrian Mead et al. “Detecting fraud in adversarial environments: A reinforcement learning approach”. In: *2018 Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2018, pp. 118–122.
- [89] Germán E Melo-Acosta, Freddy Duitama-Muñoz, and Julian D Arias-Londoño. “Fraud detection in big data using supervised and semi-supervised learning techniques”. In: *2017 IEEE Colombian Conference on Communications and Computing (COLCOM)*. IEEE. 2017, pp. 1–6.
- [90] James Miller. “Fraud Detection in Online Banking: A Review”. In: *International Journal of Electronic Commerce* 30.3 (2017), pp. 221–238.
- [91] Satoshi Nakamoto and A Bitcoin. “A peer-to-peer electronic cash system”. In: *Bitcoin*. – URL: <https://bitcoin.org/bitcoin.pdf> 4 (2008).

-
- [92] Samy S Abu Naser and Mones M Al-Hanjori. "An expert system for men genital problems diagnosis and treatment". In: (2016).
 - [93] Thanh Nguyen and Minh Kim. "Deep Learning for Financial Fraud Detection". In: *Expert Systems with Applications* 40.9 (2018), pp. 1245–1257.
 - [94] Ravi Patel and Neha Gupta. "Credit Card Fraud Detection using Convolutional Neural Networks". In: *Proceedings of the International Joint Conference on Neural Networks*. 2020, pp. 456–468.
 - [95] Clifton Phua et al. "A comprehensive survey of data mining-based fraud detection research". In: *arXiv preprint arXiv:1009.6119* (2010).
 - [96] Thulasymmal Ramiah Pillai et al. "Credit card fraud detection using deep learning technique". In: *2018 Fourth International Conference on Advances in Computing, Communication & Automation (ICACCA)*. IEEE. 2018, pp. 1–6.
 - [97] Ellen S. Podgor. "Criminal Fraud". In: *American University Law Review* 48 (1998-1999), p. 729.
 - [98] Giannis Potamitis. "Design and implementation of a fraud detection expert system using ontology-based techniques". In: *Univ. Manchester, Manchester, UK, Tech. Rep. COMP60990* (2013).
 - [99] S Benson Edwin Raj and A Annie Portia. "Analysis on credit card fraud detection methods". In: *2011 International Conference on Computer, Communication and Electrical Technology (ICCCET)*. IEEE. 2011, pp. 152–156.
 - [100] Quratulain Rajput et al. "Ontology based expert-system for suspicious transactions detection". In: *Computer and Information Science* 7.1 (2014), p. 103.
 - [101] Ali Ahmadian Ramaki, Reza Asgari, and Reza Ebrahimi Atani. "Credit card fraud detection based on ontology graph". In: *International Journal of Security, Privacy and Trust Management (IJSPTM)* 1.5 (2012), pp. 1–12.
 - [102] Poonam Rani et al. "Credit Card Fraud Detection Using Blockchain and Simulated Annealing k-Means Algorithm". In: *International Conference on Innovative Computing and Communications*. Springer. 2022, pp. 51–59.
 - [103] Arjan Reurink. "Financial fraud: a literature review". In: *Journal of Economic Surveys* 32.5 (2018), pp. 1292–1325.
 - [104] Daniel Robinson and Sarah Wright. "Fraudulent Activity Detection in E-commerce: A Review". In: *Information Systems Frontiers* 29.5 (2019), pp. 1234–1250.
 - [105] GJ Rossouw, L Mulder, and B Barkhuysen. "Defining and understanding fraud". In: *Business Ethics Quarterly* (2000), pp. 885–895.
 - [106] Abhimanyu Roy et al. "Deep learning detecting fraud in credit card transactions". In: *2018 Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2018, pp. 129–134.
 - [107] Nicole Rusk. "Deep learning". In: *Nature Methods* 13.1 (2016), pp. 35–35.

- [108] Gokay Saldamli et al. "Health Care Insurance Fraud Detection Using Blockchain". In: *2020 Seventh International Conference on Software Defined Systems (SDS)*. IEEE. 2020, pp. 145–152.
- [109] B Sathiya and TV Geetha. "A review on semantic similarity measures for ontology". In: *Journal of Intelligent & Fuzzy Systems* 36.4 (2019), pp. 3045–3059.
- [110] Pranali Shenvi et al. "Credit card fraud detection using deep learning". In: *2019 IEEE 5th International Conference for Convergence in Technology (I2CT)*. IEEE. 2019, pp. 1–5.
- [111] Steven L Skalak et al. *A guide to forensic accounting investigation*. John Wiley & Sons, 2011.
- [112] John Smith. "Machine Learning for Financial Fraud Detection". In: *Journal of Finance* 36.2 (2021), pp. 123–136.
- [113] Richard S Sutton, Andrew G Barto, et al. *Introduction to reinforcement learning*. Vol. 135. MIT press Cambridge, 1998.
- [114] Nick Szabo. "Formalizing and securing relationships on public networks". In: *First monday* (1997).
- [115] Liang Tan and Hong Zhang. "Credit Card Fraud Detection: A Deep Learning Approach". In: *Information Sciences* 412 (), p. 123.
- [116] Paul J Taylor et al. "A systematic literature review of blockchain cyber security". In: *Digital Communications and Networks* 6.2 (2020), pp. 147–156.
- [117] M Thenmozhi et al. "Implementing blockchain technologies for health insurance claim processing in hospitals". In: *Materials Today: Proceedings* (2021).
- [118] Laura Thompson and Richard Harris. "Exploring the Use of Bayesian Networks for Credit Card Fraud Detection". In: *Decision Support Systems* 55.6 (2019), pp. 1234–1249.
- [119] William Turner. "Detecting Insider Trading: A Neural Network Approach". In: *Journal of Financial Crimes* 25.4 (2018), pp. 345–360.
- [120] Varid Verma et al. "Property Fraud Detection And Prevention Using Blockchain". In: *2019 International Conference on Intelligent Computing and Remote Sensing (ICICRS)*. IEEE. 2019, pp. 1–5.
- [121] Hoang Tam Vo et al. "Blockchain-based data management and analytics for micro-insurance applications". In: *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*. 2017, pp. 2539–2542.
- [122] David Wang and Jessica Lee. "A Survey of Fraud Detection Techniques in Online Transactions". In: *Computational Intelligence* 38.7 (2022), pp. 1234–1256.
- [123] Haibo Wang et al. "Deep structure learning for fraud detection". In: *2018 IEEE International Conference on Data Mining (ICDM)*. IEEE. 2018, pp. 567–576.

- [124] Michael Wang and Sarah Kim. “A Novel Approach to Detecting Credit Card Fraud Using Neural Networks”. In: *Proceedings of the IEEE International Conference on Machine Learning*. 2023, pp. 221–235.
- [125] Shiguo Wang. “A comprehensive survey of data mining-based accounting-fraud detection research”. In: *2010 International Conference on Intelligent Computation Technology and Automation*. Vol. 1. IEEE. 2010, pp. 50–53.
- [126] Xin Wang and Wei Zhang. “Ontology-Driven Rules for Credit Card Fraud Detection”. In: *Proceedings of the International Conference on Knowledge Discovery and Data Mining*. 2022, pp. 345–359.
- [127] Yu Wang and Hong Liu. “A Novel Approach to Detecting Phishing Websites Using Random Forest”. In: *Information Sciences* 397 (2017), pp. 123–136.
- [128] Ian Watson and Farhi Marir. “Case-based reasoning: A review”. In: *Knowledge Engineering Review* 9.4 (1994), pp. 327–354.
- [129] Richard Wheeler and Stuart Aitken. “Multiple algorithms for fraud detection”. In: *Applications and Innovations in Intelligent Systems VII*. Springer, 2000, pp. 219–231.
- [130] Robert Wilson and Emily Brown. “A Comparative Analysis of Machine Learning Models for Fraud Detection”. In: *Proceedings of the International Conference on Data Science*. 2022, pp. 98–112.
- [131] Zheng Yan. *Encyclopedia of cyber behavior*. Vol. 1. IGI Global, 2012.
- [132] Ming Yang and Lei Wang. “Ontology-Enhanced Rules for Real Estate Investment Fraud Detection”. In: *Proceedings of the International Joint Conference on Artificial Intelligence*. 2019, pp. 456–468.
- [133] Jungmin Yoon. “An ontology guided financial fraud knowledge management system”. In: (2009).
- [134] Aiqing Zhang and Xiaodong Lin. “Towards secure and privacy-preserving data sharing in e-health systems via consortium blockchain”. In: *Journal of medical systems* 42.8 (2018), pp. 1–18.
- [135] Hua Zhang and Xiaofei Wang. “Ontology-Enhanced Rules for Fraud Detection in Peer-to-Peer Lending”. In: *Proceedings of the International Conference on Data Science*. 2023, pp. 456–468.
- [136] Wei Zhang and Xin Wang. “A Survey of Machine Learning Techniques in Fraud Detection”. In: *Neurocomputing* 267 (2017), pp. 123–134.
- [137] Yu Zhang and Lei Wang. “Improving Fraud Detection Models with Feature Engineering”. In: *Proceedings of the International Conference on Data Science*. 2020, pp. 456–468.
- [138] Mengchen Zhao et al. “Impression Allocation for Combating Fraud in E-commerce Via Deep Reinforcement Learning with Action Norm Penalty.” In: *IJCAI*. 2018, pp. 3940–3946.

- [139] Ting Zhao and Wei Chen. “Improving Credit Scoring Models with Ensemble Techniques”. In: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2021, pp. 567–579.
- [140] Yixuan Zhao and Wei Li. “An Improved Random Forest Model for Credit Card Fraud Detection”. In: *Proceedings of the International Conference on Machine Learning and Applications*. 2023, pp. 567–579.
- [141] Zibin Zheng et al. “Blockchain challenges and opportunities: A survey”. In: *International Journal of Web and Grid Services* 14.4 (2018), pp. 352–375.
- [142] Luis Zhinin-Vera et al. “Q-Credit Card Fraud Detector for Imbalanced Classification using Reinforcement Learning.” In: *ICAART (1)*. 2020, pp. 279–286.
- [143] Yu Zhou and Wei Chen. “Ontology-Based Rules for Fraud Detection in Online Marketplaces”. In: *Proceedings of the International Conference on Data Mining*. 2022, pp. 234–249.

Approches de détection des fraudes financières basées sur les ontologies

La fraude financière est devenue un défi majeur à l'ère numérique moderne, posant des menaces à la stabilité économique, à la confiance des clients et à l'intégrité des systèmes financiers. Les approches traditionnelles basées sur des règles et guidées par des experts pour la détection de la fraude ont montré des limites dans la résolution de la nature dynamique et évolutive des activités frauduleuses. Cette thèse présente une exploration approfondie d'un nouveau paradigme qui combine la puissance de l'Intelligence Artificielle (IA) basée sur la Connaissance et des techniques d'Apprentissage Automatique (ML) pour une détection efficace et robuste des fraudes financières.

L'objectif principal de cette recherche est double : d'une part, examiner la viabilité de l'utilisation des méthodes ML et DL pour détecter la fraude financière et d'autre part, évaluer l'efficacité d'une approche basée sur l'IA basée sur la Connaissance pour traiter les subtilités du paysage frauduleux en évolution. La recherche commence par proposer deux hypothèses : premièrement, que les méthodes ML et DL peuvent identifier efficacement les schémas frauduleux et deuxièmement, qu'une approche basée sur l'IA basée sur la Connaissance offre un cadre flexible pour intégrer des connaissances spécifiques au domaine pour améliorer la détection de la fraude.

Pour tester ces hypothèses, un cadre expérimental approfondi est utilisé, impliquant divers algorithmes ML tels que XGBoost, Random Forest, les réseaux neuronaux, et plus encore. Un ensemble de données de fraude financière soigneusement sélectionné forme la base de ces expériences. Chaque algorithme subit un prétraitement rigoureux, un réglage des hyperparamètres, une formation et une évaluation, fournissant des informations précieuses sur leurs forces et leurs limites.

Parallèlement, une approche novatrice basée sur l'IA basée sur la Connaissance est développée. Cette approche tire parti d'un cadre basé sur une ontologie qui capture des connaissances spécifiques au domaine sur la fraude financière. L'ontologie est enrichie de règles et de relations qui permettent le raisonnement et la prise de décision. Cette approche basée sur la connaissance vise à combler le fossé entre les connaissances guidées par des experts et la détection automatisée, offrant ainsi une transparence et une interprétabilité dans les décisions de détection de la fraude.

Les résultats des deux approches, ML et IA basée sur la Connaissance, sont analysés de manière critique. Les forces et les faiblesses de chaque approche sont disséquées en termes d'exactitude, d'interprétabilité, de scalabilité et d'adaptabilité. Une comparaison entre les deux paradigmes est établie, mettant en évidence leurs avantages distincts et leur potentiel d'hybridation.

En fin de compte, cette thèse contribue à l'avancement de la détection de la fraude financière en mettant en lumière la synergie entre les méthodes ML et l'IA basée sur la Connaissance. Elle présente une enquête approfondie sur les capacités de chaque approche, offrant ainsi une feuille de route pour la création d'un système hybride qui exploite les points forts des deux paradigmes. Les conclusions de cette recherche fournissent des informations précieuses aux chercheurs, aux praticiens et aux décideurs souhaitant construire des systèmes efficaces et résilients de détection de la fraude financière dans une ère de menaces en constante évolution.

Mots-clés : Fraude, Ontologie, Détection, Financier, Concepts de Connaissance, Classe, Basé sur les Règles, Scénarios, Intelligence Artificielle, Motifs, Entités, Validation, Ensemble de Données, Instantiation, K-moyennes, Centre de Gravité, Groupes, Distance

An Ontology-based Financial Fraud Detection Approaches

This thesis addresses the very timely challenge of detecting and preventing more and more numerous financial frauds.

The transition from physical/cash transactions to electronic/digital payments is highly explored by hackers and other cybercriminals. The primary targets of frauds are the banking sector, insurance, telecommunications, and e-commerce. Large number of efficient fraud detection solutions have been developed, but the traditional financial fraud detection tools are insufficient in the current dynamic context due to the incredible creativity of fraudsters, who are permanently inventing new methods, and the lack of flexibility of regulatory systems.

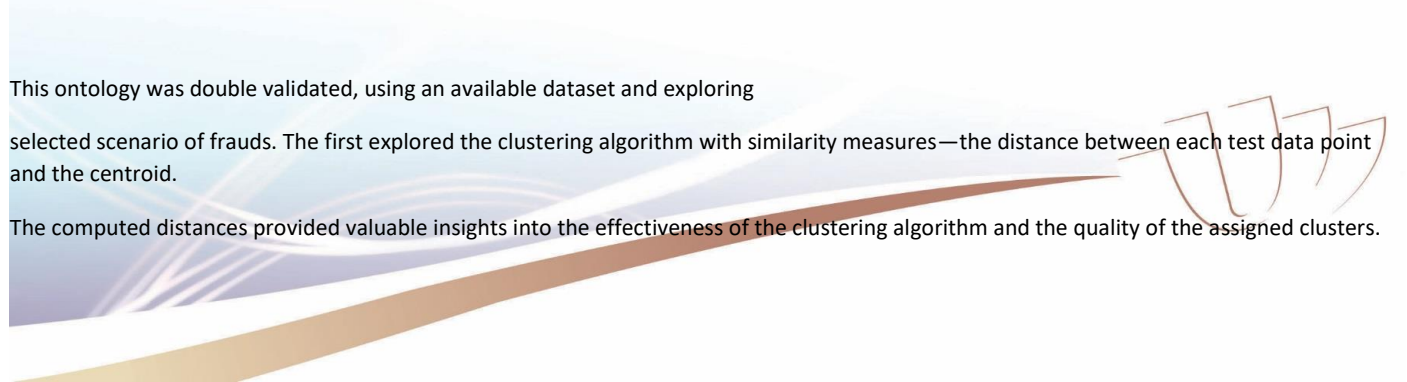
In aim to efficiently address this challenge, we studied existing fraud detection approaches based on exploring various Artificial Intelligence (AI) techniques, including connectionist (machine learning and deep learning), knowledge-based, and

blockchain, to detect frauds. After analysing fraud methods and their dynamics, and considering the availability of reliable data, we decided to build a financial-fraud ontology. %with the aim of having a "big picture" and an easy update.

This ontology was double validated, using an available dataset and exploring

selected scenario of frauds. The first explored the clustering algorithm with similarity measures—the distance between each test data point and the centroid.

The computed distances provided valuable insights into the effectiveness of the clustering algorithm and the quality of the assigned clusters.



This experiment allows for the evaluation of the model's performance in capturing the underlying patterns and characteristics of the dataset. It provides a quantitative measure of similarity, aiding in the assessment of the clustering algorithm's accuracy and the efficacy of the ontology-based approach. We also, gain valuable insights to capture fraud patterns. The obtained similarity scores obtained allowed for informed decisions and adjustments to improve the model's accuracy and reliability.

The second method consisted of exploring three typical scenarios of attacks with ontology and anti-fraud rules. These scenarios transcend traditional algorithms, intertwining structured knowledge representation and adaptive rule systems. Through meticulous exploration, we unravel the complexities of fraud within specific contexts, deciphering not just the 'what' but the 'how' and 'why' behind each malicious act. In essence, this part stands at the intersection of pioneering a new era in fraud detection. By fusing ontology and anti-fraud rules, we present a comprehensive framework. This framework not only identifies fraud but comprehensively interprets its nuances, offering a profound insight into the modus operandi of digital malefactors.

In the realm of identity theft, we unveil the hidden patterns and vulnerabilities. Through the lens of ontology and anti-fraud rules, we dissect the anatomy of stolen identities, shedding light on the subtle cues and intricate manoeuvres that often elude conventional detection methods.

While investigating investment fraud, we traverse the convoluted pathways of fraudulent investments, leveraging the power of ontology to unravel the complex web of deceit. By discerning the unique signatures of investment fraud, our research equips stakeholders with invaluable tools to safeguard investments and preserve financial integrity.

Finally, in the third scenario of banking fraud, where digital heists threaten the very core of financial institutions, our investigation takes a profound turn. By interweaving anti-fraud rules with the depth of ontological understanding, we dissect the modus operandi of cybercriminals.

The findings from the second part not only advance the field of fraud detection but also bear significant implications for financial institutions, regulatory bodies, and cybersecurity experts to bridge the gap between artificial intelligence and domain expertise.

The research findings indicate that ontology-based methods can enhance the accuracy and efficiency of financial fraud detection compared to alternative approaches in a dynamic context. In waiting for the dynamic models for ML and DL, the financial institutions can leverage the proposed framework to develop robust real-time fraud detection systems, reducing financial losses and enhancing customer trust.

Keywords : Financial Fraud, Ontology, Fraud Detection, Knowledge model, Reasoning rules, Machine learning, Dynamic model, Scenario, Clustering.

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